Industrial Structure, Technological Change, and U.S. Wage Inequality

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February 2026

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Abstract

This paper examines the determinants of U.S. wage inequality between industries. It distinguishes two channels of Skill-Biased Technological Change: a *quantity* effect, mirroring variations in the distribution of capital and labour inputs, and a *structural* effect, reflecting differences in substitution elasticities across factors of production. Counterfactual exercises shows that evolving technological heterogeneity in substitution patterns dominate wage dispersion. A quantitative decomposition reveals that structural forces, mostly reflecting substitutability of routine with non-routine labour, account for 94% of observed real wage variance, exacerbated by stronger sorting and segregation effects. Upon neutralising structural differences, the quantity channel reckons merely 6-15%. Indeed structural heterogeneity, rather than mere changes in factor quantities, is the dominant force behind U.S. wage inequality.

Jel: E24, J31, J82, L16

Keywords: wage inequality, technological change, labour force composition, elasticity of substitution

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Introduction

Wage inequality has been on the rise in the United States over the last decades, and uncovering its origins is a secular challenge in economics. One prominent explanation links inequality to employment polarization, whereby new technologies alter the task content of jobs and their relative demand (*e.g.*, Autor, Katz, et al. 2006; Acemoglu and Autor 2011). Additionally, the U.S. wage distribution reflects systematic differences across industries, largely due to self-selection of workers into workplaces (*e.g.*, Haltiwanger et al. 2024; Card, Rothstein, et al. 2024a).[1] Jointly, these perspectives suggest that technological change mediates wage inequality through differences in how factors of production interact and are organized across industries, rather than merely altering their relative allocation.

To trace how these structural and distributional patterns shape wage dispersion, this paper develops a quantitative general equilibrium framework where inequality in wages is influenced along two dimensions. If elasticities of substitution among factors of production are uniform across industries, inequality arises primarily from differences in capital and labour accumulation. If industries differ in their elasticities, even similar production inputs growth lead to diverging wage outcomes: inequality is driven less by the *quantity* of inputs and more by the *structure of technology* and its interaction with factors accumulation.

Motivated by these distinct yet interrelated mechanisms, the analysis quantifies the relative importance of each factor contributing to wage inequality. While the "quantity" channel has been examined at length, largely unexplored remains the "structural" realm – that is, cross-industry and over-time differences in the substitutability of capital and labour endowments. Key result is that inequality emerges from technological heterogeneity rather than from factor accumulation alone: what matters is not the amount of capital or labour, but how Skill Biased Technological Change (SBTC) affects industry-level elasticities of substitution.[2]

The initial empirical analysis shows that real wage inequality arises from how industries deploy technology and reorganize tasks simultaneously. As ICT technologies become pervasive, capital deepening is accompanied by systematic shifts in job tasks: the rising share of non-routine workers primarily reflects the displacement of routine workers rather than expansion in sectoral employment. Wage differentials move with this reorganization of production, as industries contributing more to U.S. wage inequality adopt ICT capital more intensively and shift their task composition toward non-routine activities.

I then construct a general equilibrium model of industries differing in substitu-

[1] Providing a sectoral perspective, Cerina et al. (2021) assert that the downward polarization of employment is characterized by the prevalence of routine tasks but it is fundamentally shaped by the type of sector (services and non-services) rather than the performed job task.

[2] SBTC posits that changes in technological endowments disproportionately benefits high-skilled workers while displacing lower-skilled ones, contributing to wage inequality. Refer, for example, to Tinbergen (1974), Katz and Murphy (1992), Card and DiNardo (2002), and Acemoglu and Autor (2011).

tion degrees among inputs of production. The market structure equips each industry with a block of monopolistically competitive firms employing two types of capital and labour: besides physical capital, non-routine labour is complementary to ICT capital, while routines are relatively substitutable (*e.g.*, Krusell et al. 2000). As for polarization, the labour market is designed around the endogenous Roy (1951)-style sorting, whereby workers allocate across firm-industry pairs according to comparative advantage. This mechanism links industry wage premium to sorting and segregation effects in a competitive labour market.[3]

Quantitatively, the model is estimated on U.S. industry-level data. The key production parameters are the elasticities of substitution between ICT capital, routine and non-routine labour tasks. The general equilibrium structure enables a clear discipline for identification, allowing production technology parameters to be recovered from sectoral labour shares. Estimates suggest "gross complementarity", ranging over 0.2 and 0.8. In the baseline cross-sectional calibration, substantial heterogeneity emerges: industries experiencing the largest wage growth exhibit the strongest impact of technological change, followed by those with minimal and moderate wage variation. The model successfully reproduces key features of the inter-industry wage structure and prevailing inequality levels.

Re-estimating the technological parameters over two sub-periods reveals that they evolved unevenly across industries both in direction and, more critically, in magnitude.[4] I use these observed shifts to conduct counterfactual simulations that decompose the industry component of U.S. wage inequality since the early 2000s, accounting for the interaction between technological change, labour force composition, and labour market concentration. To isolate the key mechanisms, I evaluate how period-variations in individual and combined technological parameters affect model outcomes relative to observed moments. In addition, sectoral changes in capital, labour, and estimated productivities are quantified once structural heterogeneity in production technology parameters is shut down.

My quantitative findings highlight that evolving elasticities of substitution are the primary determinants of U.S. between-industry wage inequality, far outweighing the role of changes in factor quantities. The counterfactuals pin down three main takeaways. Firstly, *structural effects dominate quantity effects*: allowing substitution elasticities to evolve heterogeneously across industries enables the model to account for 94% of the observed between-industry wage dispersion; by contrast, holding these elasticities uniform across industries, mere changes in production input quantities explain only 6% to 15%. Secondly, *substitution across job tasks is the dominant mechanism*: the "indirect effect" of SBTC – tasks' reorganization induced

[3] In Subsection 4.2, incorporating wage-setting power does not materially affect the model's predictions on between-industry wage inequality, consistent with the empirics by Card, Rothstein, et al. (2024a).

[4] Industries where wages increased the most (less) decreased their complementarity between technological capital and non-routine workers of about 25% (131%), while slightly increasing the complementarity between job tasks of 3% (11%). Differently, other industries undergone a reduction (20%) in substitutability of ICT capital-non-routine tuple, along a pale increase (3%) in job tasks substitution.

by technological change – is more impactful than its "direct effect" operating through capital and non-routine workers deepening, implying that inequality is tightly linked to the allocative effects on tasks across industries rather than to capital accumulation per se. Thirdly, *labour market concentration magnifies structural effects*: incorporating workforce allocation dynamics through sorting and segregation effects further amplifies the explanatory power of the model up to 98% of between-industry wage dispersion when interacted with the evolving technological structure.

As such, inspecting U.S. wage inequality requires moving beyond quantity-based narratives toward a clearer focus on structural heterogeneity and the dynamic nature of technological parameters across industries.

Related literature.– This research relates to several strands of the literature. A primary connection is with studies of wage inequality in the U.S., traditionally emphasizing *worker-level* (*e.g.*, Katz and Autor 1999; Autor, Katz, et al. 2008; Lemieux 2010) or between-firm (*e.g.*, Leonardi 2007; Card, Heining, et al. 2013; Song et al. 2019) determinants. Anyhow, recent evidence unveils how differences are clustered at the industry level. While earlier studies documented substantial but stable inter-industry wage differentials throughout much of the 20th century (*e.g.*, Slichter 1950; Cullen 1956; Krueger and Summers 1988; Allen 1995),[5] the industry component of wage inequality has grown markedly over the past three decades.[6] Central is Haltiwanger et al. 2024: wage differentials across industries drive the rise in U.S. wage inequality since the latest 1990s, with a small number of industries playing a disproportionate role.[7] Existing investigations overlook the key mechanisms of industry-level wage dispersion; this paper contributes by providing a quantitative framework that allow the role of structural variations in capital-labour substitutability to be measured separately from factor accumulation.

Second, my analysis dives into the task-based literature.[8] The role of tasks in shaping wage inequality is highlighted by Autor, Katz, et al. (2006), where technological change alters the distribution of job task demands, driving job polarization. I examine how industry-driven technological change reshapes workers' substitutability through tasks division and contributes to U.S. wage inequality.

[5] Refer to Krueger and Summers (1987) and Dickens and Katz (1987) for a discussion. A complementary literature explores the role of inter-industry dispersion in capital-labour ratios on wage inequality (*e.g.*, Montgomery 1991; Caselli 1999), while sectoral differences are important also to interpret both the racial (*e.g.*, Card, Rothstein, et al. 2024b) and the gender (*e.g.*, Fields and Wolff 1995; Blau and Kahn 2017) pay gaps, and the wage differentials from a behavioural perspective (*e.g.*, Thaler 1989).

[6] Allen (2001) notices how the "stability" argument is misleading since important within-industry factors changes over time, while the constancy depends on a persistent autocorrelation. Under a cross-country comparisons, substantial but stable inter-industry wage differentials exist among E.U. (*e.g.*, Genre, Kohn, et al. 2011; Genre, Momferatou, et al. 2005) and OECD (*e.g.*, Gittleman and Wolff 1993) countries.

[7] Rising between-industry wage inequality is well documented (*e.g.*, Davidson and Reich 1988; Bell and Freeman (1991); Howell and Wolff 1991; Allen 2001). Haltiwanger et al. (2024) show its importance on the increase in total U.S. wage inequality. Similar is Briskar et al. (2022) for Italy.

[8] Strong emphasis in measuring employment trends by job task content: rapid employment growth for non-routine tasks from early 1990s relative to routines (*e.g.*, Goos and Manning 2007; Cortes et al. 2017; Jaimovich and Siu 2020; Cerina et al. 2021; Siena and Zago 2024; Cossu et al. 2024).

Third, the paper builds on structural transformations studies (reallocation of economic activity across broadly defined sectors), in particular on the role of elasticity of substitution between capital and labour (*e.g.*, Buera and Kaboski 2012; Herrendorf, Rogerson, et al. 2014; Eden and Gaggl 2018). Sectoral differences in capital-labour elasticities of substitution, as in Herrendorf, Herrington, et al. (2015), shape factor reallocation across broadly defined sectors. This perspective intersects my results: ample differences in industry-level elasticities of substitution and their asymmetric shifts majorly explain wage inequality across industries.

Roadmap.– This introduction is succeeded by Section 1, which provides the motivating evidence at which the model presented in Section 2 refers to. Section 3 shows and assess the performance of the calibration strategy, while Section 4 quantifies the relevance of given parameters in accounting for the observed level of U.S. wage inequality. Section 5 determines the importance of capital and labour quantities and of an estimated measure of sectoral productivity. Last section concludes.

1. Motivating Evidence

To account for trends in wage dispersion I employ annual U.S. data, and the 3-digit U.S. 2017 North American Industrial Classification System (NAICS) is adopted for the industry definition. Two different sources are combined: first, information on the stocks of physical and technological capital are recovered from the Bureau of Economic Analysis (BEA), in the *Detailed Data for Fixed Assets* tables; the second source is the *Occupational Employment and Wage Statistics* (OEWS) program of the Bureau of Labor Statistics (BLS), with data on occupational employment and wage rates. Capital inputs are grouped into four categories using BEA data for 1998-2022: physical capital, digital equipment, intangible capital, and ICT capital. Occupational data from the BLS cover routine and non-routine tasks for private U.S. industries from 2003 onward. BEA and BLS data are harmonized at the 3-digit NAICS level, yielding a balanced panel of 62 private industries observed annually from 2003 to 2022. For additional details refer to Appendix A.

1.1. Validating the sample

From Haltiwanger et al. (2024), total wage variance growth across industries can be decomposed as each industry-$s \in \mathcal{S}$ contribution to wage inequality

$$\underbrace{\Delta\, var\left(w\left(s\right)-\overline{w}\right)}_{\substack{\textit{between-industry}\\ \textit{wage variance growth}}} = \sum_{s} \overbrace{\Delta \underbrace{\left(\frac{\ell\left(s\right)}{\ell}\right)}_{\substack{\textit{employment}\\ \textit{share}}} \underbrace{\left(w\left(s\right)-\overline{w}\right)^{2}}_{\substack{\textit{relative}\\ \textit{wage}}}}^{\substack{\textit{industry-}s\textit{ contribution to}\\ \textit{between-industry wage variance}}} \tag{1}$$

in which the employment share component properly weights each industry. In each year, $w\left(s\right)$ is the average real log-wage of industry-s, $\overline{w}$ the associated economy-wide period mean, while $\ell\left(s\right)$ and ℓ represent sectoral and aggregate employment,

Table 1: Contribution to between-industry wage variance

contribution	*industries*	*variance*		***share*** *employment*	***shift-share*** *wage*	*employment*
> 5%	3	.26	54%	8%	90%	10%
1% to 5%	7	.14	29%	18%	76%	24%
.05% to 1%	6	.05	10%	13%	124%	-24%
-.05% to .05%	46	.03	7%	61%		·
quantiles						
0-25	15	.24	50%	31%	91%	9%
25-50	16	.04	8%	28%	77%	23%
50-75	16	.02	4%	15%		·
75-100	15	.18	38%	26%	98%	2%

Estimates of eq. (1) for 3-digit U.S. 2017 NAICS industries. The last two columns quantify eq. (2) and, under "·", the shift-share for employment is highly less than zero. Operator Δ in the equations is $x_t - x_{t-1}$. Industries are grouped according to their contribution to between-industry wage inequality in the first part of the table while, under "quantile", by the overall change in real log-wage. Source: BEA and own calculations.

respectively. A shift-share analysis provides further insights on industry factors accounting for such variance growth, inspecting the importance of relative wage changes versus employment share changes:

$$\underbrace{\Delta\left(\frac{\ell(s)}{\ell}\right)\left(w(s)-\overline{w}\right)^2}_{\substack{\text{industry } s \text{ contribution to} \\ \text{between-industry wage variance}}} = \underbrace{\overline{\left(w(s)-\overline{w}\right)^2}\,\Delta\left(\frac{\ell(s)}{\ell}\right)}_{\text{shift share: employment}} + \underbrace{\overline{\left(\frac{\ell(s)}{\ell}\right)}\,\Delta\left(w(s)-\overline{w}\right)^2}_{\text{shift share: wage}} \tag{2}$$

The two decompositions are in Table 1. Wage inequality growth is highly concentrated across industries: 15 out of 62 sectors account for 93% of the total increase in real wage dispersion, despite employing less than 40% of the workforce. Concentration is even sharper at the top, with just three industries owing more than half of the total growth while representing only 8% of total employment. Contributions to wage dispersion are further examined through a shift-share analysis in the last two columns of Table 1. The bulk of wage variance growth stems from shifts in relative wages rather than from changes in employment composition: an expansion in industries' labour force is thus marginal, whereas their relative wage dynamics dominate.

Similarly, the second panel of Table 1 groups industries by wage growth. Top and bottom 25% together account for 88% of the growth in wage inequality while employing nearly 60% of the workforce, and relative wage movements significantly dominate: changes in sectoral employment shares contribute little to wage dispersion, reconciling with Fact 3. This aligns with Haltiwanger et al. (2024).

Fact 1 (Contribution) *A small subset of industries drives the rise in wage inequality; these are in the tails of the industry-level wage growth distribution.*

What intrinsic characteristics underpin the dispersion of wages across industries? Why does the employment size of industries seem to play a minimal role?

(A) INTERQUARTILE RANGE, 90^{th}-10^{th} (B) LOG-VARIANCE

FIGURE 1: CROSS-INDUSTRY DISPERSIONS OF CAPITAL TYPES

Note: dispersion in industry-specific real log-wage, physical, ICT, intangible capital types and digital equipment per capita. Solid red and blue lines relate to wages and ICT capital, while dashed green, purple and orange and lines to intangible and physical capital, and digital structures, respectively. Panel 1a plots the series of the difference between top and bottom 10%, while Panel 1b the log-variances. Series are standardized on the y-axis and indexed to unity in 1998. Plots are referred to 3-digit U.S. 2017 NAICS industries. *Source*: BEA and own calculations.

1.2. U.S. INDUSTRIES DIGITALIZATION

As a first step, the analysis shall examine whether cross-industry differences in capital stocks are reflected in wage differentials. To capture rising dispersion, Figure 1 traces the evolution of the gap in capital-labour intensities and real wages across industries. Panel 1a measures industry-level dispersion using the interquartile ranges – the gap between top and bottom 10% industries: dispersion in ICT capital ratios increases steadily over time, widening the gap between industries with high and low ICT capital per worker, while intangible assets and digital equipment display similar, though less uniform, patterns. By contrast, dispersion in physical capital declines, suggesting a convergence across industries.

FACT 2 (Industry capital gaps) *Dispersion in physical capital-labour ratio is decreasing across industries, while it is not that of ICT capital-labour ratio.*

Comparing inter-industry differentials in capital intensities with those in labour remuneration reveals that real wage gaps are primarily associated with disparities in ICT capital, rather than with other forms of capital. This pattern highlights the role of a macro-complementarity:[9] as shown in Panel 1b, dispersion in intangibles or digital equipment alone fails to match the evolution of wage dispersion unless the two are combined into a unified ICT capital measure: intangibles loosely track wage

[9] Emphasized by a growing empirical and methodological debate. Corrado et al. (2017) show that returns to ICT depend critically on otherwise "unmeasured" intangible assets, documenting ICT-intangible complementarity in both U.S. and EU data. Crouzet et al. (2022) note that intangible capital, lacking physical embodiment, requires digital infrastructure or storage medium to operate.

dispersion until 2012, while digital equipment becomes relevant only thereafter. The interaction between these two is, therefore, central.

Further evidence on the role of ICT complementarities on real wages is provided in Table A.1, where fixed-effects panel regressions largely confirm the preceding patterns. While increases in physical capital are paired with higher real wages, ICT capital alone is weaker (column 1), suggesting substantial heterogeneity across industries. Intangible capital is central, accounting for larger wage variations (column 2), whereas digital equipment on its own appears less influential. Crucially, their interaction (column 3) remains significant, and continues to be when all capital types are included (column 4).

1.3. Accounting for workforce composition

Since their advent, technical and technological advances have raised issues about their influence on employees' qualification requirements at both the aggregate and industry levels (*e.g.*, Horowitz and Herrnstadt 1966; Rumberger 1981; Autor 2022), with effects operating less through absolute employment levels than through relative occupational shifts.[10] A natural question is whether changes in the workforce composition across tasks reflect shifts in industry size or the reallocation of worker types within industries. Denoting the set of tasks as $\{a, a'\} \in \mathcal{A}$ and with $s \in \mathcal{S}$ that of industries, the following decomposition provides the answer to this inquiry:

$$\Delta\left(\frac{\ell(a)}{\ell(a')}\right) = \underbrace{\sum_s \overline{\left(\frac{\ell(a,s)}{\ell(s)}\right)} \Delta\left(\frac{\ell(a,s)}{\ell(a',s)}\right)}_{\textit{within-industry: substitution effect}} + \underbrace{\sum_s \overline{\left(\frac{\ell(a,s)}{\ell(a',s)}\right)} \Delta\left(\frac{\ell(a,s)}{\ell(s)}\right)}_{\textit{between-industry: size effect}} \tag{3}$$

Shifts in the economy-wide ratio of task-a to task-a' is decomposed into within- and between-industry components. The former captures the substitution and reallocation of tasks within industries, while the latter reflects changes in the relative size of task-specific employment across industries. The steady rise in the share of non-routine workers alongside the decline in routine employment (*e.g.*, Jaimovich, Zhang, et al. 2024) forms the perspective on whether this trend is driven by changes in employment size across industries or by task substitution within them. This structural change is exactly the underlying logic behind eq. (3).

Table 2 reports the decomposition. In line with Table 1, within-industry adjustments shape the rise in the non-routine-to-routine task ratio, replacing away routinary jobs rather than workforce expansion. Conversely, movements in the routine-to-non-routine ratio are small and mainly between industries, reflecting limited employment growth rather than increased reliance on routine tasks.

Fact 3 (Labour force composition) *Increases in non-routine relative share are determined by a substitution effect rather than by the employment size of industries.*

[10] Refer to Melman (1951) and Braverman (1974) for an analysis over the period 1820-1970.

Table 2: Labour force composition

	routine		***non-routine***	
interval	*within*	*between*	*within*	*between*
2003-2008	83%	17%	38%	62%
2009-2015	81%	19%	58%	42%
2016-2022	78%	22%	69%	31%

Quantification of eq. (3); changing components (Δ) are linear trends predicted by a Fixed Effects (FE) regression of the form $y_t = \beta_c + \beta_t \mathcal{X}_t + u_t$, with y representing the task ratio, and $\mathcal{X}$ a vector of industry and year fixed effects, at 3-digit U.S. 2017 NAICS industries. Source: BLS and own calculations.

Next, the analysis examines the nexus between capital accumulation and labour force composition, asking whether industries with faster real wage growth expand ICT capital and rely more on non-routine labour – unlike routine workers performing mainly manual tasks. Using industry-level fixed-effects regressions, and grouping industries by their position in the real wage growth distribution:

$$\Delta \, log \, w_{\omega t}\,(s) = \beta_{c,\omega} + \beta_{i,\omega}\mathcal{X}_{i,\omega t} + \delta_{j,\omega}\mathcal{Z}_{j,\omega t} + u_{\omega t} \tag{4}$$

where ω identifies each quantile in Table 1, $\mathcal{X}_i$ bundles percentage changes in both ICT-to-physical capital (k) and non-routine-over-routine workers (ℓ) ratios, and $\mathcal{Z}_j$ serves as controls. In Table 3 the interaction between ICT capital accumulation and task reallocation is particularly pronounced in industries at both ends of the wage growth distribution, thereby aligning with stylized Fact 1. High-growth industries exhibit large joint increases in ICT capital and non-routine employment, translating into substantial wage gains (1.56), whereas low-growth industries show milder adoption and weaker wage performance (.876). No significant effects for middle industries, contributing relatively little to overall wage inequality. Table A.2 reinforces this pattern. Column 3 shows that the interaction term $\beta_{\Delta k \times \Delta \ell}$ of joint changes in ICT capital and task composition has a large and statistically significant effect on real log-wage growth (.919), far exceeding the impact of isolated changes in either ICT capital (.009) or labour composition (.035). Therefore, wage inequality is driven primarily by the concentrated performance of industries at the extremes of the wage growth distribution, where the combined evolution of ICT capital and non-routine labour is most pronounced. The evidence underscores that their interaction, rather than isolated changes, plays a central role in shaping U.S. wage dispersion.

Fact 4 (Dynamic transformations) *Industries marked by highest changes in real wages experience a substantial rise in their non-routine workers relative share along an increasing ICT capital ratio dynamics.*

When jointly considered, the presented stylized facts reveal some important characteristics of the U.S. inter-industry wage structure: inequality of industries' real wages is shaped by a growing dispersion in technology (Fact 2) reflecting, in turn, variations in their labour force composition (Fact 3). In other words, industries con-

Table 3: Combined regressions by groups, percentage changes

	$\Delta\, log\, w\, (s)$			
	0-25 quantile	*25-50 quantile*	*50-75 quantile*	*75-100 quantile*
$\beta_{\Delta k}$	-.693	.302***	.078	-.082*
	(.618)	(.072)	(.179)	(.042)
$\beta_{\Delta \ell}$	.041***	-.297	.015***	.052*
	(.004)	(.247)	(.003)	(.025)
$\beta_{\Delta k \times \Delta \ell}$	.876***	-1.304	-.643	1.560*
	(.100)	(2.945)	(.470)	(.789)

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022. Each Fixed Effects (FE) regression – groups of industries by quantiles (ω) on their overall growth in real wage –, is of the form of eq. (4), with $\mathcal{X}_i$ representing the percentage change in both ICT-to-physical capital and non-routine-over-routine workers ratios, and $\mathcal{Z}_j$ being a set of time-varying controls. Variables are all in log format. Constant not reported to save space. Source: BEA, BLS and own calculations.*

tributing more to wage inequality (Fact 1) tend to exhibit a higher wage effect from the substitution of routine with non-routine tasks and a significant increase in their stock of ICT relative to physical capital (Fact 4). Reminiscent from the introduction, these findings are opened to two possible interpretations: wage inequality across industries could be the result of *(i)* a *quantity* effect, under different trajectories in factor of productions; or *(ii)* a *structural* effect, stemming from changes in the technological parameters that govern the relationship between capital and labour types. These general equilibrium considerations provide the motivating background for the quantitative counterfactual exercises.

2. Structural Model

Moved by the presented empirical regularities, I build and simulate a structural general equilibrium model. The households block features a skill-related heterogeneity: households are *ex-ante* divided in routine and non-routine workers, sorting into firms and industries given heterogeneous productivities. Final and sectoral outputs are competitively aggregated, while firms in each industry compete monopolistically. They employ ICT and non-ICT capital types, where non-routine labour is complementary to ICT capital, while routine workers are substitutable. Hence, the structure of production functions is analogous in all industries except for their degrees of substitutability across factors of production. To deepen the role of substitution parameters, productivity differences across industries are excluded.[11]

[11] Productivity and structural changes are theoretically analysed by Ngai and Pissarides (2007), linking the elasticity of substitution among intermediate inputs to the sectoral reallocation of employment. Section 5 inspects the role for estimated industry-level productivity series on wage dispersion.

2.1. Households

A unit mass of households is split in tasks, $a = \{rt, nrt\}$. Household-i of type-a gets utility from consumption, $\mathcal{C}_i$, paired with a known idiosyncratic factor, $\wp_h^i(a,s)$, measuring its efficiency of working in firm-$h \in [1,H]$, industry-$s \in [1,S]$:

$$\mathcal{U}_h^i(a,s) := log\ \mathcal{C}^i + log\ \mathcal{B}_h(a,s) + \wp_h^i(a,s) \tag{5}$$

Increasing utility in $\wp_h^i(a,s)$ suggests that a worker is optimally allocated to a workplace that maximizes its productivity, as households self-select into job locations by comparing the potential outcomes of various firm-industry pairs.[12] In other words, a self-selection based labour market involves individuals choosing the firm-industry combination that maximizes their utility based on their spectrum of productivities across different sectors. Borrowing from the spatial literature, a firm-h, industry-s amenity for household type-a, $\mathcal{B}_h(a,s) = \left[g_h(a/a',s)\right]^{-\varsigma}$, enters multiplicatively in the utility function (*e.g.*, Kleinman et al. 2023). Here, $g_h(a/a',s)$ reflects the relative share of task-a in task-a', negatively scaled by the elasticity parameter ς, a degree of substitutability among tasks.[13]

An household can invest either in bonds (b^i, which are in zero net supply) at the rate r, and in capital types $k^i(j), \forall j = \{phy, ict\}$, while receiving dividends $\mathcal{D}^i$ from firms, so that its expected utility problem displays the budget constraint

$$\mathcal{C}_t^i + I_t^i(phy) + I_t^i(ict) + b_{t+1}^i - (1+r_t)\, b_t^i = \\ w_{h,t}(a,s)\, \ell_h^i(a,s) + R_t\Big(k_t^i(phy) + k_t^i(ict)\Big) + \mathcal{D}_t^i$$

where the wage rate associated to type-a in firm-industry pair h,s is $w_h(a,s)$, with labour $\ell_h^i(a,s)$ inelastically supplied and set to unity. Any capital stock depreciates at a rate δ and accumulates over time according to a quantity-adjusted law of motion: it is a negative function of ζ_t^i (the relative ICT capital share) that scales new capital investment, $I_t^i(j)$: as the capital stock owned by household-i becomes more technologically advanced (higher ζ_t^i), a higher rate of capital investment is required to maintain the future stock of that particular capital type.[14] ζ_t^i thus represents the technology level of household-i's total capital stock.

Inter-temporal maximization displays a canonical Euler condition for future path of marginal utility from consumption, and the capital rental rate evolution:

$$R_t = (1+r_{t+1})\,\zeta_t - (1-\delta)\,\zeta_{t+1} \tag{6}$$

[12] Under high $\wp_h^i(a,s)$, household-i is utilitarianism better off, as it reflects its ability to achieve the maximum when working as type-a in the firm-industry h,s tuple. Due to the absence of a non-employment option and unemployment, the terms household(s) and worker(s) are interchangeable.

[13] $\mathcal{B}_h(a,s)$ characterizes the equilibrium wages as a function of the task ratio as well, as in Section 1. Its relevance in determining the endogenous labour supply of eq. (7) – *i.e.*, the outcome of the sorting choice by the part of households – will be empirically tested and validated in Subsection 3.1.

[14] Capital dynamics follows a law of motion as $k_{t+1}^i(j) = \frac{I_t^i(j)}{\zeta_t^i} + (1-\delta)\,k_t^i(j)$, $\forall j = \{phy, ict\}$. A higher ζ_t^i is an improvement in the technological sophistication of the capital bundle, symptom that any additional unit of invested capital faces diminishing marginal returns (*e.g.*, Inada 1963)

where changes across states of R_t are shaped by changes in the relative quantities across capital types, $\zeta_t = \int_i \zeta_t^i \, \mathrm{d}i$. In the spirit of Karabarbounis and Neiman (2014), eq. (6) determines that investing in capital types is profitable as long as the marginal benefit of investment is at least lower than its marginal cost.

In a market economy workers do not randomly sort across workplaces. Rather, in a discrete choice Roy (1951) model of self-selection, households locate based on how their capabilities vary across firms and industries.[15] The idiosyncratic *productivity* of household-i, task-a, working in firm-industry h, s tuple is drawn once from a multivariate Frechét-type cumulative distribution:

$$\mathcal{F}_i\left(\wp^i_{h,\ldots,H}(a,1), \ldots, \wp^i_{h,\ldots,H}(a,S)\right) = exp\left[-\sum_s \left(\int_h \wp^i_h(a,s)\, \mathrm{d}h\right)^{-\theta}\right]$$

Its shape parameter, $\theta > 1$, governs the degree of dispersion in idiosyncratic productivities' draws (*i.e.*, the labour supply elasticity), and lower values of θ imply major degrees of dispersion – *i.e.*, a more elastic labour supply response to wage differences.[16] Location and scale parameters are normalized to 1.

Central is that workers are potentially employable in any workplace, but their productivity varies across firm-industry pairs. This structure captures both absolute and comparative advantages. As for absolute advantages, any worker performing a task can, in principle, work in any location. Comparative advantage, however, determines actual employment: workers choose to be employed in the firm and industry where their relative productivity is highest. Consequently, labour allocation reflects not only wage differences but also task-specific efficiency across different environments. This mechanism generates a "segregation" effect, whereby firms increasingly hire workers whose comparative productivity aligns closely with their operational and technological requirements.

The labour market, while competitive and frictionless in a traditional sense, is structurally imperfect due to this idiosyncratic heterogeneity. The distribution of households' productivities – and thus the self-decision around employment choices – makes the labour supply endogenous, shaped by both firm-industry level wages and the resulting task-specific comparative advantages. Described sorting preferences allows to derive analytical form for the measure of each worker-type in firms and industries given the respective wage level: the labour supply curve of each a, h, s takes the form of

$$\ell_h(a,s) = \left(\frac{w_h(a,s)}{W(a)} \; \frac{\mathcal{B}_h(a,s)}{\mathcal{B}(a)}\right)^{\theta} \tag{7}$$

with $W(a) = \sum_{h,s} w_h(a,s)$ and $\mathcal{B}(a) = \sum_{h,s} \mathcal{B}_h(a,s)$. Labour supply depends not

[15] Self-selection concept dates back to Roy (1951)'s verbal-model, whose scope is to describe how the subjective choice for a given occupation would affect the distribution of wages: it is not only a function of gaps in potential wages, but rather it includes also occupational sorting choices.

[16] It measures the centrality of a change in labour supply of task-a induced by a change in its firm-h, industry-s wage level. Rogerson 2024 stresses the importance of the (aggregate) labour supply elasticity, pointing to both intensive (average hours worked) and extensive (employed people) margins.

only on the wage offered to task-specific workers but also on its relative attractiveness, captured by the "pay premia" – how the wage compares with those offered elsewhere. Additionally, to attract a larger share of workers with similar productivity (the "sorting" effect), a firm must offer higher wages.[17]

Eq. (7) summarizes the role of wage premium, sorting and segregation, pivotal in shaping wage inequality.[18] Across industries, Haltiwanger et al. (2024) highlight how wage dispersion is increasingly driven by sorting associated to wage premium, while in Card, Rothstein, et al. (2024a) the correlation between workers sorting and the industry wage premium is high.[19] In other words, self-selection directly enhances a mechanism on the critical interaction of the workplaces' labour force composition with sorting and segregation effects, both associated to the firm-industry wage premium from not perfectly elastic labour supplies.

Remark 1 (Labour market elasticity, sorting and segregation) *Selected specification of households' productivities allows to characterize firms and industries workforce composition of tasks as determined by a not fully elastic labour supply elasticity, shaped by comparative advantage, interacting with equilibrium wage premium designed through perfect competition in the labour market.*

Labour supply curves in eq. (7) are upward-sloping not due to frictions, but due to the structure of comparative advantage and the imperfect substitutability of workers across jobs. In this sense, workers in the same task are "highly rival factors" (Hicks 1932) since they can be freely substituted for one another, yet ensuring that high-wage workers do sort in high-wage firms and industries, and also that more efficient people would be preferred to get a job in high-wage workplaces than ones who are less productive.

Assumption 1 *Since households' productivities are Frechét-distributed, workers are perfectly mobile across firms within an industry, and immobile across industries.*

Parameter θ governs the elasticity of labour supply, and it works as a structural measure of workers' concentration. As idiosyncratic productivities become more dispersed, the strength of sorting and segregation mechanisms diminishes, reducing labour market polarization. Figure 2 provides a conceptual illustration of how variations in productivity dispersion shape employment allocations.

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[17] Endogenous sorting is determined by a comparing wages and productivities across workplaces, that is $prob\left[w_h(a,s)\,\mathcal{B}_h(a,s)\,\wp^i_h(a,s) > w_h(a,s')\,\mathcal{B}_h(a,s')\,\wp^i_h(a,s')\right], \forall s' \in \mathcal{S}$.

[18] In continuity with Card, Heining, et al. (2013) for Germany, several empirical works (*e.g.*, Alvarez et al. 2018 for Brazil; Morin 2023 for Denmark; Briskar et al. 2022 for Italy; Card, Cardoso, et al. 2016 for Portugal; Håkanson et al. 2020 for Sweden; Barth et al. 2016, Song et al. 2019 and Haltiwanger et al. 2023 for United States) detect how these major effects are increasingly impacting the distribution of wages.

[19] Using a cross-sectional approach at sectoral level, Gibbons et al. 2005 show how the high-skill occupational sorting is determined by differentials in return to skills, and the comparative advantage motive, rather than learning, seems to affect the resulting wage premiums.

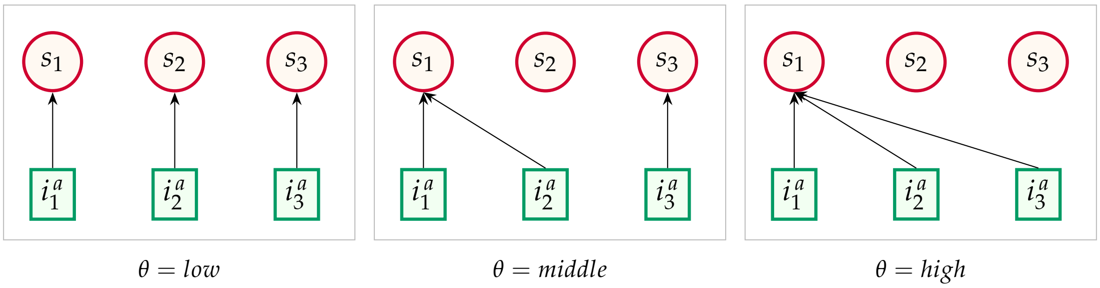

FIGURE 2: PRODUCTIVITY DISPERSION, SORTING AND SEGREGATION

Note: stylized examples on how productivity dispersion of households relates to labour market concentration at industry level. Each i^a represents an household of task-a, while s is a specific industry. The higher is the value of θ, the lower is the dispersion of households' productivities, so that the larger are the effects of sorting and segregation, thus the higher is the degree of labour market concentration of workers across industries.

EXAMPLE 1 (Labour market concentration) *Consider a stylized economy with $\{s_1, s_2, s_3\} \in \mathcal{S}$ industries, each represented by a single firm, and three households of unique type-a. Households' productivities may acquire only three values, $\wp^i(a,s) = \{low, middle, high\}$ depending where employed, with parameter θ determining even dispersion. When $\theta = low$, productivity levels are highly dispersed across households and industries, leading each household to sort into a different industry, and resulting in low labour market concentration. Conversely when $\theta = high$: productivities result to be very clustered, with all households concentrating in the same industry, thereby generating strong polarization. An intermediate case occurs when $\theta = middle$, yielding to partial sorting pattern and reflecting moderate labour market concentration.*

Under a broader perspective, I am interpreting type-a workers' concentration at industry level to be directly proportional to the degree of variability in households' productivities, and therefore to the labour supply elasticity.

2.2. INDUSTRIES

The supply side is made of a countable set of industries $s \in \mathcal{S}$, each producing a specific variety $y(s)$. A perfectly competitive final good producer combines intermediate outputs from industries using a Constant Elasticity of Substitution (CES) technology

$$Y = \left(\sum_s y(s)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$$

where $\eta > 1$ is the elasticity of substitution among varieties of differentiated goods/services from industries. Final producer's optimizing behaviour gives birth to a relative demand of each industry-s output, given by $\frac{y(s)}{Y} = \left(\frac{p(s)}{P}\right)^{-\eta}$, with aggregate price index $P = \left(\sum_s p(s)^{1-\eta}\right)^{\frac{1}{1-\eta}}$ since final good is competitively supplied.

A unitary mass of monopolistically competitive homogeneous firms, indexed with $h \in \mathcal{H}$, is comprised in all industries. Given market power of firms, aggregation for industry-s bundles together different varieties by a new CES aggregating function

$$y(s) = \left(\int_h y_h(s)^{\frac{\epsilon-1}{\epsilon}} \, \mathrm{d}h \right)^{\frac{\epsilon}{\epsilon-1}}$$

where $\epsilon > 1$ is the sectoral elasticity of substitution among firms' varieties. In a way analogous to final producer, the conditional demand of firm h, s arises from competitive profit maximization at industry level:

$$y_h(s) = \left(\frac{p_h(s)}{p(s)} \right)^{-\epsilon} y(s) \tag{8}$$

with the sectoral price index as the *numeraire*, $p(s) = \left(\int_0^1 p_h(s)^{1-\epsilon} \, \mathrm{d}h \right)^{\frac{1}{1-\epsilon}} = 1$. Firm-$h$ in industry-s comprises two types of workers, $a = \{rt, nrt\}$: non-routine (nrt) workers are complementary to ICT capital, while routines (rt) are substitutable with the ICT composite good produced under the non-routine-ICT capital complementarity. Its production function exploits that in Krusell et al. (2000):

$$y_h(s) = \left(k_h(phy, s) \right)^{\alpha} \left[\mu \left(\ell_h(rt, s) \right)^{\varsigma} + (1-\mu) \left(q_h(s) \right)^{\varsigma} \right]^{\frac{1-\alpha}{\varsigma}} \tag{9}$$

with

$$q_h(s) = \left[\lambda \left(k_h(ict, s) \right)^{\varrho} + (1-\lambda) \left(\ell_h(nrt, s) \right)^{\varrho} \right]^{\frac{1}{\varrho}}$$

where $k_h(phy, s)$ and $k_h(ict, s)$ are non-ICT and ICT capital, $\ell_h(rt, s)$ and $\ell_h(nrt, s)$ measures routine and non-routine workers, with μ and λ being weights governing output share of each firm h, s; parameter α is the physical capital share of output. Elasticity indicators (ϱ, ς) are $\varrho = \frac{\rho-1}{\rho}$ and $\varsigma = \frac{\sigma-1}{\sigma}$.

ASSUMPTION 2 *Substitution elasticities, $\rho_s, \sigma_s \in [0, \infty)$, physical capital and distributive shares, $\alpha_s, \mu_s, \lambda_s \in (0,1)$, are industry-specific. Since elasticities are finite, each industry adopts strictly positive levels for capital and labour quantities.*

Parameter ρ is the elasticity of substitution between ICT capital and non-routine labour input (*i.e.*, the CES composite), while σ identifies the elasticity of substitution between routine labour input and the CES composite. The CES structure imposed above yields a symmetry constraint in the interpretation of σ: the elasticity of substitution between routine and non-routine workers is the same as that between routine labour force and the ICT composite. Capital-task complementarity requires $\sigma > \rho$: this condition resorts the idea that technological process (namely an increase in ICT capital relative to non-ICT one) widens the relative demand of non-routine labour force, thus lessening that of routines.

Given the market structure, wages are set at the firm level under monopolistic competition within industries. Profit maximization implies the Cobb-Douglas nested CES production function in eq. (9) to be subject to the conditional demand in eq. (8).

$$\max_{p_h(s), k_h(j,s), \ell_h(a,s)} \left[\mathcal{D}_h(s) \mid y_h(s) \right]$$

with $\mathcal{D}_h(s)$ being profits. Since firms within an industry are homogeneous, the wage choice of firm h, s does not affect the sectoral wage level.[20]

ASSUMPTION 3 *Firms are atomistic, so that any wage level offered by a firm h, s only does not affect the sectoral one:* $\frac{\partial w(a,s)}{\partial w_h(a,s)} = 0 \; \forall a, h, s.$

Under this assumption, equilibrium wages paid by firm h, s are set, for routine and non-routine workers, respectively, to

$$\begin{aligned} w(rt,s) &= \left[\Lambda(s)\, \chi(rt,s)\, \left(k(phy,s)\right)^{\alpha} \mathcal{V}(s)^{\frac{1-\alpha-\varsigma}{\varsigma}} \left(\frac{\mathcal{B}(rt,s)}{\mathcal{WB}(rt)} \right)^{\theta(\varsigma-1)} \right]^{\frac{1}{1+\theta-\theta\varsigma}} \\ w(nrt,s) &= \left[\Lambda(s)\, \chi(nrt,s)\, \left(k(phy,s)\right)^{\alpha} \mathcal{V}(s)^{\frac{1-\alpha-\varsigma}{\varsigma}}\, \mathcal{Q}(s)^{\frac{\varsigma-\varrho}{\varrho}} \left(\frac{\mathcal{B}(nrt,s)}{\mathcal{WB}(nrt)} \right)^{\theta(\varrho-1)} \right]^{\frac{1}{1+\theta-\theta\varrho}} \end{aligned} \tag{10}$$

where industry-specific composite parameters are given by $\chi(rt,s) = (1-\alpha)\,\mu$ and $\chi(nrt,s) = (1-\alpha)(1-\mu)(1-\lambda)$, while $\Lambda(s) = p(s)\; \mathcal{M}^{\epsilon,-1}$ is an indicator combining the industry-specific price level multiplied by the price mark-up, $\mathcal{M}^{\epsilon} = \frac{\epsilon}{\epsilon-1}$, and the aggregate element $\mathcal{WB}(a) = \sum_s w(a,s)\, \mathcal{B}(a,s)$; refer to Appendix B for the definition of the other components. Aggregate industry wage is thus found by averaged aggregation among job tasks: $w(s) = (\max \mathcal{A})^{-1} \sum_a w(a,s)$.

Before delving into the intuition behind the equations, it is important to establish a clear understanding of why wages are at industry level: gaining this foundational insight will help to better contextualize eq. (10).

PROPOSITION 1 (Firm and industry layers) *In an economy characterized by sorting and segregation effects and a not perfectly elastic labour supply, as long as (i) firms within an industry have the same size; or (ii) workers are perfectly mobile across firms within an industry,*

$$\frac{\partial \ell_h(a,s)}{\partial w_h(a,s)} = -\frac{\partial \ell_{h'}(a,s)}{\partial w_h(a,s)} \quad \textit{and} \quad \frac{\partial w_h(a,s)}{\partial \ell_h(a,s)} = \frac{\partial w_{h'}(a,s)}{\partial \ell_{h'}(a,s)} \quad ,$$

profit-maximizing wages set by firms in a specific industry are equal. Moreover, (iii) workers are immobile across firms between industries.

Proof in Appendix B.

Worker mobility across firms and industries is governed by the distribution of households' idiosyncratic productivity $\wp_h^i(a,s)$. The *max-stability* of the Fréchet distribution ensures that, once a workplace is chosen, workers do not move across firms or industries, as each household selects its utility-maximizing productivity $\wp_h^i(a,s)_{max}$. Note that, under within-industry firm homogeneity, productivity is identical across firms in the same industry.

[20] This assumption is a natural outcome in the proof of Proposition 1 in Appendix B.

Remark 2 (Max stability and workers' movement) *As from eq. (5), a worker has no incentive to choose a workplace where performing worse since*

$$\mathcal{U}_h^i\left(a,s\right)\ \mid \wp_h^i\left(a,s\right)_{max} \gg \mathcal{U}_h^i\left(a,s\right)\ \mid \forall\, \wp_h^i\left(a,s\right) \in \left[\wp_h^i\left(a,s\right)_{min}\,,\, \wp_h^i\left(a,s\right)_{max}\right)$$

is ensured by each worker's selection of its maximal productivity for the h,s tuple. In addition, since firms are symmetric within an industry, the sorting choice is uniquely driven by the dispersion of households' productivities between industries so that workers are free to move across firms within an industry while getting the same utility.

Interpreting eq. (10), both wages rise with non-ICT capital and the price level, and each is function of the economy-wide wage for the corresponding task. Due to the production structure, the wage of non-routine workers is directly influenced by the joint evolution of ICT capital and non-routine employment, while routine wages are not. Task differences also manifest through the distinct roles of the two elasticities of substitution, ρ and σ: routine wages are primarily affected by substitution among themselves and non-routine tasks, whereas non-routine wages respond mainly to the elasticity governing the ICT composite, – *i.e.*, the substitution between ICT capital and non-routine labour. Both elasticities further affect wages indirectly through total industry output, $\mathcal{V}\left(s\right)$. Consistently, an increase in the industry-specific amenity of a task, $\mathcal{B}\left(a,s\right)$, driven by substitution effects (ς) following a higher relative share of the other task, raises the wage of that task: for instance, a rise in $\mathcal{B}\left(rt,s\right)$ increases the ratio of non-routine to routine workers, thereby increasing routine wages. Finally, wages are non-decreasing in their aggregate task wage $\mathcal{W}\left(a,\mathcal{S}\right)$, but rather increasing if $\varrho,\varsigma < 1$, since $\left[\sum_s \mathcal{W}_{\mathcal{H}}\left(a,s\right)\right]^{-\theta\left(\left[\varrho,\varsigma\right]-1\right)}$ with $-\theta\big(\left[\varrho,\varsigma\right]-1\big) > 0$.

Cross-sectional differences in ρ and σ not only reflect divergences across comparable tasks, but also differences across industries. In fact, if the distribution of capital and worker types were identical across industries, wage premia would be the same. This highlights that industry-specific characteristics, particularly their production plans and the degree of complementarity or substitutability among factors of production, play a central role in explaining real wage differentials

Parameter θ is common across all industries, capturing the strength of sorting and segregation forces and serving as a proxy for economy-wide labour market concentration (Figure 2). This is a single structural distortion: as sorting and segregation intensify, the workforce composition within each industry becomes less diverse, generating a systematic allocation of tasks toward selected industries.

Closing the model, optimal stocks of ICT and non-ICT capital are in order:

$$k_h\left(j,s\right)\ :\ p_h\left(s\right) f_{k_h\left(j,s\right)} = \mathcal{M}^{\epsilon}\, R \tag{11}$$

where $\mathcal{M}^{\epsilon} = \frac{\epsilon}{\epsilon-1}$ is the (constant) price mark-up, $f_{k_h(j,s)}$ being marginal products of both capital types from the production function specified in eq. (9).

According to the structure of the model, equilibrium conditions read as follows.

(Equilibrium) *An equilibrium for this economy is defined as an households' choice of job place, a combination of factors' prices $\{w\left(a,s\right),R\}$, and a set of aggregate quan-*

tities $\Omega = \{Y, K(phy), K(ict), L(rt), L(nrt)\}$, such that: (a) each household picks the firm-industry tuple that maximizes eq. (5); (b) according to the occupational choice, each household maximizes its expected-utility version of the utility in eq. (5); (c) final and sectoral good producers maximize their revenues; (d) given the availability of workers in each job task as in eq. (7), optimal wages are determined by the equilibrium of labour demand and supply; (e) firms choose also capital bundles to maximize their profits; and (f) all markets clear, shaping Ω.

Proof in Appendix B.

3. From Theory to Data

In this section I quantitatively evaluate the model and bring it to the data. To gauge the performance and the consistency of the model in addressing salient features of the inter-industry wage structure of the United States economy, I split the universe of industries into three groups (following the "contribution" result, Fact 1 of the data motivating section), namely the top 25%, the middle 50% and the bottom 25% of industries according to the overall growth in their real log-wage. The time window considered is the usual, spanning from 2003 to 2022.

3.1. Validating the model implications

As a first step to validation, I quantify the relevance of the labour force composition on the industry-specific wage level. Beside the usual role of capital and labour types in both $\mathcal{V}(s)$ and $\mathcal{Q}(s)$, the industry-specific wage level in eq. (10) considers a sizeable role of the industry benefit of each task, namely $\mathcal{B}(a, s)\ \forall s$, which points directly to the substitution among job tasks at the industry layer. A set of panel Fixed Effects (FE) regressions provides reduced-form evidence on how it impacts industry wages:

$$log\, w_t(s) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_{j,t} \mathcal{Z}_{j,t} + u_t$$

where $\mathcal{X}_i$ comprises the regressors – task ratio and its inverse – and $\mathcal{Z}_j$ is a set of employment-related controls. Results are presented in Table 4. Column 1 presents the baseline estimation with industry FE only. A positive effect of increases in both ratios is observed, with the non-routine-to-routine worker ratio exerting a stronger role on industry real log-wages, consistent with stylized Fact 4. Column 3 examines cross-sectional variation by omitting industry FE: an increase in the routine-to-non-routine ratio reduces wages, contrasting with the industry-aggregated prediction of eq. (10). Nonetheless, the positive impact of the relative non-routine task ratio remains robust and quantitatively substantial compared with the baseline.

Employment and relative wages.– The second test examines whether tasks' relative wages predict task-level employment, as in eq. (7). Table C.1 shows strong empirical support: employment in both routine and non-routine tasks increases with relative wages, with larger effects for non-routine workers, implying that higher-paying industries attract more workers to a given task.

TABLE 4: INDUSTRY WAGE AND RELATIVE TASK SIZE

	$\log w(s)$		
	(1)	(2)	(3)
$g(rt/nrt, s)$	.016*** (.006)	.001 (.007)	-.020*** (.002)
$g(nrt/rt, s)$	.129*** (.036)	.250*** (.084)	.318*** (.043)
Industry FE	✓	✓	✗
Time FE	✗	✓	✗

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on* $N = 1240$ *observations. The Fixed Effects (FE) regressions are of the form* $\log\ w_t(s) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$*, with* $\mathcal{X}_i$ *being the regressors, and* $\mathcal{Z}_j$ *a set of controls. All series are in logs. Constant not reported to save space. Source: BEA, BLS and own calculations.*

3.2. CALIBRATION

The vector of structural parameters to calibrate is $\Theta = (\alpha_s, \epsilon, \lambda_s, \mu_s, \theta, \rho_s, \sigma_s)_{\forall s}$, and comprises features of both households and industries. A total of 17 parameters is computed, and the strategy can be summarized in three steps: *(i)* some parameters are calibrated directly from the data, and some are externally taken; *(ii)* the elasticities of substitution are directly and individually estimated via panel robust regressions at industry level; finally, *(iii)* the left-outside parameters are internally estimated by moment-matching of key features of the U.S. economy.

Data and external calibration.– The elasticity of substitution among different varieties (ϵ) is externally set to 6 (to have an annual mark-up of 20%). Differently, the production function's share of non-ICT capital (α) is directly taken by manipulating data from BEA: as in Arvai and Mann (2022), for each group of industries, I compute one minus the share of labour in total output, adjusted by considering the weight of non-ICT capital in the total capital stock. This procedure reports $\alpha = \{0.263, 0.195, 0.514\}$, order from bottom to top groups.[21]

Elasticities of substitution.– The key model parameters, namely ρ_s, σ_s, are estimated through the general equilibrium properties of the model as in Karabarbounis and Neiman (2014). Labour, capital, and profits are expressed as income shares of output and blended with the F.O.C.s in eq. (11) to exploit the evolution of the capital rental rate (eq. 6). The resulting condition is expressed in differences between two periods. Using a linear approximation around a zero over-time trend, this approach yields two separate estimating equations that both relate the industry-level labour shares, $s_\ell(a, s)$ and $s_\ell(s)$, to technological capital:

$$\frac{s_{\ell,t}(nrt, s)}{1 - s_{\ell,t}(nrt, s)} \widehat{s}_\ell(nrt, s) = \beta_c + (\rho - 1)\,\widehat{\zeta}(s) + u_t \tag{12}$$

[21] Estimates are consistent with the argument against the usual "alpha equal one-third" rule – that the elasticity of output with respect to capital (*i.e.*, the capital share of output in a neoclassical production function) is 0.33 – by Vollrath (2024) for the U.S. economy in 1948-2018.

and

$$\frac{s_{\ell,t}(s)}{1-s_{\ell,t}(s)}\widehat{s}_{\ell}(s) = \beta_c + (\sigma-1)\,\widehat{\zeta}\,(s) + \beta_k\left(\widehat{\frac{\ell\,(nrt,s)}{k\,(ict,s)}}\right) + u_t \tag{13}$$

where *hatted* variables identify their own percent change between arbitrary t and $t-1$ periods, and $\zeta\,(s)$ is the industry-related quantity of ICT capital relative to physical capital. The first equation identifies the elasticity of substitution between ICT capital and non-routine labour (ρ) by treating the ICT composite as a nested intermediate input. By contrast, eq. (13) estimates the elasticity of substitution between routine and non-routine workers (σ), derived from the full production structure in eq. (9). Estimated coefficients do not map into $\beta_{\zeta}^{(\rho,\sigma)} = \left([\rho,\sigma]-1\right)$; rather, the elasticity parameters ρ,σ are directly identified.

A negative relationship between trends in labour shares and trends in relative capital quantities emerges only when the estimated elasticities imply sufficient complementarity, that is, when $\rho,\sigma < 1$. An increase in the relative stock of ICT capital then translates into a decline in the labour share of each occupational group.[22] Across 3-digit U.S. 2017 NAICS industries, estimated correlations between trends in labour shares and relative ICT capital stocks are strongly negative over 2003-2022, with $corr_{\rho} = -0.76$ and $corr_{\sigma} = -0.75$.

Given the relatively short timing, the estimation may be sensitive to outliers. A robust regression is therefore exploit, iteratively down-weighting observations that lie far from the fitted regression line.[23] Estimates of equations (12) and (13) are reported in Table 5. A key result is that all industries but the middle one exhibit ICT capital-non-routine complementarity, as indicated by $\sigma_s > \rho_s$ for $s = \{bot, top\}$. This pattern reflects a broadly similar adoption of technological change across industries, coupled with heterogeneous effects on labour force adjustments, as the magnitude of the ρ,σ gap varies across groups. Consistent with the evidence in Section 1, industries at the top of the distribution benefit the most from task reallocation driven by technological change.[24]

Interpreting ρ, top industries exhibit the strongest complementarity between ICT capital and non-routine workers, followed by bottom and then middle industries. Values of $\rho \approx 1$ indicate greater gross substitutability, implying weaker complementarity. Accordingly, top industries sustain larger complementarity of stocks of ICT

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[22] The approach differs from Karabarbounis and Neiman (2014), where elasticities exceed unity so that a fall in the user cost of capital – from a declining investment prices – reduces the labour share. By contrast, I exploit variation in quantities, as capital stocks are directly observable in the BEA data.

[23] It mitigates potential violations of OLS assumptions arising from small samples by reducing the influence of high-residual observations. At each iteration, observations with Cook's distance greater than one – indicative of high leverage – are removed, so that the weight of each n^{th}-observation is no longer uniform at $\frac{1}{n}$.

[24] Estimating the elasticity of substitution between capital and labour is inherently challenging, as it reflects both demand- and supply-side forces. No consensus exists on its precise value: aggregate and sectoral estimates range from 0.3 to 0.9 (*e.g.*, Arrow et al. 1961; Klump et al. 2007; Herrendorf, Herrington, et al. 2015; Alvarez-Cuadrado et al. 2018, Oberfield and Raval 2021), from unity (*e.g.*, Berndt 1976), to between 1.25 and 1.6 (*e.g.*, Piketty 2013; Karabarbounis and Neiman 2014).

TABLE 5: BASELINE ESTIMATION

	ρ	*Std.Err.*	*95% CI*	σ	*Std.Err.*	*95% CI*
bottom	.329	.04	[.250, .407]	.634	.08	[.482, .785]
middle	.420	.02	[.375, .466]	.400	.05	[.310, .491]
top	.249	.03	[.188, .310]	.766	.06	[.656, .877]

Estimation of the elasticities of substitution from eqs. (12)-(13), for 3-digit U.S. 2017 NAICS industries over the period 2003-2022. ρ refers to the degree of substitutability between non-routine workers and ICT capital; σ relates to the degree of substitutability between routine and non-routine workers.

capital with employed non-routine workers than other industry groups. Turning to σ, the elasticity of substitution between routine workers and the ICT composite (or, by symmetry, between routine and non-routine tasks), values closer to one imply a lower propensity to retain routine workers as ICT capital and non-routine employment expand. This precisely emerges in Table 5: top and bottom industries are least likely to employ routine workers alongside non-routine workers, while middle industries respond only weakly. The estimated elasticities thus reveal pronounced heterogeneity in structural heterogeneity across industries, pointing to an uneven reallocation of economic activity toward selected sectors. This reallocation is driven by cross-industry differences in capital-labour prices and quantities, as manifested in labour share trends.

Matching moments.– o conclude the calibration, the remaining parameters – *i.e.*, the industry-specific weights μ_s, λ_s in the production function, as well as θ, governing the dispersion of households' idiosyncratic productivities and, in turn, the strength of sorting and segregation effects – are estimated by matching salient moments in the data using the Method of Simulated Moments (MSM). Given the estimates thus far, the calibration leaves seven parameters to be identified by matching seven moments. Each λ_s is pinned down by the ICT capital share of the corresponding industry group, while each μ_s is identified using the group-specific share of routine workers, exploiting its theoretical definition in eq. (7). Finally, since θ captures productivity dispersion and the resulting degree of labour market polarization, it is calibrated to match the real log-wage premium of routine workers in the top industry group relative to the bottom group.[25]

For $s = \{bot, mid, top\}$, the specified method estimates the vector of parameters $\widetilde{\Theta} = \{\lambda_s\,,\, \mu_s\,,\, \theta\}$, by minimizing the *loss function*

$$\mathcal{L}\left(\widetilde{\Theta}\right) = \left(\widehat{m}(\widetilde{\Theta}) - \widetilde{m}\right)' \mathbf{W} \left(\widehat{m}(\widetilde{\Theta}) - \widetilde{m}\right)$$

to reckon $\widetilde{\Theta}$ that is, to minimize the distance between the estimated moments $\widehat{m}(\widetilde{\Theta})$ and the data moments counterpart, $\widetilde{m}$. Element $\mathbf{W}$ is an *efficient weighting* matrix that allows to implement an efficient estimator.[26] Table C.2 in **??** summa-

[25] The dynamics of labour market concentration in aggregate employment are closely mirrored by those of routine workers at the 3-digit U.S. 2017 NAICS industry level (Figure 4).

[26] Practically, I first set it to be an identity matrix whose diagonal elements are the mean values of each

TABLE 6: SUMMARY OF CALIBRATION

	parameter	*bottom*	*middle*	*top*	*global*	***source***
		value				
α	*physical capital, share of $y(s)$*	0.263	0.195	0.514		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in $y(s)$*	0.676	0.495	0.337		*MSM*
λ	*ICT capital share in $Q(s)$*	0.457	0.465	0.451		*MSM*
θ	*households' productivities dispersion*				11.3	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.329	0.420	0.249		*estimation*
σ	*EoS, routine and ICT composite*	0.634	0.400	0.766		*estimation*

Parameters by industry groups. Under "data" the values are directly computed from the data sources, while "external" sets standard values from the literature. "MSM" refers to the Methods of Simulated Moments, and "estimation" reports the previously estimated values from Table 5.

rizes the resulting values, and compares them to the data.

Of particular interest is the estimate of θ. As extensively discussed, a higher value indicates less heterogeneity in individual productivity and, consequently, way more stronger incentives for workers to sort and segregate into specific industries rather than move across them. Put differently, the farther θ lies from unity, the more polarized the labour market becomes in terms of workforce characteristics, reflecting stronger sorting and segregation forces. The estimated value, $\theta = 11.3$, points to a highly concentrated labour market (as in Figure 2), consistent with recent evidence for the U.S. labour market in Yeh et al. (2022).[27]

Comparative discussion of the calibration.– A summary of the parametrization of the model is in Table 6. Top industries – those experiencing the largest increases in real wages – are characterized by the strongest complementarity between ICT capital and non-routine labour (lowest ρ) and, simultaneously, by a greater propensity to substitute routine workers with non-routine tasks (highest σ). By contrast, bottom industries also display meaningful ICT-non-routine complementarity, with weaker substitutability between routine and non-routine tasks, limiting the reallocation away from routine labour. These results closely align with the stylized facts documented in Section 1: while the non-routine-to-routine task ratio evolves primarily within industries (Fact 3), real wage growth is muted unless this shift is accompanied by an increase in the ICT-to-non-ICT capital ratio (Fact 4). Moreover, top industries feature a substantial share of physical capital (α) in their production structure, alongside the lowest weight assigned to routine labor (μ). Overall, the MSM calibration is internally consistent and mirrors the estimated elasticities of substitution across industry groups.

moment in the data, to then update these values using the estimated vector valued function with the distance between data and simulated moments.

[27] Concentration – due to employer market power from aggregate markdown (ratio between wages and marginal revenue product of labour) –, decreased between 1977 and 2002, but thereafter it has been sharply increased. Such turn is well captured even at industry level (refer to Figure 4).

TABLE 7: IMPLIED FIT OF WAGE VARIANCES

moment	***data***	***model***
routine wage variance, across industries	2.285	2.289
non-routine wage variance, across industries	2.314	2.311
between-industry wage variance	2.299	2.300

Variances computed according to eq. (1) over the period 2003-2022. The first two rows compute the measure for routine and non-routine tasks separately, while the last row directly reports the wage dispersion across industries.

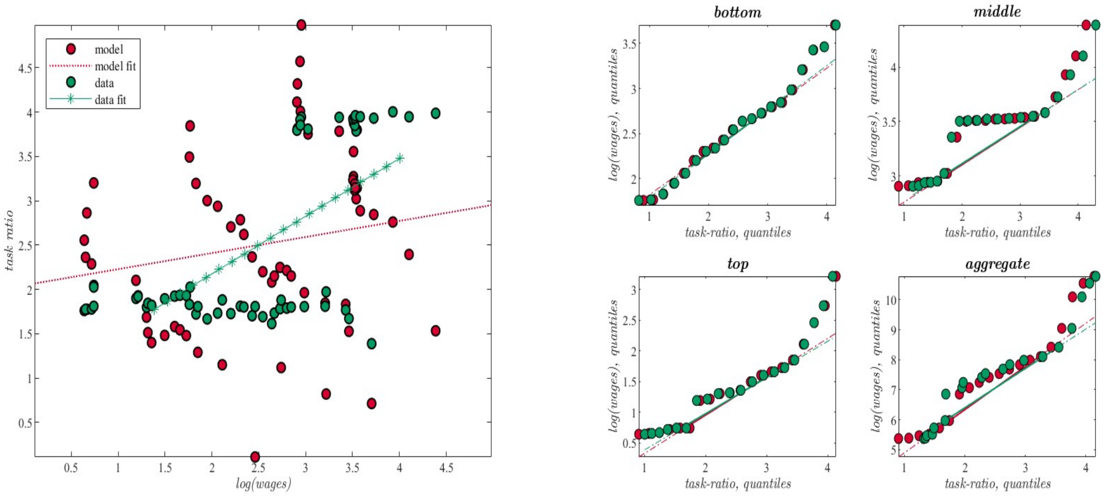
3.3. MODEL FIT

The primary scope of the model is to detect the industry's determinants that account for between-industry wage inequality, moment not directly targeted in the MSM parameters' estimation. Table 7 reports the fitting of real log-wage variance across industries for routine and non-routine workers, as well as that for industry-specific wage rates. Tasks' specific variances are computed according to eq. (1) given the analytical wage series sparking from eq. (10), while the between-industry wage variance is found by averaging these two dispersion indicators. Given the calibrated values of parameters, the model built has the virtue of well targeting all the considered measures of inequality on real log-wages.

Another central hypothesis to validate in the cross-section is the model's ability to capture an increase in industry real log-wage following a rise in the task ratio, as discussed in Subsection 3.1. To this end, Panel 3a plots the correlation between the share of non-routine workers over the mass of routine workers and the industry wage as both in the data and in the model. Imposed model structure is able to successfully replicate the upward-sloping correlation implied by the data.

Finally, the validity of the calibration strategy can be assessed by checking how the model performs in addressing some other data moments, in particular those not targeted throughout the estimation procedure. I pick all the moments related to the wage structure in the economy, and evaluate the fitting also at the industry-group levels. Table C.3, Appendix C, reports the outcome. The direction of the untargeted moments in the data is captured by the model. An important observation is that almost all the appraised wage-moments have negative slope, thus linking my model to the secular decline in real wages (*e.g.*, Massenkoff and Wilmers 2023). The only increasing moments are the "top/bottom wage ratio" and the "aggregate task premium" (reflecting the wage premium of non-routine over routine tasks), making the model suitable for studying the evolution of this wage gap.

To conclude, the model performs well in replicating both targeted and untargeted moments related to the wage structure of the U.S. economy over the period 2003-2022: fitting is detected either at the aggregate and industry levels, and also for differences among routine and non-routine workers.



(A) TASK RATIO VS. WAGES

(B) QUANTILES

FIGURE 3: MODEL AND DATA COMPARISON

Note: comparison between the model series and the related series in the data. Panel 3a represents the correlation in the baseline equilibrium between the task ratio (non-routine over routine tasks) with the empirical and model-implied real log-wage level, and the corresponding lines show the linear fit of the correlation; Panel 3b plots the parametric curves associated to quantiles of the considered HP-filtered series (model vs. data) one against each other. Series are scaled to be in the same range for graphical comparison. All plots consider industries to be grouped in terms of bottom, middle, and top industries' groups. Red circles refer to the model, while red ones to the data.

4. COUNTERFACTUAL ANALYSIS

The primary goal of the paper is to identify the key drivers of wage variance across U.S. industries and quantify the share of observed wage inequality explained by the "structural" or by the "quantity" effects. Specifically, the model's parameters are fully re-estimated over two equal periods, and the associated moments are computed by sequentially "switching on" changes in one or more parameters or series while holding the others fixed.[28]

In the set of counterfactual exercises the core parameters are three: the elasticity of substitution between ICT capital and non-routine labour, ρ; the elasticity of substitution between routine and non-routine labour, σ; and the household productivity dispersion, θ, which captures sorting, segregation, and overall labour market concentration. Their contributions are evaluated both individually and jointly. Variations in industry-specific elasticities, ρ_s, σ_s, isolate the role of structural heterogeneity across industries, independent of changes in input quantities. Changes in θ reflect the degree of labour market concentration: with imperfectly elastic labour supplies, higher value implies stronger sorting and segregation tuple, making labour supply more industry-specific.

[28] Tables 9 and C.8 report the results for overall between-industry wage inequality. In Appendix C, the same is applied separately to routine (Table C.9) and non-routine (Table C.10) workers.

TABLE 8: TIME-VARYING ESTIMATION

	2003-2012		***2013-2022***	
	ρ	σ	ρ	σ
bottom	.355 [.12]	.366 [.12]	.819 [.05]	.326 [.06]
middle	.431 [.04]	.429 [.08]	.345 [.04]	.438 [.09]
top	.408 [.04]	.367 [.12]	.508 [.06]	.358 [.05]

Estimation of the elasticities of substitution in eqs. (12)-(13) over different time spans (2003-2012 and 2013-2022), in absolute values, for 3-digit U.S. 2017 NAICS industries. Standard errors in parenthesis, and 95% confidence interval significant but not reported. ρ refers to the degree of substitutability between non-routine workers and ICT capital; σ relates to the degree of substitutability between routine and non-routine workers.

4.1. SELECTING THE PARAMETERS

Table 8 summarizes the evolution of ρ, σ across two periods, 2003-2012 and 2013-2022. Top industries display a pronounced change in only one elasticity: the complementarity between ICT capital and non-routine workers declines, while the "gross substitution" between routine and non-routine workers shows only a modest increase. In the middle group, non-routine workers strengthen their complementarity with ICT capital, accompanied by a slight reduction in complementarity with routine workers. The bottom group experiences a simultaneous drop in both the substitutability across job types and the complementarity between ICT capital and non-routine labour. These patterns highlight heterogeneous structural adjustments across industries over time.

Figure 4 provides intuition on the evolution of labour market concentration by plotting the Herfindahl-Hirschman Index (HHI) for both worker types and aggregate employment. Since θ is estimated via MSM, it reflects not only a cross-sectional measure but also the variation of sorting and segregation – similar workers cluster more tightly within industries when θ is higher. A clear pattern emerges: labour market concentration rises steadily in the first half of the sample, to then flattening thereafter (yellow line) from declining concentration among non-routine workers (green line). Re-estimation over the two sub-periods delivers $\theta_{2003\text{-}2012} = 7.26$ and $\theta_{2013\text{-}2022} = 7.79$, with stronger variation in industry-level concentration in the first period while moderate in the second. Hence, tracking concentration of routine tasks suffices to capture the economy-wide pattern, consistent with using the top/bottom industry wage ratio for routine jobs to calibrate θ. Together, heterogeneous changes in ρ, σ, coupled with variations in θ, effectively capture the reallocation of capital and labour types across industries, reflecting both structural heterogeneity and rising labour market concentration for routine and non-routine tasks.

Key then is to assess the role of differential effects of these structural changes. As a preview of the results, heterogeneous shifts in both elasticities matter: changes in ρ and σ individually explain a substantial portion of real wage variance, but it is their joint evolution that accounts for the largest share of U.S. wage inequality. In addition, higher labour market concentration – reflected in stronger sorting and

FIGURE 4: LABOUR MARKET CONCENTRATION AS A STRUCTURAL CHANGE

Note: evolution in labour market concentration as measured by the Herfindahl-Hirschman Index (HHI) – on the y-axis – by routine and non-routine (red and green lines, respectively) job tasks, and by total employment (yellow line). Series are normalized relative to their initial value in 2003, set to 1. *Source*: BLS and own calculations.

segregation – amplifies the inequality captured by these structural differences.

4.2. INSPECTING THE MECHANISM: MODEL RE-CALIBRATION

The core exercise consists in re-estimating the full set of model parameters over two sub-periods (2003-2012 and 2013-2022) and quantifying the effects of allowing one – or a subset – of parameters to change at a time, providing a direct bridge between data-driven shifts in structural heterogeneity across U.S. industries and the observed evolution of wage inequality, as disciplined by the model. The new estimates are reported in Table C.6, Appendix C. The objective is to isolate the contribution of each structural parameter by computing counterfactual wage variances in which only a given parameter(s) moves from its first- to its second-period value, while all remaining parameters are held fixed at their initial levels. Formally, given the re-estimated parameters with $\Psi = \{\rho, \sigma, \theta\}$ as the key ones (those selected in Subsection 4.1), consider the following model:

$$\Delta var\Big(w\left(s\right)-\overline{w}\Big) = f\Big(\Phi_s\left(t_1\right), \Theta_s(t_1), \Phi_s\left(t_2\right) \mid \Delta\Theta_s\left(\Psi_{t_2}, -\Psi_{t_1}\right)\Big) \qquad \text{(Model A)}$$

where the change in between-industry wage variance is a function of the input factor series in both periods, $\Phi_s\left(t_1\right)$ and $\Phi_s\left(t_2\right)$, the whole set of parameters in period one, $\Theta_s\left(t_1\right)$, and a subset of parameters in the second period keeping fixed the others, $\Theta_s\left(\Psi_{t_2}, -\Psi_{t_1}\right)$, for any industry-$s$. Practically, given the change in capital and labour types, wage levels and variances are estimated under two set of parameters: *(i)* full set related to period one, and *(ii)* full set related to period one with one (or a combination) related to its period two level.

Results are reported in Table 9. Column 2 displays the between-industry vari-

TABLE 9: MODEL COUNTERFACTUAL, CHANGE

		Δ ***model*** $\mid \Delta\, m\Big(\Phi\,(t_2)\,, \Theta = \{\Psi_{t_2}, -\Psi_{t_1}\}\Big)$		
industry wages	*var* $(w)_{t_2}$	*level*	*share, model*	*share, data*
DATA	1.18			
MODEL	1.09			
$\Delta\sigma$		2.37	2.18	1.99
$\Delta\rho$		.88	.81	.74
$\Delta(\sigma, \rho)$		1.12	1.03	.94
$\Delta\theta$		1.34	1.23	1.13
$\Delta(\sigma, \rho, \theta)$		1.16	1.07	.98

Quantification of Model A. Model implied between-industry real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(t_2)\,\mid\,\Theta(\Psi,t_2),\Theta(-\Psi,t_1)\right]}{m\left[\Phi(t_2)\,\mid\,\Theta(\Psi,t_2)\right]}$*, where $\Phi\,(t_2)$ identifies the series in the second period, and Θ the set of parameters where some of them, (Ψ, t_2), are taken in the second period, while $(-\Psi, t_1)$ reflects the set of all the parameters in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

ance in the second period (t_2) for both the data and the model specification described in column 1. To assess the contribution of individual structural parameters, column 3 reports the model-implied wage variance obtained by holding all parameters at their first-period values, except for the parameter(s) of interest, which are set at their second-period levels. The resulting counterfactuals highlight the central role of industry-heterogeneous changes in the substitutability of production factors in shaping U.S. between-industry wage inequality. In particular, the joint evolution of the two substitution elasticities – ρ, governing complementarity between ICT capital and non-routine labour, and σ, governing substitution between routine and non-routine workers – emerges as pivotal. Isolated changes in ρ alone would have generated a non-negligible increase in wage dispersion (.88), while changes in σ alone would have implied an even larger surge in inequality (2.37). Strikingly, when both elasticities are allowed to change simultaneously, their opposing magnitudes partially offset each other, yielding an implied between-industry real log-wage variance (1.12) that closely aligns with both the full model calibration (1.09) and the observed data (1.18). This evidence underscores that it is the coordinated – and uneven – restructuring of task substitutability across industries, rather than movements in any single elasticity in isolation, that accounts for the observed evolution of wage inequality.

Column 5 quantifies the contribution of each parameter to the observed rise in U.S. wage inequality by reporting the share of the empirical real log-wage variance explained by the variance implied by parameter shifts in the model. Isolated changes in ρ account for about 74% of the observed between-industry wage dispersion, while heterogeneous and joint shifts in both elasticities explain as much as 94% of the data-implied variance (corresponding to 103% of the variance generated by the fully calibrated model for the 2013-2022 period). Hence, large wage differentials across industries are driven primarily by industry-specific differences in the substitutability between routine and non-routine workers, and not by variations in the substitutability between ICT capital and non-routine labour alone. Crucially, it is the interaction of these two structural dimensions – how industries reorganize tasks internally and how they combine labour with technology – that accounts for the bulk of the observed increase in wage inequality.

These conclusions naturally extend to the interpretation of changes in θ. Taken in isolation, its increase substantially amplifies (1.34) real log-wage dispersion (*e.g.*, Rinz 2022). Once these sorting/segregation forces interact with the heterogeneous shifts in substitution elasticities documented in Table 8, their effect becomes more nuanced: labour market concentration slightly sharpens the economy-wide between-industry wage inequality, raising the model-implied variance (1.16). Put it differently, when industries differ in how they reorganize tasks and combine labour with technology, a more concentrated labour market magnifies existing wage gaps rather than creating them on its own. When variations in industry-specific structural characteristics – captured by joint changes in both substitution elasticities – are considered together with changes in labour market concentration, the combined component $\Delta(\sigma, \rho, \theta)$ accounts for nearly the entire observed increase in U.S. between-industry real *log*-wage variance, explaining about 98% of its data-implied level.

Sensitivity.– So far the focus is on changes in the three core parameters of the model, ρ_s, σ_s, θ, while holding fixed the production function parameters $\alpha_s, \mu_s, \lambda_s$. The results are further strengthened by the complementary, reverse exercise. Formally, the counterfactual asks what the level of between-industry wage variance would have been had one parameter remained fixed at its initial (period-1) value, while all other parameters were allowed to evolve from period one to period two:

$$\Delta var\Big(w(s) - \overline{w}\Big) = f\Big(\Phi_s(t_1), \Theta_s(t_1), \Phi_s(t_2) \mid \Delta\Theta_s(\Psi_{t_1}, -\Psi_{t_2})\Big) \qquad \text{(Model B)}$$

whose results are presented in Table C.8. Absent any change in σ the model would explain about 79% of the observed rise in wage inequality. By contrast, fixing both elasticities simultaneously, $\Delta\Theta_{|(\sigma,\rho)}$, leads the model to over-predict dispersion: the implied variance (1.31) exceeds that observed in both the data and the baseline model. A similar overstatement emerges when all structural parameters are held fixed, leaving only changes in income shares to operate. Conversely, holding constant the productivity dispersion parameter θ while allowing both substitution elasticities and production weights to vary captures roughly 88% of the data-implied between-industry wage variance (96% in the model).

Investigations highlight a central feature of the U.S. wage structure: structural heterogeneity across industries is the dominant source of wage inequality. When the model is disciplined by the data, most of the observed dispersion is traced back to industry-specific differentials in the elasticities of substitution between capital and labour types (94%, Table 9), and this share remains large when combined with heterogeneity in production weights (88%, Table C.8). Conditional on these structural differences, the increase in labour market concentration acts as an amplifier of inequality. Once both mechanisms interact, the model explains up to 98% of the observed between-industry variance in real log wages over the past two decades (Table 9).[29] Overall, the evidence points to uneven structural transformation across industries – rather than quantity-based technological change – as the key force for the evolution of U.S. wage inequality.

The role of labour market power.– odel specification allows to inspect whether aggregate monopsony power plays a role on between-industry wage inequality. Labour market failures due to employer market power in wage-setting is widely discussed (*e.g.*, Dodini et al. 2021), though Card, Rothstein, et al. 2024a show that, in many manufacturing industries, wage premiums are not positively correlated with wage markdowns. Appendix D replicates the counterfactual analysis in a setting where monopolistically competitive firms optimally exploit monopsonistic power: eq. (10) would display an average wage markdown over the marginal revenue product of labour, $\mathcal{M}^{\theta} = \frac{\theta}{1+\theta}$, which declines whenever the labour supply elasticity $\theta \to 1$. Results for $\Delta\theta$ and $\Delta(\sigma, \rho, \theta)$ remain largely unchanged in both direction and magnitude: changes in between-industry wage inequality are not primarily driven by variations in aggregate wage markdowns.

5. Factor Quantities and Productivity Changes

As a concluding step I examine the extent to which divergences in a production input-x matter when industries share the same structural parameters. Taking t_0 as the initial year (2003) of the sample:

$$var\Big(w\,(s) - \overline{w}\Big) = f\Big(\Delta\Phi_s\,(x)\,, \Phi_s\,(-x, t_0)\;\Big|\;\Theta_{=\forall s}\Big) \tag{Model C}$$

This exercise conjectures what the cross-industry variance in real log-wages would have been if only a selected set of inputs – capital types, labour types, or their combinations – had evolved over time, $\Delta\Phi_s\,(x)$, while all remaining inputs were held fixed at their initial levels, $\Phi_s\,(-x, t_0)$, and all industries were constrained to share common structural parameters, $\Theta_{=\forall s}$. In the counterfactuals presented below, these parameters are set to their economy-wide means: $\mu = .503$, $\lambda =,458$, and $\theta = 11.3$, as estimated via the MSM procedure, together with a physical-capital income share of $\alpha = .324$ and a price-markup parameter fixed at $\epsilon = 6$. Treating all industries as a unique broader group, the elasticities of substitution are re-estimated and set to $\rho = .333$ and $\sigma = .595$.[30]

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[29] Robust when wage dispersion is examined separately for routine and non-routine tasks (Tables C.9 and C.10, Appendix C). Substitution elasticities exert a more balanced effect for routine workers, while sorting and segregation forces are relatively more important for non-routine workers.

[30] Standard errors are $0.2_{|\rho}$ and $0.3_{|\sigma}$, with 95% confidence bands of $[.301, .366]_{\rho}$ and $[.545, .645]_{\sigma}$.

TABLE 10: MODEL VS. DATA COUNTERFACTUAL, SERIES AND MEAN PARAMETERS

		model $\mid \Delta\,\Phi\,(x)$						
							$\Delta\,(tfp)$	
	data	$\Delta\ell\,(rt)$	$\Delta\ell\,(nrt)$	$\Delta\,(\ell)$	$\Delta k\,(ict)$	$\Delta\,(tech)$	*mean*	*baseline*
WAGES, VARIANCE								
routine	2.285	.35	.16	.22	.24	.15	.25	.22
non-routine	2.314	.32	.12	.17	.23	.13	.21	.23
industry	2.299	.33	.14	.19	.23	.14	.23	.23

Quantification of Model C. Changes in variances in real log-wages in the model induced by variations in one or more series, keeping fixed the others, and imposing the mean parameters to be homogeneous across industries. $\Delta\,(\ell)$ *refers to joint variations in routine and non-routine series,* $\Delta\,(tech)$ *is associated to simultaneous changes in both ICT capital and non-routine workers, while changes in estimated industry-specific Hicks-neutral exogenous total factor productivity (TFP), given eq. (14), are captured by* $\Delta\,(tfp)$*; TFP series are taken under the same (mean) parameters' values, or given the baseline calibration (Table 3.2). In all the columns, the variation is computed throughout the period-by-period percentage differential thus identifying overall changes in empirical trends implied both by the data and the model; same values differ in terms of decimals.*

Table 10 presents a model-based decomposition of Table 7, following the framework in Model C. The counterfactual analysis reveals that shifts in industry-specific routine workers explain the largest portion of the variation, while dynamics in non-routine employment, $\Delta\ell\,(nrt)$, contribute little – even when combined with changes in ICT capital, $\Delta\,(tech)$. Divergent ICT capital across industries has a non-negligible effect, but it remains quantitatively minor.

Productivity differentials.– So far all the analysis excluded total factor productivity (TFP), whose sectoral series can be recovered exploiting eq. (9). The product-augmenting Hicks-neutral TFP series for industry-s is then

$$log\, z_t\,(s) = log\, y_t\,(s) - \alpha_s\, log\, k_t\,(phy, s) - (1 - \alpha_s)\;\; log\, v_t\,(s) \tag{14}$$

where $v_t\,(s)$ comprises the relation among ICT capital, routine and non-routine workers.[31] Using industry-by-industry BEA data on the annual value added to estimate the yearly sectoral productivities, the right-hand side of Table 10 reports the magnitude of their shifts for wage inequality. As for factor quantities, isolated differential changes in TFP have little explanatory power.

Overall, holding all structural parameters constant, individual and combined shifts in sectoral capital, labour, and productivity account for only 6-15% of the observed wage inequality, notably reinforcing the conclusions from Section 4.

CONCLUSIONS

Wage inequality is largely driven by structural differences across industries: evolving heterogeneity in the substitution elasticities between routine and non-routine workers emerge as the primary determinant of between-industry wage variance,

[31] Neutrality in the sense of Hicks (1932) implies that the marginal rate of substitution between capital and labour inputs is not altered, and hence including productivity does not pose threads to the parameters' identification strategy of Subsection 3.2, which may be confounded by other forms of neutrality (labour-augmenting Harrod-neutrality, or capital-augmenting Solow-neutrality).

whereas variation in the substitutability between ICT capital and non-routine labour plays a secondary role. These technological channels account for 94% of the observed variance. Upon neutralizing differences in these structural parameters, changes in capital and labour inputs, or sectoral productivity explain only 6-15% of the observed wage variance. The quantitative analysis also illuminates broader labour market mechanisms: adding worker sorting and labour concentration, nearly 98% of between-industry wage variance is rationalized.

Two elements emerge as fundamental: *(i)* structural technology parameters, and *(ii)* the composition of capital and labour within industries. While both are essential for wage dynamics, persistent inequality is largely driven by an "indirect" effect of Skill Biased Technological Change operating through evolving substitution across job tasks and the associated reallocation of workers.

Overall, the analysis underscores that wage inequality is primarily shaped by the organization of economic activity across industries, and by differences in how technologies interact with the composition of labour. Shifts in capital-labour substitutability are secondary, suggesting that structural changes on the labour side – how tasks are organized and how workers are employed – emerge as the dominant forces behind U.S. wage inequality. These results strongly elevate technological heterogeneity and differences in structural parameters as the key forces governing the between-industry wage inequality, and remain robust when the analysis is conducted separately for routine and non-routine workers.

References

Acemoglu, D. and D. H. Autor, “Skills, tasks and technologies: implications for employment and earnings”, in: *Handbook of Labor Economics*, vol. 4B, pp. 1043-1171, edited by Orley C. Ashenfelter and David Card, Amsterdam: North Holland 2011. 1.

Allen, S. G., “Updated notes on the interindustry wage structure, 1890-1990”, *Industrial and Labor Relations Review*, 1995, 48(2):305-321. 3.

— “Technology and the wage structure”, *Journal of Labor Economics*, 2001, 19(2):440-483. 3.

Alvarez, J., F. Benguria, N. Engbom, and C. Moser, “Firms and the decline in earnings inequality in Brazil”, *American Economic Journal: Macroeconomics*, 2018, 10(1):149-189. 12.

Alvarez-Cuadrado, F., N. Van Long, and M. Poschke, “Capital-labor substitution, structural change and the labor income share”, *Journal of Economic Dynamics and Control*, 2018, 87:206-231. 19.

Arrow, K. J., H. B. Chenery, B. S. Minhas, and R. M. Solow, “Capital-labor substitution and economic efficiency”, *Review of Economics and Statistics*, 1961, 43(3):225-250. 19.

Arvai, K. and K. Mann, “Consumption inequality in the digital age”, *Banque de France Working Paper Series n. 890*, 2022. 18.

Autor, D. H., “The labor market impacts of technological change: from unbridled enthusiasm to qualified optimism to vast uncertainty”, *NBER Working Paper n. 30074*, 2022. 7.

Autor, D. H., L. F. Katz, and M. S. Kearney, “The polarization of the U.S. labor market”, *American Economic Review*, 2006, 96(2):189-194. 1, 3.

— “Trends in U.S. wage inequality: revising the “revisionists””, *Review of Economics and Statistics*, 2008, 90(2):300-323. 3.

Barth, E., A. Bryson, J. Davis, and R. Freeman, “It’s where you work: increases in the dispersion of earnings across establishments and individuals in the United States”, *Journal of Labor Economics*, 2016, 34(S2):pp. S67-S97. 12.

Bell, L. A. and R. Freeman, “The causes of rising interindustry wage dispersion in the United States”, *Industrial and Labor Relations Review*, 1991, 44(2):275-287. 3.

Berndt, E. R., “Reconciling alternative estimates of the elasticity of substitution”, *Review of Economics and Statistics*, 1976, 58(1):59-68. 19.

Blau, F. D. and L. M. Kahn, “The gender wage gap: extent, trends, and explanations”, *Journal of Economic Literature*, 2017, 55(3):789-865. 3.

Braverman, H., *Labor and Monopoly Capital: The Degradation of Work in the Twentieth century*, Monthly Review Press, United States 1974. 7.

BRISKAR, J., J. RODRIGUEZ MORA, E. DI PORTO, and C. TEALDI, “Decomposition of Italian inequality”, *WorkINPS papers n. 53*, 2022. 3, 12.

BUERA, F. J. and J. P. KABOSKI, “The rise of the service economy”, *American Economic Review*, 2012, 102(6):2540-2569. 4.

CARD, D., A. R. CARDOSO, and P. KLINE, “Bargaining, sorting, and the gender wage gap: quantifying the effect of firms on the relative pay of women”, *Quarterly Journal of Economics*, 2016, 131(2):633-686. 12.

CARD, D. and J. E. DINARDO, “Skill-Biased Technological Change and rising wage inequality: some problems and puzzles”, *Journal of Labor Economics*, 2002, 20(4):733-783. 1.

CARD, D., J. HEINING, and P. KLINE, “Workplace heterogeneity and the rise of West German wage inequality”, *Quarterly Journal of Economics*, 2013, 128(3):967-1015. 3, 12.

CARD, D., J. ROTHSTEIN, and M. YI, “Industry wage differentials: a firm-based approach”, *Journal of Labor Economics*, 2024, 42(S1):S11-S59. 1, 2, 12, 28.

— “Reassessing the spatial mismatch hypothesis”, *American Economic Review: Papers & Proceedings*, 2024, 114:221-225. 3.

CASELLI, F., “Technological revolutions”, *American Economic Review*, 1999, 89(1):78-102. 3.

CERINA, F., A. MORO, and M. RENDALL, “The role of gender in employment polarization”, *International Economic Review*, 2021, 62(4):1655-1691. 1, 3.

CORRADO, C., J. HASKEL, and C. JONA-LASINIO, “Knowledge spillovers, ICT and productivity growth”, *Oxford Bulletin of Economics and Statistics*, 2017, 79(4):592-618. 6.

CORTES, G. M., N. JAIMOVICH, and H. E. SIU, “Disappearing routine jobs: who, how, and why?”, *Journal of Monetary Economics*, 2017, 91:69-87. 3.

COSSU, F., A. MORO, and M. RENDALL, “Training time, robots and technological unemployment”, *CEPR Discussion Papers n. 19343*, 2024. 3.

CROUZET, N., J. C. EBERLY, A. L. EISFELDT, and D. PAPANIKOLAOU, “The economics of intangible capital”, *Journal of Economic Perspectives*, 2022, 36(3):29-52. 6.

CULLEN, D. E., “The interindustry wage structure, 1899-1950”, *American Economic Review*, 1956, 46(3):353-369. 3.

DAVIDSON, C. and M. REICH, “Income inequality: an inter-industry analysis”, *Industrial Relations*, 1988, 27(3):263-286. 3.

DICKENS, W. and L. F. KATZ, “Interindustry wage differences and industry characteristics”, IN: *Unemployment and the Structure of Labor Markets*, pp. 48-89, edited by Kevin Lang and Jonathan S. Leonard, London: Basil Blackwell 1987. 3.

DODINI, S., K. G. SALVANES, and A. WILLÉN, “The dynamics of power in labor markets: monopolistic unions versus monopsonistic employers”, *CESifo Working Paper n. 9495*, 2021. 28.

EDEN, M. and P. GAGGL, “On the welfare implications of automation”, *Review of Economic Dynamics*, 2018, 19:15-43. 4.

FIELDS, J. and E. N. WOLFF, “Interindustry wage differentials and the gender wage gap”, *Industrial and Labor Relations Review*, 1995, 49(1):105-120. 3.

GENRE, V., K. KOHN, and D. MOMFERATOU, “Understanding inter-industry wage structures in the euro area”, *Applied Economics*, 2011, 43(11):1299-1313. 3.

GENRE, V., D. MOMFERATOU, and G. MOURRE, “Wage diversity in the euro area - an overview of labour cost differentials across industries”, *ECB Occasional Paper n. 24*, 2005. 3.

GIBBONS, R., L. F. KATZ, T. LEMIEUX, and D. PARENT, “Comparative advantage, learning and sectoral wage determination”, *Journal of Labor Economics*, 2005, 23(4):681-724. 12.

GITTLEMAN, M. and E. N. WOLFF, “International comparison of inter-industry wage differentials”, *Review of Income and Wealth*, 1993, 39(3):295-312. 3.

GOOS, M. and A. MANNING, “Lousy and lovely jobs: the rising polarization of work in Britain”, *Review of Economics and Statistics*, 2007, 89(1):118-133. 3.

HÅKANSON, C., E. LINDQVIST, and J. VLACHOS, “Firms and skills: the evolution of worker sorting”, *Journal of Human Resources*, 2020, 56(2):512-538. 12.

HALTIWANGER, J., H. R. HYATT, and J. R. SPLETZER, “Increasing earnings inequality: reconciling evidence from survey and administrative data”, *Journal of Labor Economics*, 2023, 41(S1):pp. S61-S93. 12.

— “Rising top, falling bottom: industries and rising wage inequality”, *American Economic Review*, 2024, 114(10):3250-3283. 1, 3-5, 12.

HERRENDORF, B., C. HERRINGTON, and Á. VALENTINYI, “Sectoral technology and structural transformation”, *American Economic Journal: Macroeconomics*, 2015, 7(4):104-133. 4, 19.

HERRENDORF, B., R. ROGERSON, and Á. VALENTINYI, “Growth and structural transformation”, IN: *Handbook of Economic Growth*, vol. 2, pp. 855-941, edited by Philippe Aghion and Steven N. Durlauf, Amsterdam: North Holland 2014. 4.

HICKS, J. R., *The Theory of Wages*, Palgrave Macmillan, London 1932. 12, 29.

HOROWITZ, M. A. and I. L. HERRNSTADT, “Changes in the skill requirements of occupations in selected industries”, IN: *The Employment Impact of Technological Change*, pp. 223-287, edited by National Commission on Technology, Automation, and Economic Progress, Washington, D.C.: G.P.O. 1966. 7.

HOWELL, D. R. and E. N. WOLFF, “Skills, bargaining power and rising interindustry wage inequality since 1970”, *Review of Radical Political Economics*, 1991, 23(1-2):30-37. 3.

INADA, K.-I., “On a two-sector model of economic growth: comments and a generalization”, *Review of Economic Studies*, 1963, 30(2):119-127. 10.

JAIMOVICH, N. and H. E. SIU, "Job polarization and jobless recoveries", *Review of Economics and Statistics*, 2020, 102(1):129-147. 3.

JAIMOVICH, N., M. B. ZHANG, and N. VINCENT, "Under the hood of the routine share decline", *Economic Letters*, 2024, 234:111437. 7.

KARABARBOUNIS, L. and B. NEIMAN, "The global decline of the labor share", *Quarterly Journal of Economics*, 2014, 29(1):61-103. 11, 18, 19.

KATZ, L. F. and D. H. AUTOR, "Changes in the wage structure and earnings inequality", IN: *Handbook of Labor Economics*, vol. 3A, pp. 1463-1555, edited by Orley C. Ashenfelter and David Card, Amsterdam: North Holland 1999. 3.

KATZ, L. F. and K. M. MURPHY, "Changes in relative wages, 1963-1987: supply and demand factors", *Quarterly Journal of Economics*, 1992, 70(1):35-78. 1.

KLEINMAN, B., E. LIU, and S. J. REDDING, "Dynamic spatial general equilibrium model", *Econometrica*, 2023, 91(2):385-424. 10.

KLUMP, R., P. MCADAM, and A. WILLMAN, "Factor substitution and factor-augmenting technical progress in the United States: a normalized supply-side system approach", *Review of Economics and Statistics*, 2007, 89(1):183-192. 19.

KRUEGER, A. and L. SUMMERS, "Reflections on the inter-industry wage structure", IN: *Unemployment and the Structure of Labor Markets*, edited by Kevin Lang and Jonathan S. Leonard, London: Basil Blackwell 1987. 3.

— "Efficiency wages and the inter-industry wage structure", *Econometrica*, 1988, 56(2):259-293. 3.

KRUSELL, P., L. E. OHANIAN, J.-V. RÍOS-RULL, and G. L. VIOLANTE, "Capital-skill complementarity and inequality: a macroeconomic analysis", *Econometrica*, 2000, 68(5):1029-1053. 2, 14.

LEMIEUX, T., "What do we really know about changes in wage inequality?", IN: *Labor in the New Economy*, pp. 17-62, edited by Katharine G. Abraham, James R. Spletzer, and Michael Harper, National Bureau of Economic Research: Studies in Income and Wealth, The University of Chicago Press 2010. 3.

LEONARDI, M., "Firm heterogeneity in capital-labour ratios and wage inequality", *The Economic Journal*, 2007, 117(518):375-398. 3.

MASSENKOFF, M. and N. WILMERS, "Wage stagnation and the decline of standardized pay rates, 1974-1991", *American Economic Journal: Applied Economics*, 2023, 15(1):474-507. 22.

MELMAN, S., "The rise of administrative overhead in the manufacturing industries of the United States 1899-1947", *Oxford Economic Papers*, 1951, 3(1):62-112. 7.

MONTGOMERY, J. D., "Equilibrium wage dispersion and interindustry wage differentials", *Quarterly Journal of Economics*, 1991, 106(1):163-179. 3.

MORIN, A., “Workplace heterogeneity and wage inequality in Denmark”, *Journal of Applied Econometrics*, 2023, 38(1):123-133. 12.

NGAI, R. L. and C. A. PISSARIDES, “Structural change in a multisector model of growth”, *American Economic Review*, 2007, 97(1):429-443. 9.

OBERFIELD, E. and D. RAVAL, “Micro data and macro technology”, *Econometrica*, 2021, 89(2):703-732. 19.

PIKETTY, T., *Le Capital au XXIe Siècle*, Éditions du Seuil, Seuil, 25 boulevard Romain-Rolland, Paris 14 2013. 19.

RINZ, K., “Labor market concentration, earnings, and inequality”, *Journal of Human Resources*, 2022, 57(S):S251-S283. 27.

ROGERSON, R., “Why labor supply matters for macroeconomics”, *Journal of Economic Perspectives*, 2024, 38(2):137-158. 11.

ROY, A. D., “Some thoughts on the distribution of earnings”, *Oxford Economic Papers*, 1951, 3(2):135-146. 2, 11.

RUMBERGER, R. W., “The changing skill requirements of jobs in the U.S. economy”, *Industrial and Labor Relations Review*, 1981, 34(4):578-590. 7.

SIENA, D. and R. ZAGO, “Job polarisation, labour market fluidity and the flattening of the Phillips curve”, *The Economic Journal*, 2024, 134(661):2141-2174. 3.

SLICHTER, S. H., “Notes on the structure of wages”, *Review of Economics and Statistics*, 1950, 32(1):80-91. 3.

SONG, J., D. J. PRICE, F. GUVENEN, N. BLOOM, and T. VON WACHTER, “Firming up inequality”, *Quarterly Journal of Economics*, 2019, 134(1):1-50. 3, 12.

THALER, R. H., “Anomalies: interindustry wage differentials”, *Journal of Economic Perspectives*, 1989, 3(2):181-193. 3.

TINBERGEN, J., “Substitution of gradute by other labour”, *Kylos*, 1974, 27(2):217-226. 1.

VOLLRATH, D., “The elasticity of aggregate output with respect to capital and labor”, *American Economic Journal: Macroeconomics*, 2024, 16(4):470-504. 18.

YEH, C., C. MACALUSO, and B. HERSHBEIN, “Monopsony in the U.S. labor market”, *American Economic Review*, 2022, 112(7):2099-2138. 21.

Appendix

(Outline) *In this appendix I report all the material complementary to the main text; it is made of additional tables and figures, and of discussions on further analytical results. Section A complements and enriches the exposition of the empirical findings in Section 1; Section B derives all the salient elements of the model outlined in Section 2, while Section C consolidates and further discusses the calibration strategy and the counterfactual analysis of the model of Sections 3-4. A replication is in Section D under both monopsony and monopolistically competitive power of firms.*

A. Motivating evidence: data, figures and tables

(Data details) *Data are for United States (US). For what concerns data from Bureau of Economic Analysis (BEA), private non-residential capital types are in net stocks evaluated at current costs by detailed industry dating back from 1925 to 2022.*[I] *Gross categories are "total equipment", "total structures", and "total intellectual property products", and each contains different types of assets according to their own National Income and Product Accounts (NIPA) asset type code; in total there are 96 different asset types. Data are provided for 74 industries at different layers.*[II] *Among all these types of capital assets I am going to classify them accordingly: digital equipment comprises all the electronic structures that are useful to process technological issues, that is "Mainframes", "PCs", "DASDs", "Printers", "Terminals", "Tape Drives", "Storage Devices", and "System Integrators"; the stock of intangible capital coincides with that of "Total Intellectual Property Products" (IPP);*[III] *while physical (or non-ICT) capital is the sum of all the remaining asset types. In addition to the aggregated capital types, I further define industry-specific ICT capital to be the combination of intangibles and digital equipment.*[IV]

BEA also collects data on wage levels of industries, together with their total employment. Herby, I extract also annual data on (seasonally adjusted) value added, wages and salaries, and of persons engaged in production (effective workforce) for

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[I] Starting year is 1998 so that industries are classified with the latest available system, the U.S. 2017 NAICS; before, classification follows the U.S. Standard Industrial Classification (SIC) system. Assets' value is expressed in millions of U.S. dollars, and last update was in November 3, 2023.

[II] Classified according the BEA industry code system. Most are classified at 3-digit U.S. 2017 NAICS level (some 3-digit industries may fall in the same BEA code), four at 2-digit ("construction", "management of companies and enterprises", "educational services", and "other services, except government"), while five industries related to finance and insurance are at 4-digit level.

[III] A recent discussion analyses how the BEA collects data on IPP. Koh et al. (2020) argue that the effects of IPP capital on the U.S. labour share only emerged since 2013, when the BEA revised its methodology by re-classifying the notion of capital; now, the capital income comprises the rents arising from IPP investment. The authors compare the IPP effect on the labour share using both pre- and post-2013 data, finding out a negative effect of the rise in IPP on the U.S. labour share.

[IV] Similar classification in Arvai and Mann (2022), as well coherent with Eden and Gaggl (2018).

each industry. Complementing the analysis with real wages, data on annual (seasonally adjusted) Consumer Price Index (CPI) for all items (indexed at 2015 = 100) is taken from Federal Reserve Economic Data (FRED) database. Not exploiting BLS data throughout, the time window considered is 1998-2022. Both capital stocks and real wages are taken in industry per capita log-terms.

Production does not occur with capital only. A thorough examination of wage dispersion necessitates a careful consideration of the composition of the labour force within each industry. To this end, reference is made to the U.S. Bureau of Labor Statistics (BLS)' Occupational Employment and Wage Statistics (OEWS) programme, which provides comprehensive data covering over one hundred occupational categories. Regarding the data extracted from BLS, information are available for a total of more than 80 industries at 3-digit U.S. 2017 NAICS level starting from 2003 onwards.[V] *At workers' level, each industry displays information on a set of several and granular occupations classified according to the U.S. Office of Management and Budget's Standard Occupational Classification (SOC) system, i.e., occupations are categorized based on the type of job and on required skills, and employees are assigned to an occupation based on the work they perform and not on their education or training. However, since detailed occupations may vary across industries, for my purposes I classify occupations by considering their "major" group membership:*[VI] *I divide such major occupations in both routine and non-routine tasks. The latter group considers occupations such as "Management", "Business and Financial Operations", "Computer and Mathematical", "Architecture and Engineering", "Life, Physical, and Social Science", "Community and Social Service", "Legal", "Educational Training and Library", and "Arts, Design, Entertainment, Sports, and Media", while the left-outside occupations are comprised in the group of routine worker tasks.*

In the merging process (BEA and BLS), the 4-digit industries in the BEA data have been clustered into 3-digit category. The final dataset employed in the conducted analysis comprises information on a complete sample of 62 private-sector industries at the 3-digit U.S. 2017 NAICS system over an annual time window spanning from 2003 to 2022. This dataset forms the empirical foundation for the conduction of the main empirical analysis and subsequent investigation.

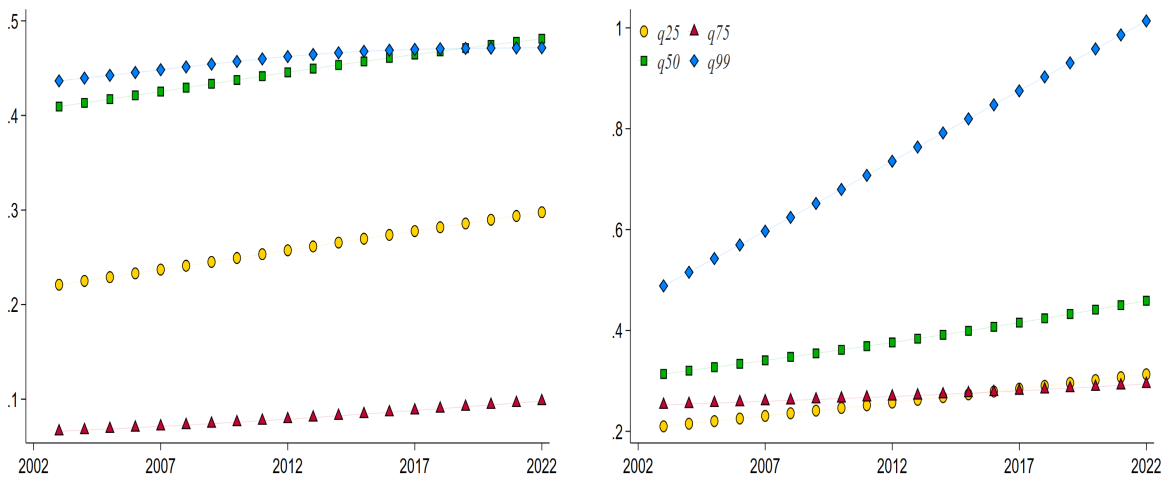
[V] The number of industries is variable year by year due to issues of data sampling. Moreover, prior 2003, these are classified according to the Standard Industrial Classification (SIC) system.

[VI] This choice arises from the fact that there are missing group definitions for some more detailed occupations. In fact, the number of occupations in each industry is not fixed: there are from 80 to 180 types of occupations for each industry, and these are clustered in common 22 major groups.

TABLE A.1: REGRESSIONS, CAPITAL STOCKS

	$\log w\,(s)$			
	(1)	(2)	(3)	(4)
$\beta_{k^{phy}}$	.134*** (3.48)	.122** (2.94)	.164*** (9.05)	.114*** (3.52)
$\beta_{k^{ict}}$	.052* (2.00)			.049* (2.27)
$\beta_{k^{int}}$		.057* (2.03)		
$\beta_{k^{deq}}$		-.003 (-.26)		
$\beta_{k^{int}} \times \beta_{k^{deq}}$			.009** (2.99)	.009** (2.80)
R^2	.461	.491	.287	.497

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). t-statistics in parentheses. Analysis at 3/4-digit U.S. 2017 NAICS industries over 1998-2022 on* $N = 1650$ *observations. Fixed Effects (FE) regressions are of the form* $\log\, w_t\,(s) = \beta_c + \beta_i \mathcal{X}_{i,t} + u_t$, *with* $\mathcal{X}_i$ *representing the different capital-labour ratios considered. Results are robust when controlling for the log size of industries, or when taking capital series directly in levels. Variables are in log format. Constant not reported to save space. Source: BEA and own calculations.*



(A) ICT CAPITAL RATIO

(B) TASK RATIO

FIGURE A.1: CHANGES BY PERCENTILES

Note: each subplot draws the HP-filtered trend in ICT capital ratio (ICT capital stock in physical capital quantity, Panel A.1a), and in task ratio (fraction of non-routine workers of routine ones, Panel A.1b). Series are divided according the growth in group-specific industry wage (*i.e.*, total $\Delta_{\%}$ in industry wage per worker). Industries are classified at 3-digit U.S. 2017 NAICS level. *Source*: BEA, BLS and own calculations.

(A) REAL WAGES VS. ICT CAPITAL RATIO

(B) REAL WAGES VS. TASK RATIO

FIGURE A.2: INDUSTRY CORRELATIONS

Note: each subplot of the figure represents the correlation of overall percentage change in industry-specific real log-wage per capita with both ICT capital ratio (ICT over non-ICT capital) in Panel A.2a, and task ratio (non-routine over routine workers) in Panel A.2b. The oblique (dashed-grey) line represents the fitting curve with its associated degree of correlation, while the horizontal (solid-black) line identifies the mean value of all industries' overall percentage change in real log-wage per capita. For a better graphical visualization, the ICT ratio takes constant the initial level of physical capital. Each circle is referred to a specific industry, and I report the label only for more and less virtuous (top and bottom 10%) industries; circles' size captures different groups of industries, each expressing the group's position in the distribution of overall percentage change in their real log-wage. Plots are referred to 3-digit U.S. 2017 NAICS industries over the period 2003-2022. *Source*: BEA, BLS and own calculations.

TABLE A.2: COMBINED REGRESSIONS, PERCENTAGE CHANGES

	$\Delta \log w(s)$		
	(1)	(2)	(3)
$\beta_{\Delta k}$	.054*** (.002)		.009 (.008)
$\beta_{\Delta \ell}$	.035* (.018)		.035*** (.007)
$\beta_{\Delta k \times \Delta \ell}$		.982 (.649)	.919*** (.152)
R^2	.030	.021	.037
N	1178	1178	1178

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). t-statistics in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries over 2003-2022. Fixed Effects (FE) regressions are of the form* $\Delta \log w_t(s) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$, *with* $\mathcal{X}_i$ *representing the percentage change in both ICT-to-physical capital and non-routine-over-routine workers ratios, and* $\mathcal{Z}_j$ *being a set of time-varying controls. Variables are in log format. Constant not reported to save space. Source: BEA, BLS and own calculations.*

Table A.3: Combined regressions, levels

	log w (s)				
	Fe	*25th quantile*	*50th quantile*	*75th quantile*	*100th quantile*
β_k	.102**	.084***	.102***	.119***	.171*
	(.040)	(.029)	(.022)	(.027)	(.074)
β_ℓ	.299***	.269***	.300***	.328***	.416***
	(.078)	(.040)	(.030)	(.037)	(.104)
$\beta_{k\times\ell}$	.031	.029***	.031***	.034***	.041
	(1.89)	(.010)	(.008)	(.009)	(.025)

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on* $N = 1240$ *observations. The Fixed Effects (FE) regression is of the form* $\log w_t(s) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$, *with* $i = k, \ell$, *and* $\mathcal{Z}_j$ *being a set of controls, and associated with* $R^2 = .348$. *Analogously, the conditional quantile regressions are then* $\mathcal{Q}_\omega\left(\log\ w_{\omega t}(s)\right) = \beta_{i,\omega}\mathcal{X}_{i,\omega t} + \delta_{j,\omega}\mathcal{Z}_{j,\omega t} + u_{\omega t}$, *where* ω *represents each quantile (defined on the independent variable). Variables are all in log format. Constant not reported to save space, while quantile regressions do not have the constant term. Source: BEA, BLS and own calculations.*

Table A.4: Combined regressions, standard deviations

	sd [log w (s)]				R^2	
	(1)		(2)			
	$\beta_{sd(k)}$	β_ℓ	β_k	$\beta_{sd(\ell)}$	(1)	(2)
25th quantile	.256***	.070***	.008***	.403***		
	(.054)	(.015)	(.002)	(.010)		
50th quantile	.121***	.051***	.008***	.404***		
	(.034)	(.009)	(.002)	(.008)		
75th quantile	.031	.038***	.008***	.405***		
	(.039)	(.011)	(.002)	(.010)		
100th quantile	-.127	.015	.008	.408***		
	(.112)	(.024)	(.005)	(.021)		
Fe	.149***	.055***	.008***	.404***	.324	.793
	(.031)	(.010)	(.002)	(.007)		

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on* $N = 1240$ *observations. The Fixed Effects (FE) regression is of the form* $sd[\log\ w_t(s)] = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$, *with* $i = k, \ell$, *and* $\mathcal{Z}_j$ *being a set of controls. Analogously, the conditional quantile regressions are then* $\mathcal{Q}_\omega\left(sd[\log\ w_{\omega t}(s)]\right) = \beta_{i,\omega}\mathcal{X}_{i,\omega t} + \delta_{j,\omega}\mathcal{Z}_{j,\omega t} + u_{\omega t}$, *where* ω *represents each quantile (defined on the independent variable). Variables are all in log or sd format. Constant not reported to save space, while quantile regressions do not have the constant term. Source: BEA, BLS and own calculations.*

(Group decomposition) *Total change in wage inequality can be written also as*

$$
\begin{aligned}
\underbrace{\Delta\,\widetilde{var}\left(w_t(s)-\overline{w}_t\right)}_{\textit{total, wages}} = & \underbrace{\left(\frac{\ell_0(g)}{\ell_0}\right)\left[\Delta\,\widetilde{var}\left(w_t(s\in g)-\overline{w}_t(g)\right)\right]}_{\textit{within-group, wages}} \\
& + \underbrace{\sum_g \left[\Delta\,\widetilde{var}\left(\ell_t(s)-\overline{\ell}_t(g)\right)\right]\widetilde{var}\left(w_0(s)-\overline{w}_0(g)\right)}_{\textit{between-groups, employment}} \\
& + \underbrace{\sum_g \left[\Delta\,\widetilde{var}\left(w_t(s)-\overline{w}_t(g)\right)\right]\left[\Delta\,\widetilde{var}\left(\ell_t(s)-\overline{\ell}_t(g)\right)\right]}_{\textit{between-groups, interaction}} \qquad \text{(A.1)} \\
& + \underbrace{\underbrace{\sum_{\mathcal{G}_{/g}}\left(\frac{\ell_0\left(\mathcal{G}\right)}{\ell_0}\right)\left[\Delta\,\widetilde{var}\left(w_t(s)-\overline{w}_t\left(\mathcal{G}\right)\right)\right]}_{\textit{within-other groups, wages}}
+ \underbrace{\sum_g \left[\Delta\left(\frac{\ell_t(g)}{\ell_t}\right)\widetilde{var}\left(\overline{w}_t(g)-\overline{w}_t\right)\right]}_{\textit{between groups, wages}}}_{\textit{residual}}
\end{aligned}
$$

where industries are partitioned in $g \in \mathcal{G}$ groups, and variances are employment-weighted. Subscript $t = 0$ indicates the initial level (starting year of the sample). Total change in variance can be decomposed in several components: (i) rise in variance for a particular group g; (ii) reallocation of employment across groups, keeping constant the variance of each group at its base level; (iii) cross-changes of wages and employment; (iv) rising variance within all other groups but g; and (v) rising variance between groups. Quantification of each component is in Table A.5; this decomposition is borrowed from Kleinmann (2023).

Table A.5: Decomposition of the rise in wage inequality

	industry group g				
share of the increase, wage variance	(1) *tails*	(2) *middle*	(3) *services*	(4) *manuf.*	(5) *other*
rising variance within the group	79%	32%	58%	11%	27%
employment reallocation across groups	34%	34%	17%	51%	10%
comovement (variance, employment)	7%	7%	3%	4%	5%
residual	-20%	27%	-22%	-34%	58%
total change across all industries	100%	100%	100%	100%	100%

Estimates of each component in eq. (A.1) for 3-digit U.S. 2017 NAICS industries between 2003 and 2022 related to $\log w(s)$. *Operator* Δ *in the equation is* $x_t - x_{t-1}$, *and not a percentage change. The first row shows the share of total increase in variance due to rising variance in the group of industries; the second row shows the share due to changes in employment between that group and the other industries in the sample (employment reallocation), holding constant the change in variance in each group; the third row shows the share that is due to the cross-product of rising variance and rising employment share; the fourth row is so that the sum for each column is 100%. "tails" and "middle" are referred to overall percentage changes distribution in industry real wage per capita; "manuf." stands for manufacturing industries, while "other" does not consider services and manufacturing industries. Source: BEA and own calculations.*

(A) Routine workers

(B) Non-routine workers

Figure A.3: Labour shares pattern

Note: the figure shows the fitted values and the associated linear trend for the labour share of both routine (Panel A.3a) and non-routine (Panel A.3b) workers. The fitted values are extracted from a Fixed Effects (FE) regression for both the labour share measures with year and industry fixed effects. *Source*: BLS and own calculations.

TABLE A.6: INDUSTRY CONTRIBUTION TO WAGE VARIANCE GROWTH

industry	***contribution share*** (%)	***group***
Administrative and Support Services	0.01	*top*
Agriculture	0.12	*top*
Ambulatory Health Care Services	0.23	*bottom*
Amusement, Gambling, and Recreation industries	0.62	*bottom*
Chemical Manufacturing	-0.26	*bottom*
Computer and Electronic Product Manufacturing	0.94	*top*
Fabricated Metal Product Manufacturing	0.14	*bottom*
Federal Reserve banks, Credit Intermediation, and related activities	0.47	*top*
Food Services and Drinking Places	2.36	*top*
Food and Beverage Stores	1.34	*bottom*
Food and Beverage and Tobacco Products	0.39	*bottom*
Funds, Trusts, and Other Financial Vehicles	0.10	*top*
General Merchandise Stores	1.29	*bottom*
Information and Data Processing Services	5.90	*top*
Machinery Manufacturing	-0.02	*bottom*
Management of Companies and Enterprises	2.71	*top*
Oil and Gas Extraction	0.26	*top*
Other Services, except government	0.43	*top*
Other Transportation and Support Activities	0.67	*bottom*
Paper Manufacturing	-0.02	*bottom*
Performing Arts, Spectator Sports, Museums, and related activities	-0.02	*top*
Printing and Related Support Activities	0.08	*bottom*
Professional, Scientific, and Technical Services	11.7	*bottom*
Publishing industries (including software)	2.23	*top*
Rail Transportation	-0.10	*bottom*
Real Estate	-0.04	*top*
Rental and Leasing Services and Lessors of Intangible Assets	-0.08	*top*
Securities, Commodity Contracts, and Other Financial Investments and Related Activities	8.59	*top*
Transportation Equipment Manufacturing	1.17	*bottom*
Warehousing and Storage	0.86	*bottom*

Contribution to between-industry wage variance growth of industries comprised in top and bottom (in terms of overall growth in real log-wage per capita, as reported by Table 1) groups over the period 2003-2022. Variance contribution follows the definition proposed in eq. (1). Industries are at 3-digit U.S. 2017 NAICS level. Source: BEA and own calculations.

B. Model derivation

(Household inter-temporal problem) *The household utility problem in eq. (5) can be rewritten inter-temporally, future-discounted by factor β, as*

$$\max_{\mathcal{C}_t^i,\ b_{t+1}^i,\ \{k_{t+1}^i(j)\}_{\forall j}} \mathcal{U}_{h,t}^i(a,s) = \sum_{t=0}^{\infty}\left[\beta^t \log \mathcal{C}_t^i\right] + \sum_{t=0}^{\infty}\left[\log \mathcal{B}_{h,t}(a,s)\right] + \wp_h^i(a,s)$$

$$\begin{aligned} s.t.\quad & \mathcal{C}_t^i + I_t^i(phy) + I_t^i(ict) + b_{t+1}^i - (1+r_t)\, b_t^i = \\ & = w_{h,t}(a,s)\,\ell_h^i(a,s) + R_t\Big(k_t^i(phy) + k_t^i(ict)\Big) + \mathcal{D}_t^i \\ with\quad & k_{t+1}^i(phy) = \frac{I_t^i(phy)}{\zeta_t^i} + \left(1-\delta_{phy}\right) k_t^i(phy) \\ and\quad & k_{t+1}^i(ict) = \frac{I_t^i(ict)}{\zeta_t^i} + \left(1-\delta_{ict}\right) k_t^i(ict) \end{aligned}$$

where the investment schedule comprises both assets, b_t^i (which are in zero net supply, thus exchanged only across households), and physical and ICT capital holdings, $k_t^i(j)$, with $j = \{phy, ict\}$. Household-specific productivity when working as type-a for firm-h in industry-s is denoted with $\wp_h^i(a,s)$. Each household inelastically supplies one unit of work, so that $\ell_h^i(a,s) = 1$. The firm benefit of household-a is defined to be $\mathcal{B}_{h,t}(a,s) = \left[g_{h,t}(a,s)\right]^{-\varsigma}$, where $g_{h,t}(a,s)$ reflects the relative share (i.e., the task ratio) of a given task, which is negatively scaled by the elasticity parameter ς, while $\mathcal{D}_t^i$ is the share of firms' profits that goes to household-i. Capital stock of household-i depreciates at a rate $\delta_{(j)}$ and accumulates over time by a law of motion which is function of $\zeta_{i,t}$, namely the quantity of ICT capital relative to that of non-ICT capital, that enters negatively in new capital investment, $I_t^i(j)$, under the idea that as the stock of capital owned by household-i becomes more sophisticated (larger ICT relative to non-ICT capital when ζ_t^i increases), it is required an higher rate of capital investment to keep constant the future stock of given capital type. Modelling the dynamics of capital in this way serves only to derive analytical solutions for the equations (12) and (13) estimating the pair ρ, σ of Section 3.

Utility maximization implies the Lagrangian function to be

$$\begin{aligned} \mathcal{L}_{\mathcal{C}_t^i,\ b_{t+1}^i,\ \{k_{t+1}^i(j)\}_{\forall j}} = & \sum_t \beta^t\left[\log \mathcal{C}_t^i\right] + \sum_t\left[\log \mathcal{B}_{h,t}(a,s)\right] + \wp^i(a,s)\ + \\ & - \sum_t \beta^t \psi^t\Big[\mathcal{C}_t^i + I_t^i(phy) + I_t^i(ict)\ + \\ & + b_{t+1}^i - (1+r_t)\, b_t^i - w_{h,t}(a,s)\ + \\ & - R_t\Big(k_t^i(phy) - k_t^i(ict)\Big) - \mathcal{D}_t^i\Big] \end{aligned}$$

with ψ^t being the penalty multiplier. Optimality conditions are in order

$$\frac{1}{\mathcal{C}_t^i} = \beta^t \left(1 + r_{t+1}\right) \frac{1}{\mathcal{C}_{t+1}^i}$$

$$R_t = \left(1 + r_{t+1}\right) \zeta_t - (1-\delta)\zeta_{t+1}$$

The first is the usual Euler condition, which displays future path of consumption. The second implies that, aggregating across households, the path on interest rates on capital types, $R_t\,(phy) = R_t\,(ict) \equiv R_t$, is linked to the path of aggregate relative quantity of ICT capital, $\zeta_t = \int_i \zeta_t^i\,\mathrm{d}i$, when $\delta_{phy} = \delta_{ict} \equiv \delta$;[I] *in other words, given the constancy of both the discount factor and the capital depreciation rate, changes across states of the capital rental rate are determined by changes in the relative quantities across capital types. In the spirit of Karabarbounis and Neiman (2014), eq. (6) determines that investing in capital types is profitable as long as the marginal benefit of investment (the capital rental rate) is at least lower than its marginal cost (interest rate, r_t and depreciation rate, δ).*

□

(Labour supply derivation) *The probability of worker-a choosing firm-$h = 1$ in industry-$h = 1$ can be written as*

$$\left(\mathcal{P}_1\,(a,1) \equiv\right) \Longleftrightarrow \mathcal{P}\,(a,1) \qquad \textit{for notation purposes, abstract from firm subscript}$$

$$\begin{aligned}\mathcal{P}\,(a,1) &= prob\Big[w\,(a,1)\,\mathcal{B}\,(a,1)\,\wp^i\,(a,1) > w\,(a,s)\,\mathcal{B}\,(a,s)\,\wp^i\,(a,s)\Big],\ \forall s \neq 1 \\ &= prob\left[\frac{w\,(a,1)\,\mathcal{B}\,(a,1)}{w\,(a,s)\,\mathcal{B}\,(a,s)}\wp^i\,(a,1) > \wp^i\,(a,s)\right],\ \forall s \neq 1\end{aligned}$$

For $s \in [2, S]$, the partial derivative of $\mathcal{P}\,(a,1)$ with respect to $\wp^i\,(a,1)$ is

$$\left\{\wp^i\,(a,1)\,,\ \frac{w\,(a,1)\,\mathcal{B}\,(a,1)}{w\,(a,2)\,\mathcal{B}\,(a,2)}\wp^i\,(a,1)\,,\ \ldots,\ \frac{w\,(a,1)\,\mathcal{B}\,(a,1)}{w\,(a,S)\,\mathcal{B}\,(a,S)}\wp^i\,(a,1)\right\}$$

so that $\mathcal{P}\,(a,1)$ can be re-written as

$$\begin{aligned}\mathcal{P}\,(a,1) &= \mathcal{F}_{\wp^i(a,1)}\Big(\wp^i\,(a,1)\,,\ \alpha_2\wp^i\,(a,1)\,,\ \ldots,\ \alpha_s\wp^i\,(a,1)\Big),\ \forall\wp^i\,(a,s) \\ &= \int \mathcal{F}_{\wp^i(a,1)}\Big(\wp^i\,(a,1)\,,\ \alpha_2\wp^i\,(a,1)\,,\ \ldots,\ \alpha_s\wp^i\,(a,1)\Big)\mathrm{d}\wp^i\,(a,1)\end{aligned}$$

for $\alpha_s = \frac{w(a,1)\mathcal{B}(a,1)}{w(a,s)\mathcal{B}(a,s)}$. Now, recall that the parameter $\wp^i\,(a,s)$ is drawn from a multivariate Frechét-type cumulative distribution,

$$\mathcal{F}_i\Big(\wp^i_{h,\ldots,H}\,(a,1)\,,\ldots,\wp^i_{h,\ldots,H}\,(a,S)\Big) = exp\left[-\sum_s\left(\int_h \wp^i_h\,(a,s)\ \mathrm{d}h\right)^{-\theta}\right]$$

[I] Alternatively, the common depreciation rate might be a weighted average of all the capital-types' depreciation rates.

which becomes, in this framework (i.e., not considering firm's notation),

$$\mathcal{F}\Big(\alpha_1 \wp^i(a,1), \ldots, \alpha_s \wp^i(a,s), \ldots, \alpha_S \wp^i(a,S)\Big) = exp\left[-\sum_s \alpha_s^{-\theta} \sum_s \wp^i(a,s)^{-\theta}\right]$$

$$= exp\left[-\sum_s \Big(\alpha_s \, \wp^i(a,s)\Big)^{-\theta}\right]$$

Taking its derivative with respect to $\wp^i(a,s)$ turns to write that

$$\mathcal{F}_{\wp^i(a,1)}\Big(\wp^i(a,1),\ \alpha_2 \wp^i(a,1),\ \ldots,\ \alpha_s \wp^i(a,1)\Big) = -(-\theta)\, \wp^i(a,s)^{-\theta-1} exp\left[\overline{\alpha}\wp^i(a,s)^{-\theta}\right]$$

with $\overline{\alpha} = \sum_s \alpha_s^{-\theta}$. Evaluating the integral in $\mathcal{P}(a,1)$ yields to

$$\mathcal{P}(a,1) = \int \theta \underbrace{\wp^i(a,s)^{-\theta-1}}_{A} \underbrace{exp\left[\overline{\alpha}\wp^i(a,s)^{-\theta}\right]}_{B} \mathrm{d}\wp^i(a,s)$$

By multiplying and dividing by $\overline{\alpha}$ (so that A can be integrated with B), one gets

$$\begin{aligned}
\mathcal{P}(a,1) &= \frac{\overline{\alpha}}{\overline{\alpha}} \int \theta \wp^i(a,s)^{-\theta-1} exp\left[\overline{\alpha}\wp^i(a,s)^{-\theta}\right] \mathrm{d}\wp^i(a,s) \\
&= \frac{1}{\overline{\alpha}} \int \overline{\alpha}\theta \wp^i(a,s)^{-\theta-1} exp\left[\overline{\alpha}\wp^i(a,s)^{-\theta}\right] \mathrm{d}\wp^i(a,s) \\
&= \frac{1}{\overline{\alpha}} \int \mathrm{d}\mathcal{F}\Big(\wp^i(a,1), \ldots, \wp^i(a,s), \ldots, \wp^i(a,S)\Big) \\
&= \frac{1}{\overline{\alpha}} \\
&= \frac{1}{\sum_s \alpha_s^{-\theta}}
\end{aligned}$$

By recalling that $\alpha_s = \frac{w(a,1)\mathcal{B}(a,1)}{w(a,s)\mathcal{B}(a,s)}$, it is possible to obtain that, $\forall s \neq 1$,

$$\mathcal{P}(a,1) = \frac{\Big(w(a,1)\,\mathcal{B}(a,1)\Big)^{\theta}}{\Big(\sum_s w(a,s)\,\mathcal{B}(a,s)\Big)^{\theta}}$$

where, including firm's subscript-h, it becomes

$$\mathcal{P}_1(a,1) = \frac{\Big(w_1(a,1)\,\mathcal{B}_1(a,1)\Big)^{\theta}}{\Big(\sum_{h,s} w_h(a,s)\,\mathcal{B}_h(a,s)\Big)^{\theta}}$$

Taking in general notation, $\forall h \in [1,H]$ and $\forall s \in [1,S]$, it can be written as

$$\mathcal{P}_h(a,s) = \frac{\Big(w_h(a,s)\,\mathcal{B}_h(a,s)\Big)^{\theta}}{\Big(\sum_{h,s} w_h(a,s)\,\mathcal{B}_h(a,s)\Big)^{\theta}} \quad , \quad \text{with} \quad \mathcal{B}_h(a,s) = \left[g_h(a,s)\right]^{-\varsigma}$$

which denotes the fraction of type-a households choosing to work in firm-h, industry-s as in eq. (7).

To interpret the role of θ – the shape parameter of the Frechét distribution –, Figure B.1 plots the distribution of productivities under different values of θ, interpreted as being the degree of dispersion of households-specific efficiencies in working in firm-h in industry-s. Basically, the larger is the value of θ the lower is the variability of the productivity distribution (i.e., the lower is the dispersion of households' productivities), thus the higher is the degree of labour market concentration of workers across firms and industries.

Moreover, as explained in the main text to interpret eq. (7), "the number of workers of a given task-a willing to be employed in a certain firm-industry pair is determined by different optimal wage levels by firms in industries (wage premium), jointly with workforce composition in terms of both the firm-specific relative amount of workers of a certain type (sorting) and workers having the same efficiency in conducting that job task content (segregation). In this sense, workers in the same task are "highly rival factors" (Hicks 1932) since they can be freely substituted for one another, and this possibility ensures that high-wage workers do sort in high-wage firms and industries, and also that more efficient people would be preferred to get a job in high-wage workplaces than ones who are less productive".[II]

□

(Firm optimization) *Given $y_h(s)$ being the production function in eq. (9), and the firm's conditional demand given by eq. (8), the problem of a monopolistically competitive firm (h, s) choosing capital and labour endowments is*

$$\max_{p_h(s), \{k_h(j,s)\}_{\forall j}, \{\ell_h(a,s)\}_{\forall a}} p_h(s)\, y_h(s) - \left(R \sum_j k_h(j,s) + \sum_a w_h(a,s)\; \ell_h(a,s) \right)$$

$$\text{s.t.} \quad y_h(s) = \left(\frac{p_h(s)}{p(s)} \right)^{\epsilon} y(s)$$

[II] To build intuition, such idea can be formalized using an Horvath (2000)'s aggregator: the aggregate type-a workforce of industry-s can be taught as a CES aggregator of firm-specific labour measures of eq. (7),

$$\ell(a,s) = \left[\int_h \left(\ell_h(a,s) \right)^{\frac{1+\psi}{\psi}} \mathrm{d}h \right]^{\frac{\psi}{1+\psi}}$$

since each worker-i is endowed with $\ell_h^i(a,s) = 1$ unit inelastically supplied. Here, as $\psi \to \infty$, type-a workers become perfect substitutes; conversely, for any given value of parameter ψ comprised in $[0, \infty)$ workers won't be freely substituted across firms in the same industry. As a natural consequence, each worker would work in the firm h, s paying the highest wage according to its $\wp_h^i(a,s)$. More, at the intensive margin, firms in the same industry would pay the same wage level.

This is a crucial aspect to consider when moving from firm to industry dimension: free labour mobility within an industry – materialized when $\psi = \infty$ – allows to impose firms' optimal wages to be even, so to have a unique industry-s wage in equilibrium; refer to Proposition 1 and Remark 2.

Analogously, worker-a wage can be seen as $w(a,s) = \left[\int_h \left(w_h(a,s) \right)^{\frac{1+\psi}{\psi}} \mathrm{d}h \right]^{\frac{\psi}{1+\psi}}$ with $\psi = \infty$.

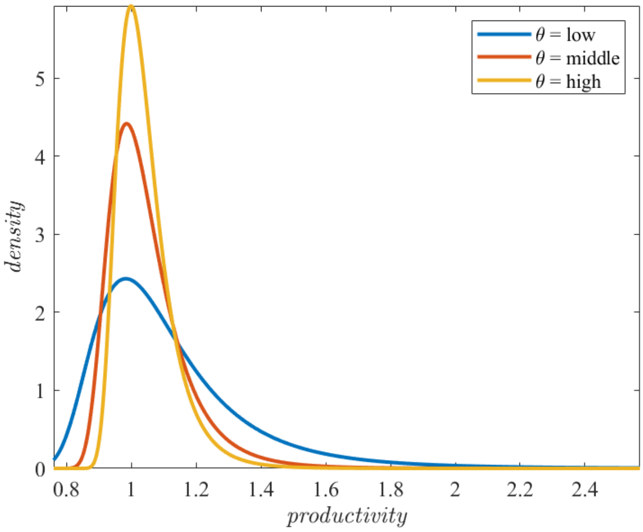


FIGURE B.1: VARIABILITY AND THE SHAPE PARAMETER

Note: the plot represents how the distribution of an arbitrary variable (*e.g.*, productivity) changes along different values of the shape parameter, keeping unchanged the scale parameter; lower shape parameter results in major variability.

with $a = \{rt, nrt\}$ and $j = \{phy, ict\}$. The inter-temporal Lagrangian function for firm-h industry-s takes the form of

$$\begin{aligned}
\underset{p_h(s),\{k_h(j,s),\ell_h(a,s)\}_{\forall j,a}}{\mathcal{L}} = p_h(s)\, y_h(s) \; - & \Big(R\, k_h(phy,s) + R\, k_h(ict,s) \; + \\
& + w_h(rt,s)\;\ell_h(rt,s) + w_h(nrt,s)\;\ell_h(nrt,s)\Big) + \\
& - \psi^t \Big[y_h(s)\, p_h(s)^{\epsilon} - y(s)\Big]
\end{aligned}$$

with ψ^t being the penalty multiplier. Optimality conditions are in order

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial p_h(s)} &: \; \psi = \frac{1}{\epsilon} p_h(s)^{1-\epsilon} \\
\frac{\partial \mathcal{L}}{\partial k_h(phy,s)} &: \; p_h(s)\, f_{k_h(phy,s)} = \mathcal{M}^{\epsilon} R \\
\frac{\partial \mathcal{L}}{\partial k_h(ict,s)} &: \; p_h(s)\, f_{k_h(ict,s)} = \mathcal{M}^{\epsilon} R
\end{aligned}$$

with the price mark-up being $\mathcal{M}^{\epsilon} = \frac{\epsilon}{\epsilon-1}$, and first order conditions of firm-industry specific output relative to capital-types are

$$\begin{aligned}
f_{k_h(phy,s)} &= \alpha \Big(k_h(phy,s)\Big)^{\alpha-1} \left[y_h(s) \Big(k_h(phy,s)\Big)^{-\alpha}\right] \\
f_{k_h(ict,s)} &= (1-\alpha)(1-\mu)\lambda \left[\Big(k_h(phy,s)\Big)^{\alpha} \mathcal{V}_h(s)^{\frac{1-\alpha-\varsigma}{\varsigma}}\, \mathcal{Q}_h(s)^{\frac{\varsigma-\varrho}{\varrho}}\right] \Big(k_h(ict,s)\Big)^{\varrho-1}
\end{aligned}$$

with

$$\mathcal{V}_h\left(s\right) = \mu\Big(\ell_h\left(rt,s\right)\Big)^{\varsigma} + \left(1-\,\mu\right)\mathcal{Q}_h\left(s\right)^{\frac{\varsigma}{\varrho}}$$

and

$$\mathcal{Q}_h\left(s\right) = \lambda\Big(k_h\left(ict,s\right)\Big)^{\varrho} + \left(1-\lambda\right)\Big(\ell_h\left(nrt,s\right)\Big)^{\varrho}$$

Note how aggregate capital rental rates, $R_t\left(j\right)$, are obtained by aggregating marginal product of capital types, $f_{k_h(j,s)}$, across firms and industries.

Optimality conditions for wages are found by deriving directly for $\ell_h\left(a,s\right)$; this results in computing what follows.

$$\begin{aligned}
\frac{\partial\mathcal{L}}{\partial\ell_h\left(rt,s\right)} &: \; p_h\left(s\right)f_{\ell_h(rt,s)} = \mathcal{M}^{\epsilon}\,w_h\left(rt,s\right)\\
\frac{\partial\mathcal{L}}{\partial\ell_h\left(nrt,s\right)} &: \; p_h\left(s\right)f_{\ell_h(nrt,s)} = \mathcal{M}^{\epsilon}\,w_h\left(nrt,s\right)
\end{aligned}$$

with the price mark-up still being $\mathcal{M}^{\epsilon} = \frac{\epsilon}{\epsilon-1}$, and first order conditions of firm h,s's output relative to labour-types are

$$\begin{aligned}
f_{\ell_h(rt,s)} &= (1-\alpha)\mu\Big(k_h\left(phy,s\right)\Big)^{\alpha}\mathcal{V}_h\left(s\right)^{\frac{1-\alpha-\varsigma}{\varsigma}}\Big(\ell_h\left(rt,s\right)\Big)^{\varsigma-1}\\
f_{\ell_h(nrt,s)} &= (1-\alpha)(1-\mu)(1-\lambda)\Big(k_h\left(phy,s\right)\Big)^{\alpha}\mathcal{V}_h\left(s\right)^{\frac{1-\alpha-\varsigma}{\varsigma}}\mathcal{Q}_h\left(s\right)^{\frac{\varsigma-\varrho}{\varrho}}\Big(\ell_h\left(nrt,s\right)\Big)^{\varrho-1}
\end{aligned}$$

with

$$\mathcal{V}_h\left(s\right) = \mu\Big(\ell_h\left(rt,s\right)\Big)^{\varsigma} + \left(1-\,\mu\right)\mathcal{Q}_h\left(s\right)^{\frac{\varsigma}{\varrho}}$$

and

$$\mathcal{Q}_h\left(s\right) = \lambda\Big(k_h\left(ict,s\right)\Big)^{\varrho} + \left(1-\lambda\right)\Big(\ell_h\left(nrt,s\right)\Big)^{\varrho}$$

Now, note that labour supplies for both types of tasks $a = \{rt,nrt\}$ are all determined by eq. (7), namely $\ell_h\left(a,s\right) = f\Big(w_h\left(a,s\right), \mathcal{B}_h\left(a,s\right), \mathcal{W}_{\mathcal{H}}(a,\mathcal{S}), \mathcal{B}_{\mathcal{H}}(a,\mathcal{S})\Big)$. Therefore, equating demand and supply of labour for each task-a results in deriving the associated optimal wage level in general equilibrium, that is

$$\begin{cases}
p_h\left(s\right)f_{\ell_h(rt,s)} = \mathcal{M}^{\epsilon}\,w_h\left(rt,s\right)\\[2em]
\qquad\ell_h\left(rt,s\right) = \left(\dfrac{w_h\left(rt,s\right)\mathcal{B}_h\left(rt,s\right)}{\sum_{h,s}w_h\left(rt,s\right)\mathcal{B}_h\left(rt,s\right)}\right)^{\theta}
\end{cases}$$

and

$$
\begin{cases}
p_h(s)\, f_{\ell_h(nrt,s)} = \mathcal{M}^{\epsilon}\, w_h(nrt,s) \\
\\
\ell_h(nrt,s) = \left(\dfrac{w_h(nrt,s)\, \mathcal{B}_h(nrt,s)}{\sum_{h,s} w_h(nrt,s)\, \mathcal{B}_h(nrt,s)} \right)^{\theta}
\end{cases}
$$

By writing down the extensive forms for each derivative of $\ell_h(a,s)$ and exploiting the calculations to solve for $w_h(a,s)$, together with proposition 1, optimal wages are those reported in eq. (10). Note that $\mathcal{V}_h(s)$ expresses the substitutability between routine workers $\left(\ell_h(rt,s)\right)$ and the ICT composite good, while $\mathcal{Q}_h(s)$ identifies the substitutability between ICT capital $\left(k_h(ict,s)\right)$ and non-routine workers $\left(\ell_h(nrt,s)\right)$, considering each firm-$h \in \mathcal{H}$ in industry-s.

Finally, firm h,s profits can be found by including equilibrium optimality conditions for both capital and wages in

$$
\mathcal{D}_h(s) = p_h(s)\, y_h(s) - \left(R \sum_j k_h(j,s) + \sum_a w_h(a,s)\, \ell_h(a,s) \right)
$$

where $j = \{phy, ict\}$ identifies the types of capital in the economy.

$\square$

(Proof of Proposition 1) *This heuristic proof is centred around the definition of the workers' measures as given by eq. (7) with no worker-a benefit, $\mathcal{B}_h(a,s) = 1$, $\forall a,h,s$. Imagine an economy in which there is only one industry-$h \in \mathcal{S} = 1$ populated by two firms, $\{h,h'\} \in \mathcal{H}$, and define $\mathcal{W}_{\mathcal{H}}(a,s) = w_h(a,s) + w_{h'}(a,s)$. Then:*

(a) *when firms are assumed to be homogeneous in their size, i.e., when $\ell_h(a,s) \equiv \ell_{h'}(a,s)$, then they should set the same optimal wage level since*

$$
\left(\frac{w_h(a,s)}{\mathcal{W}_{\mathcal{H}}(a,s)} = \right) \ \ell_h(a,s) \equiv \ell_{h'}(a,s) \ \left(= \frac{w_{h'}(a,s)}{\mathcal{W}_{\mathcal{H}}(a,s)} \right)
$$

holds only if $w_h(a,s) = w_{h'}(a,s)$.

(b) *if firms are heterogeneous in size eq. (7) predicts how, in order to have $\ell_h(a,s) \neq \ell_{h'}(a,s)$, it is necessary that wage levels are different, $w_h(a,s) \neq w_{h'}(a,s)$. Assume that firm h',s sets a wage rate higher than firm h,s, so that the former is larger than the latter. Assume further an increase in the wage chosen by firm h',s, labelling the new level as $w_{h'}(a,s)'$, while the other wage remains unchanged. Henceforth, it must be the case that*

$$
w_{h'}(a,s) \to w_{h'}(a,s)', \quad \text{so that} \quad \mathcal{W}_{\mathcal{H}}(a,s) < \mathcal{W}_{\mathcal{H}}(a,s)'.
$$

For such firm, clearing the condition

$$
\mathcal{W}_{\mathcal{H}}(a,s)' = \frac{w_{h'}(a,s)'}{\ell_{h'}(a,s)'} \tag{B.1}
$$

means that there should be a related increase in its workforce, $\frac{\partial \ell_{h'}(a,s)}{\partial w_{h'}(a,s)} > 0$, so *that $\ell_{h'}(a,s)' > \ell_{h'}(a,s)$. Different scenario happens to the employees level of firm h,s: it turns out that $\ell_h(a,s)' < \ell_h(a,s)$ when $w_{h'}(a,s) \to w_{h'}(a,s)'$ since $\frac{\partial \ell_h(a,s)}{\partial w_{h'}(a,s)} < 0$. The implied mechanism is just*

$$\frac{\partial \ell_{h'}(a,s)}{\partial w_{h'}(a,s)} = -\frac{\partial \ell_h(a,s)}{\partial w_{h'}(a,s)}$$

However, the given change in the wage of firm h',s not only has an impact on $\ell_{h'}(a,s)$, but it indirectly translates to the wage level of firm h,s due its negative effect on $\ell_h(a,s)$. The firm h,s version of the condition in eq. (B.1)

$$\mathcal{W}_{\mathcal{H}}(a,s)' = \frac{w_h(a,s)'}{\ell_h(a,s)'} \tag{B.2}$$

implies that, after an increase in $\mathcal{W}_{\mathcal{H}}(a,s)$ and a decrease in $\ell_h(a,s)$ due to a positive change in $w_{h'}(a,s)$, then the wage level of firm h,s must increase as well. It turns out that an increase in the wage level of firm h',s must determine an equal increase in the right-hand-side of both eqs. (B.1)-(B.2), which means

$$\frac{w_h(a,s)'}{\ell_h(a,s)'} = \frac{w_{h'}(a,s)'}{\ell_{h'}(a,s)'}$$

so that $\frac{\partial w_h(a,s)}{\partial \ell_h(a,s)} = \frac{\partial w_{h'}(a,s)}{\partial \ell_{h'}(a,s)}$.

To sum up, when firms in a specific industry are of different size in terms of employed workers of type-a, an increase in the wage level of a given firm causes: (i) an increase in the number of employed people in that firm, $\frac{\partial \ell_{h'}(a,s)}{\partial w_{h'}(a,s)} > 0$; (ii) a related decrease in the number of workers of the other firms, $\frac{\partial \ell_h(a,s)}{\partial w_{h'}(a,s)} < 0$, and thus an increase in other firms' wage levels, $\frac{\partial w_h(a,s)}{\partial w_{h'}(a,s)} > 0$. This is true as long as workers of each type-a are free to move across firms in the same industry at no cost and firms can hire new workers from the other firms;

(c) *by applying the same reasoning of point (b), if workers were perfectly mobile across industries, then it would have been the case that $\frac{w(a,s')}{\ell(a,s')} = \frac{w(a,s)}{\ell(a,s)}$. This chance is ruled out by assuming a very high cost of transition from one industry to another such that no one among workers is keen to move.*

□

(Equilibrium characterization) *In equilibrium, the model should specify the clearing conditions of labour, capital, and goods markets. Starting from the labour market, since each household inelastically supplies one unit of labour, then it must hold that the total number of workers of type-a in firm h,s is $\ell_h(a,s) = \int_0^1 \ell_h^i(a,s)\,\mathrm{d}i$, so that the total labour supply is $L^S = \sum_a \sum_h \sum_s \ell_h(a,s)$. The measure of type-$a$ worker in firm h,s is given by eq. (7). Aggregating it across tasks and firms results in obtaining the industry-specific labour supply, $L(s) = \sum_{a,h} \ell_h(a,s)$. Analogously, aggregate labour supply of task-a is found by aggregating across firms and industries,*

$L(a) = \sum_{h,s} \ell_h(a,s)$. *It follows that, considering* $a = \{rt, nrt\}$, *aggregate labour demand for this economy is just* $L^D = \sum_a L(a) = L(rt) + L(nrt)$. *Labour market clearing requires that* $L^D = L^S$.

For what concerns equilibrium in the capital market(s), total physical and ICT capital demands from industries are, respectively, $K^D(phy, s) = \sum_h k_h(phy, s)$ *and* $K^D(ict, s) = \sum_h k_h(ict, s)$, *so that aggregate demands are simply determined:* $K^D(phy) = \sum_s K^D(phy, s)$ *and* $K^D(ict) = \sum_s K^D(ict, s)$. *By the part of supply, aggregating capital quantities over households would results in aggregate physical and ICT capital supplies:* $K^S(phy) = \int_i k^i(phy)\,\mathrm{d}i$ *and* $K^S(ict) = \int_i k^i(ict)\,\mathrm{d}i$. *Equilibrium in both markets requires* $K(phy) \equiv K^D(phy) = K^S(phy)$ *and* $K(ict) \equiv K^D(ict) = K^S(ict)$, *while market clearing in the capital market implies* $K^D(phy) + K^D(ict) = K^S(phy) + K^S(ict)$.

Finally, aggregate profits to be given to households are $\mathcal{D}^D = \int_i \mathcal{D}_i\,\mathrm{d}i$, *while* $\mathcal{D}^S = \sum_s \mathcal{D}(s)$ *are the total profits computed by aggregating industry-specific profits,* $\mathcal{D}(s) = \sum_h \mathcal{D}_h(s)$. *Equilibrium requires* $\mathcal{D} \equiv \mathcal{D}^D = \mathcal{D}^S$. *This implies that, by aggregating the households' inter-temporal budget constraints and imposing the clearing conditions so far, including also total quantities for*

$$
\begin{aligned}
\mathcal{C}^i &= \int_i \mathcal{C}^i\,\mathrm{d}i \\
I(j) &= \int_i I^i(j)\,\mathrm{d}i\,, \quad \forall j \\
b &= \int_i b^i\,\mathrm{d}i \\
w &= \sum_a \sum_h \sum_s w_h(a,s) \\
\mathcal{B} &= \sum_a \sum_h \sum_s \mathcal{B}_h(a,s)
\end{aligned}
$$

where $\mathcal{B} = 1$ *since it is the sum of relative quantities, the aggregate resource constraint for this economy at time* t *reads as*

$$\mathcal{C}_t + I_t(phy) + I_t(ict) + b_{t+1} - (1+r_t)b_t = w_t\mathcal{B}_t L_t + R_t\Big(K_t(phy) + K_t(ict)\Big) + \mathcal{D}_t$$

which equals the total output as defined by the final output CES aggregator, Y. *Equilibrium conditions are described in Section 2.*

□

C. Model estimation and simulation

(Estimating equations) *To derive the equations to estimate the elasticity of substitution between ICT capital and non-routine workers (ρ), and the elasticity of substitution between routine workers and ICT composite good (σ), i.e., that shown in eqs. (12) and (13), I apply the procedure implemented by Karabarbounis and Neiman (2014). The main steps to be implemented are:*[I]

1. *define a CES production function, $y(\cdot)$ and compute the related F.O.C.s.;*

2. *define the following income shares. For a given labour force (ℓ), a given capital stock (k), and given profits $(\mathcal{D})$,*

$$\phi_\ell = \left(\frac{1}{\mathcal{M}^\epsilon}\right)\left(\frac{w(\ell)\,\ell}{w(\ell)\,\ell + R\,k}\right),\ \phi_k = \left(\frac{1}{\mathcal{M}^\epsilon}\right)\left(\frac{R\,k}{w(\ell)\,\ell + R\,k}\right),\ \phi_{\mathcal{D}} = 1 - \frac{1}{\mathcal{M}^\epsilon};$$

3. *by combining the F.O.C. for capital (either for labour) with all the above shares, one gets an equation whose left-hand side is $1 - \phi_\ell \mathcal{M}^\epsilon$. Then, this should be written in changes between two arbitrary periods, whose resulting elements should be "hatted", that is $\widehat{\cdot}$;*

4. *use eq. (6) to substitute $\widehat{R}$;*

   $\rightarrow$ *use the Euler condition (from the households side) expressed in deviation between two arbitrary periods get $\widehat{(1+r)} = \frac{1}{\beta}$ so that, under constant β and δ, it holds that $\widehat{R} = \widehat{\zeta}$;*

5. *once substituting $\widehat{R}$ in eq. (6), take a linear approximation of the resulting equation around $\widehat{\zeta} = 0$, thereby obtaining the estimating equation.*

Apply this procedure for whatever CES functional form of the production function $y(\cdot)$. Note that, to carry out this procedure in the framework I propose it is necessary to assume equal marginal product of each type of capital. In fact, in eq. (6), trends in both capital rental rates are tied with trends in the quantity of ICT capital relative to non-ICT one, after aggregating optimal conditions in the firm problem. Here, the two capital rates are given by capitals' marginal productivity: thus, to have a unique capital rental rate, such that $R(phy) = R(ict) \equiv R$, equal marginal product of capital types are in order.

(Targeting moments for MSM) *Start from the weighting parameters, λ and μ. For each industry-$s \in \mathcal{S}$, the weight of ICT capital (λ) in the ICT composite is matched with the industry-specific ICT capital in the aggregate stock in the data.*

Differently, the weight of routine workers (μ) in the production function is used to bridge the share of routine workers in the data with that predicted by the model,

[I] Please refer to Karabarbounis and Neiman (2014) for further details and discussion.

i.e., I implement the following identity using eq. (7): $\ell_{model}(a,s) = \left(\frac{w(a,s)}{\mathcal{W}(a)} \frac{\mathcal{B}(a,s)}{\mathcal{B}(a)}\right)^{\theta} \approx \ell_{data}(a,s)$. *Of course, since the model's measures of employment are determined by relative wages, some slight differences in the estimated matched moments are in order.*

Finally, I consider productivity dispersion parameter, θ*, which directly relates to the between-industry wage difference for worker-*a*. Given* $a = rt$*, I use the wage premium of type-*a *working in top industry-*s *relative to its counterpart in the bottom industry-*s'*:*

$$\frac{w(rt,s)}{w(rt,s')} = \frac{\left[\Lambda(s)\,\chi(rt,s)\left(k(phy,s)\right)^{\alpha(s)} \mathcal{V}(s)^{\frac{1-\alpha(s)-\varsigma(s)}{\varsigma(s)}} \mathcal{B}(rt,s)^{\theta(\varsigma(s)-1)}\, \mathcal{WB}(rt)^{\theta(1-\varsigma(s))}\right]^{\frac{1}{1+\theta-\theta\varsigma(s)}}}{\left[\Lambda(s')\,\chi(rt,s')\left(k(phy,s')\right)^{\alpha(s')} \mathcal{V}(s')^{\frac{1-\alpha(s')-\varsigma(s')}{\varsigma(s')}} \mathcal{B}(rt,s')^{\theta(\varsigma(s')-1)}\, \mathcal{WB}(rt)^{\theta(1-\varsigma(s'))}\right]^{\frac{1}{1+\theta-\theta\varsigma(s')}}}$$

If considering changes over time, only $k(phy,\cdot)$*,* $\mathcal{V}(\cdot)$*,* $\mathcal{B}(\cdot)$*, and* $\mathcal{WB}(\cdot)$ *are time-varying, with all the other parameters previously calibrated, targeted and fixed: this leaves* θ *as the only free parameter to match this moment. Note how I choose to pin down labour market concentration for routine workers in accordance with Figure 4.*

TABLE C.1: EMPLOYMENT MEASURES AND TASKS' RELATIVE WAGES

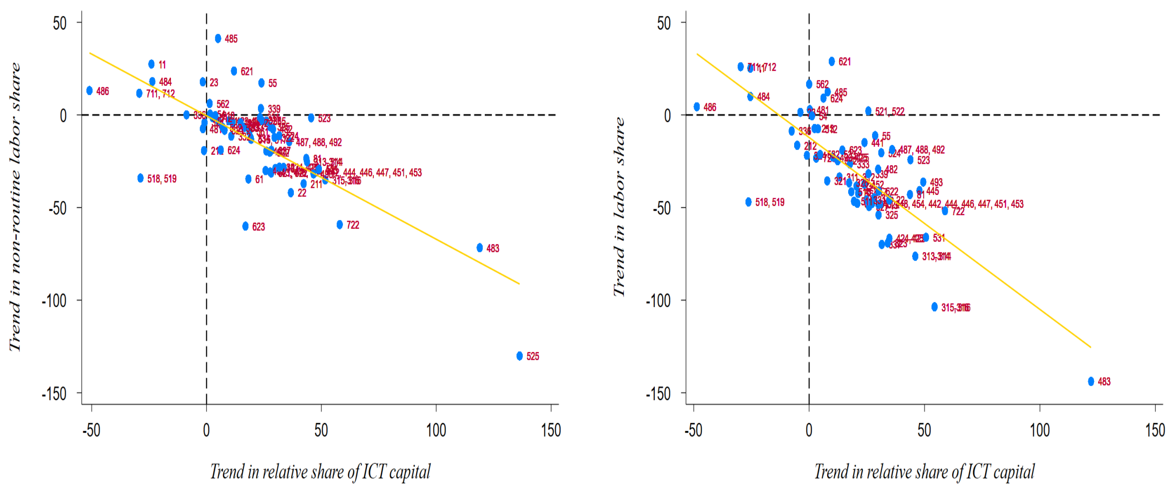
	$\log \ell(rt,s)$			$\log \ell(nrt,s)$		
	(1)	(2)	(3)	(1)	(2)	(3)
$w(rt,s)\,\mathcal{B}(rt,s)\,/W(rt)\,\mathcal{B}(rt)$	.765*	.657*	3.55***			
	(.32)	(.28)	(.17)			
$w(nrt,s)\,\mathcal{B}(nrt,s)\,/W(nrt)\,\mathcal{B}(nrt)$				5.76***	3.71***	29.4***
				(1.9)	(1.1)	(2.2)
Industry FE	✓	✓	✗	✓	✓	✗
Time FE	✗	✓	✗	✗	✓	✗

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on* $N = 1240$ *observations. All the regressions are of the form* $y_t = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$*, with* $\mathcal{X}_i$ *being the regressors, and* $\mathcal{Z}_j$ *a set of controls. All series are in logs. Constant not reported to save space. Source: BLS and own calculations.*

Table C.2: Method of Simulated Moments

				fit	
	parameter	*value*	***moment to match***	*data*	*model*
μ_{bot}	weight of routines in $y(bot)$	0.6763	routine share, bottom	.0858	.0434
μ_{mid}	weight of routines in $y(mid)$	0.4953	routine share, middle	.1357	.0458
μ_{top}	weight of routines in $y(top)$	0.3366	routine share, top	.0985	.0199
λ_{bot}	weight of ICT in $\mathcal{Q}(bot)$	0.4565	ICT share, bottom	.3968	.3968
λ_{mid}	weight of ICT in $\mathcal{Q}(mid)$	0.4645	ICT share, middle	.3042	.3042
λ_{top}	weight of ICT in $\mathcal{Q}(top)$	0.4514	ICT share, top	.2990	.2990
θ	productivity dispersion	11.302	wage premium, $w\,(a,\{s,s'\})$	.9945	.9945

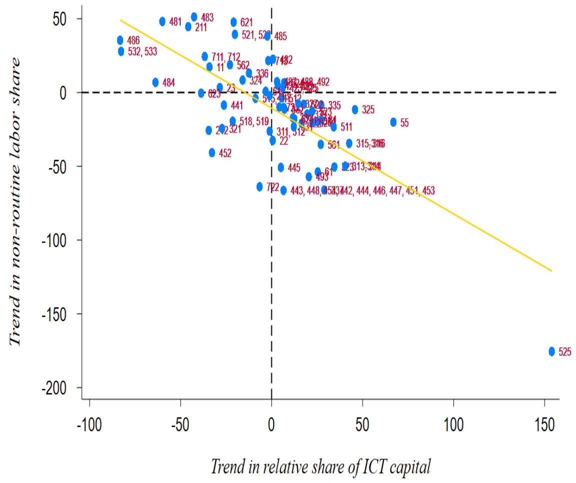
Estimated values and related matched moment using the Methods of Simulated Moments.



(A) Estimates for ρ (B) Estimates for σ

Figure C.1: Correlation between elasticities and relative ICT

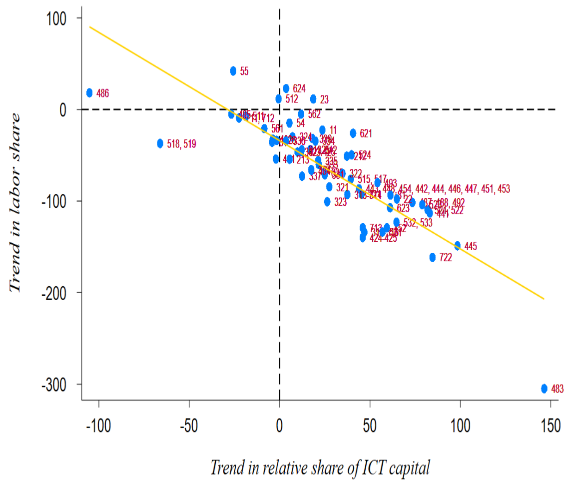
Note: these scatterplots compute the correlation of trends in labour share with trends in the stock of ICT capital relative to physical capital, namely the left- and right-hand sides estimated through the elasticities of substitution as in eqs. (12) and (13), respectively. Each y-axis report the labour share, while common x-axis is referred to ICT capital. Panel C.1a refers the correlation of the elasticity of substitution between ICT capital and non routine workers, while Panel C.1b plots that of the entire labour share (considering both routine and non routine workers). Solid-gold negatively-sloped line is the estimated linear trend from robust regression. Labels to each point refer to the 3-digit U.S. 2017 NAICS code of the related industry.
Source: BEA, BLS and own calculations.



(A) ESTIMATES FOR ρ, 2003-2012

(B) ESTIMATES FOR σ, 2003-2012

(C) ESTIMATES FOR ρ, 2013-2022



(D) ESTIMATES FOR σ, 2013-2022

FIGURE C.2: CORRELATION BETWEEN IN-TIME ELASTICITIES AND RELATIVE ICT

Note: these scatterplots compute the correlation of trends in labour share with trends in the stock of ICT capital relative to physical capital, namely the left- and right-hand sides estimated through the elasticities of substitution as in eqs. (12) and (13), respectively; first row displays correlation in the period 2003-2012, while the second row that in the period 2013-2022. Each y-axis report the labour share, while common x-axis is referred to ICT capital. Panels C.2a and C.2c refer the correlation of the elasticity of substitution between ICT capital and non routine workers, while Panels C.2b and C.2d plot that of the entire labour share (considering both routine and non routine workers). Solid-gold negatively-sloped line is the estimated linear trend from robust regression. Labels to each point refer to the 3-digit U.S. 2017 NAICS code of the related industry. *Source*: BEA, BLS and own calculations.

Table C.3: Model fit, untargeted moments

moment	***fit***	
	data	*model*
aggregate task-premium	.001	.005
aggregate wage	-.008	-.073
routine wage, bottom	-.016	-.070
routine wage, middle	-.001	-.051
routine wage, top	-.007	-.079
non-routine wage, bottom	-.019	-.060
non-routine wage, middle	.003	-.074
non-routine wage, top	-.010	-.046

Untargeted moments to match to validate the calibration strategy. All moments, referred to real log-wages, are taken as percentage changes throughout the series.

(Market concentration) *To evaluate the pattern in concentration at industry level, the standard measure Herfindahl-Hirschman Index (HHI) is computed to account for market concentration of labour force for industries. To compute the dynamics in each year, labour market-level concentration for task-a is defined as*

$$HHI_{\ell(a)} = \sum_{s_{|g}} \left(\frac{\ell(a, s_{|g})}{\ell(a)} \right)^2 \tag{C.1}$$

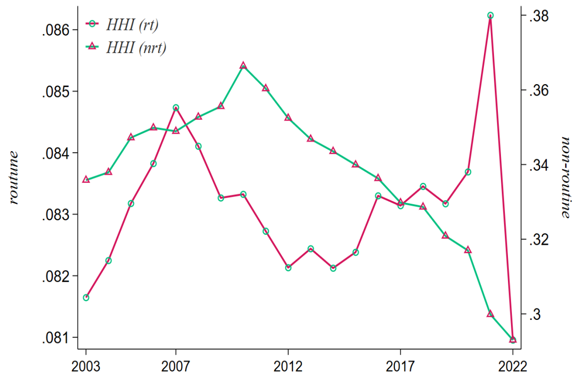
where the sum is over individual industries (s), or over a group of industries ($s_{|g}$). Analogously, total employment concentration is defined by $HHI_{\ell} = \sum_a HHI_{\ell(a)}$. This measure is included in the range $[0, 1]$: a value of 1 identifies maximum market concentration, namely a single monopsonist in the labour market; conversely, a value of 0 results in a perfectly competitive environment. By definition, if industries have equal labour force size, the index would converge to the number of industries. An index below 0.15 points for an un-concentrated labour market, an index between 0.15 and 0.25 indicates a moderate concentration, while a value higher than 0.25 results in an highly concentrated labour market.

Each subplot of Figure C.3 depicts the evolution in HHI score for both routine and non-routine workers. As a general observation, labour market concentration for both tasks has increased for top and bottom groups, but it has decreased for middle. With respect to bottom group, Panel C.3a indicates a joint increase in market concentration for both job tasks, with a very high concentration for non-routine workers, $HHI_{\ell(nrt)} > 0.25$; routine workers are approaching to a moderate concentration.

In relation to middle group, after a steady increase, labour market concentration for non-routine workers constantly drops, even if concentration is still high since $HHI_{\ell(nrt)} > 0.25$. The score for routine workers is low, but if follows cyclical fluctuations, with a steady increase before a recession, and thereafter a substantial drop.

Finally, top group has seen a marked and steady increased in market concentration for non-routine workers, which is approaching to a moderate concentration; for the part of routine workers, concentration is high and, after a steady rise in the early years, then a flat pattern has occurred.

(A) BOTTOM



(B) MIDDLE

(C) TOP

FIGURE C.3: LABOUR MARKET CONCENTRATION BY GROUPS OF INDUSTRIES

Note: the figure represents the evolution in labour market concentration as measured by the Herfindahl-Hirschman Index (HHI). Each subplot measures the dynamics in each broadly defined group of industries by routine (red line, left axis) and non-routine (green line, right axis) tasks. *Source*: BLS and own calculations.

Table C.4: Summary of calibration, 2003-2012

	parameter	***value***				***source***
		bottom	*middle*	*top*	*global*	
α	*physical capital, share of* $y(s)$	0.131	0.091	0.254		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in* $y(s)$	0.196	0.486	0.902		*MSM*
λ	*ICT capital share in* $Q(s)$	0.650	0.530	0.185		*MSM*
θ	*households' productivities dispersion*				7.26	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.355	0.431	0.408		*estimation*
σ	*EoS, routine and ICT composite*	0.366	0.429	0.367		*estimation*

Parameters by industry groups, second-half of the sample. Under "data" the values are directly computed from the data sources, while "external" sets standard values from the literature. "MSM" refers to the Methods of Simulated Moments, and "estimation" reports the previously estimated values from Table 8.

Table C.5: Summary of calibration, 2013-2022

	parameter	***value***				***source***
		bottom	*middle*	*top*	*global*	
α	*physical capital, share of* $y(s)$	0.131	0.104	0.260		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in* $y(s)$	0.415	0.506	0.608		*MSM*
λ	*ICT capital share in* $Q(s)$	0.786	0.490	0.327		*MSM*
θ	*households' productivities dispersion*				7.79	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.819	0.345	0.508		*estimation*
σ	*EoS, routine and ICT composite*	0.326	0.438	0.357		*estimation*

Parameters by industry groups, second-half of the sample. Under "data" the values are directly computed from the data sources, while "external" sets standard values from the literature. "MSM" refers to the Methods of Simulated Moments, and "estimation" reports the previously estimated values from Table 8.

TABLE C.6: METHOD OF SIMULATED MOMENTS, SUB-PERIODS

	moment to match	*2003-2012* value	data	model	*2013-2022* value	data	model
μ_{bot}	routine share, bottom	0.196	0.087	0.097	0.415	0.085	0.051
μ_{mid}	routine share, middle	0.486	0.132	0.057	0.506	0.140	0.030
μ_{top}	routine share, top	0.902	0.100	0.013	0.608	0.097	0.083
λ_{bot}	ICT share, bottom	0.650	0.399	0.399	0.786	0.395	0.395
λ_{mid}	ICT share, middle	0.530	0.314	0.314	0.490	0.295	0.295
λ_{top}	ICT share, top	0.185	0.287	0.287	0.327	0.311	0.311
θ	wage premium, $w(a,\{s,s'\})$	7.259	0.994	0.994	7.787	0.995	0.995

Estimated values and related matched moment using the Methods of Simulated Moments for the first and the second half of the sample.

TABLE C.7: VARIATIONS IN THE SERIES AND STRUCTURAL HETEROGENEITY

	data	**model**	**model** $\vert\,\Delta\Phi(x)$: $\Delta\ell(rt)$	$\Delta\ell(nrt)$	$\Delta(\ell)$	$\Delta k(ict)$	$\Delta(\ell, ict)$	$(all)_{t_0}$
WAGES, VARIANCE								
routine	2.285	2.289	2.55	2.42	2.24	2.78	2.45	2.75
non-routine	2.314	2.311	2.78	2.44	2.37	2.78	2.39	2.83
industry	2.299	2.300	2.66	2.43	2.30	2.78	2.42	2.79

Changes in variances in real log-wages in the model induced by variations in one or more series, keeping fixed the others, and imposing the parameters to be that in the baseline calibration (i.e., quantification of Model C under industry-specific parameters). $\Delta(\ell)$ refers to joint variations in routine and non-routine series, while $\Delta(\ell, ict)$ is associated to simultaneous changes in both ICT capital and both types of workers. $(all)_{t_0}$ keeps fixed all the series (physical capital included), at their initial level, $t_0 = 2003$. In all the columns, the variation is computed throughout the period-by-period percentage differential thus identifying overall changes in empirical trends implied both by the data and the model; same values differ in terms of decimals.

(Further results of Subsection 4.2) *As in the main text, below I have performed an opposite exercise of Table 9 where the main structural parameters in the model, namely* $\{\rho_s, \sigma_s, \theta\}_{\forall s \in \{bot,mid,top\}}$ *are keeping fixed one or more at time, and let to change the non-ICT capital share and the weighting parameters in the production function, respectively* $\{\alpha_s, \mu_s, \lambda_s\}_{\forall s \in \{bot,mid,top\}}$*; the output of such exercise is shown in Table C.8.*

The two analysis performed are conducted by considering industries as a whole, that is, keeping the total employment in terms of both routine and non-routine workers, in order to address the determinants of between-industry wage inequality. However, as noticed, these two exercises can be performed also by distinguish the effects of changing parameters on separate job tasks: Table C.9 reports the two for routine workers, while Table C.10 the same but for non-routine workers.

Below it is possible to find a detailed comment for each employment group and, further below, a comparative comment that shows how the patterns holding for industry with aggregate employment hold also for routine and non-routine workers separately. Visually, these results are in Figures C.4a and C.5a for aggregate industries, and in Figures C.4b and C.5b by job task categories.

TABLE C.8: MODEL COUNTERFACTUAL, FIXING

		Δ ***model*** $\mid \Delta\, m\left(\Phi\left(t_2\right), \Theta = \{\Psi_{t_1}, -\Psi_{t_2}\}\right)$		
industry wages	$var\,(w)_{t_2}$	*level*	*share, model*	*share, data*
DATA	1.18			
MODEL	1.09			
$\Delta\Theta_{\mid\sigma}$		.94	.86	.79
$\Delta\Theta_{\mid\rho}$		2.21	2.04	1.87
$\Delta\Theta_{\mid(\sigma,\rho)}$		1.31	1.21	1.11
$\Delta\Theta_{\mid\theta}$		1.04	.96	.88
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		1.30	1.20	1.10

Quantification of Model B. Model implied between-industry real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by fixing a specified parameter, $\frac{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_1),\Theta(-\Psi,t_2)\right]}{m\left[\Phi(t_2)\,\middle|\,\Theta(-\Psi,t_2)\right]}$*, where* $\Phi\left(t_2\right)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* $(-\Psi, t_2)$*, are taken in the second period, while* (Ψ, t_1) *reflects the set of all the parameters in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

TABLE C.9: MODEL COUNTERFACTUALS FOR ROUTINE WORKERS

routine wages	*var* $(w)_{t_2}$	Δ ***model*** $\mid \Delta\, m\left(\Phi\left(t_2\right), \Theta=\left\{\Psi_{t_2},-\Psi_{t_1}\right\}\right)$ *level*	*share, model*	*share, data*
		CHANGE		
DATA	1.14			
MODEL	.95			
$\Delta\sigma$		.99	1.04	.87
$\Delta\rho$		.35	.37	.30
$\Delta(\sigma,\rho)$		.66	.70	.58
$\Delta\theta$		.57	.60	.50
$\Delta(\sigma,\rho,\theta)$		.71	.76	.63
		FIXING		
DATA	1.14			
MODEL	.95			
$\Delta\Theta_{\mid\sigma}$		.71	.75	.62
$\Delta\Theta_{\mid\rho}$		1.15	1.22	1.01
$\Delta\Theta_{\mid(\sigma,\rho)}$		.84	.89	.74
$\Delta\Theta_{\mid\theta}$		.90	.95	.79
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		.81	.86	.71

Quantification of Model A and Model B. Model implied between-industry routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by fixing a specified parameter, $\frac{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2),\Theta(-\Psi,t_1)\right]}{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2)\right]}$*, where* $\Phi\left(t_2\right)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (Ψ,t_2)*, are taken in the second period, while* $(-\Psi,t_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Opposite notation for "fixing". Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

TABLE C.10: MODEL COUNTERFACTUALS FOR NON-ROUTINE WORKERS

non-routine wages	$var\,(w)_{t_2}$	Δ ***model*** $\mid \Delta\, m\Big(\Phi\,(t_2)\,,\Theta = \{\Psi_{t_2}, -\Psi_{t_1}\}\Big)$ level	share, model	share, data
		CHANGE		
DATA	1.21			
MODEL	1.22			
$\Delta\sigma$		3.74	3.06	3.09
$\Delta\rho$		1.41	1.15	1.16
$\Delta(\sigma,\rho)$		1.57	1.28	1.30
$\Delta\theta$		2.11	1.73	1.75
$\Delta(\sigma,\rho,\theta)$		1.61	1.31	1.33
		FIXING		
DATA	1.21			
MODEL	1.22			
$\Delta\Theta_{\vert\sigma}$		1.16	.95	.96
$\Delta\Theta_{\vert\rho}$		3.27	2.67	2.70
$\Delta\Theta_{\vert(\sigma,\rho)}$		1.78	1.45	1.47
$\Delta\Theta_{\vert\theta}$		1.18	.97	.98
$\Delta\Theta_{\vert(\sigma,\rho,\theta)}$		1.79	1.46	1.48

Quantification of Model A and Model B. Model implied between-industry non-routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by fixing a specified parameter, $\frac{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2),\Theta(-\Psi,t_1)\right]}{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2)\right]}$, *where* $\Phi\,(t_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (Ψ, t_2), *are taken in the second period, while* $(-\Psi, t_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Opposite notation for "fixing". Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

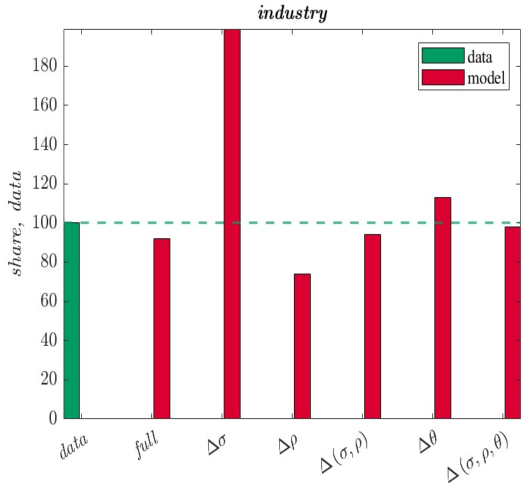
(Comparative comment) *Investigations on the contribution of structural parameters outline an important feature of the U.S. wage structure: observed structural differences among industries account for a sizeable fraction of wage inequality. All the effects can be summarized as follows:*

(a) ***routines****. Most of the share is accounted by trends in industry-heterogeneous differentials in elasticities of substitution among capital and worker types (58%, upper part of Table C.9), also taken in combination to different weights of factor inputs in production (79%, lower part of Table C.9). Considering these differences across industries, the rise in labour market concentration in terms of workers' complementarities (sorting and segregation effects) is an amplifier of U.S. wage inequality across routine job tasks in the last two decades, explaining 63% of the total between-industry routine real log-wage variance (upper part of Table C.9);*

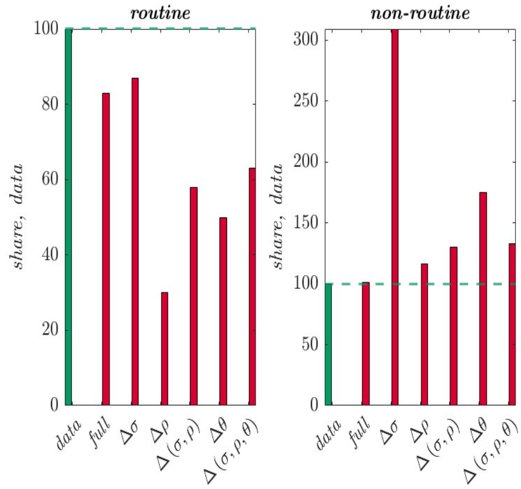
(b) ***non-routines****. Most of the share is accounted by trends in industry-heterogeneous differentials in elasticities of substitution among capital and worker types (130%, upper part of Table C.10), also taken in combination to different weights of factor inputs in production (98%, lower part of Table C.10). Considering these differences across industries, the rise in labour market concentration in terms of workers' complementarities (sorting and segregation effects) is an amplifier of U.S. wage inequality across non-routine job tasks in the last two decades, explaining 133% of the total between-industry non-routine real log-wage variance (upper part of Table C.10);*

(c) ***industries****. Most of the share is accounted by trends in industry-heterogeneous differentials in elasticities of substitution among capital and worker types (94%, Table 9), also taken in combination to different weights of factor inputs in production (88%, Table C.8). Considering these differences across industries, the rise in labour market concentration in terms of workers' complementarities (sorting and segregation effects) is a amplifier of U.S. wage inequality across industries in the last two decades, explaining 98% of the total between-industry real log-wage variance (Table 9).*

Under a comparative perspective, routine workers wage inequality behaves similarly from that of non-routine and industries. All these variances are mostly explained by the substitution elasticity between routine and non-routine workers (σ) only and, jointly, the huge negative effect of σ on wage inequality is reduced due to changes in ρ. On the role of θ, again a clear division does not emerge: real wage dispersion increases after an increase in labour market concentration either for routine and non-routine workers and for industries. Henceforth, stronger workers' complementarities through stronger sorting and segregation effects result in negatively impacting all the considered categories, thus increasing real log-wage dispersion.



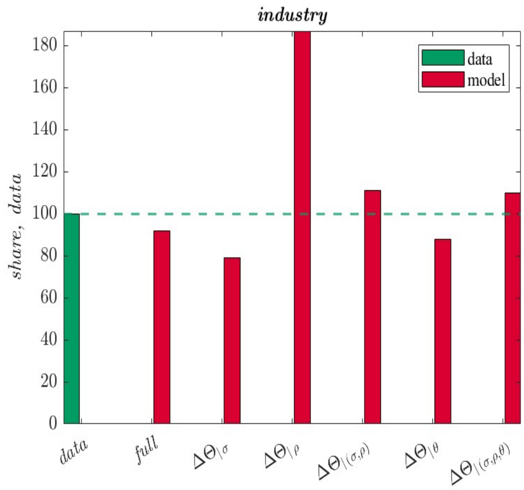
(A) INDUSTRY



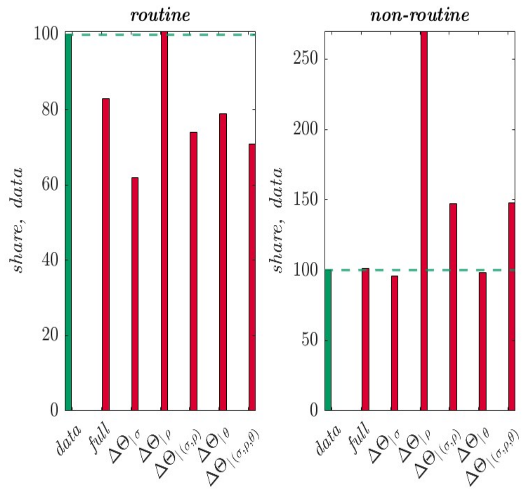
(B) TASKS

FIGURE C.4: MODEL COUNTERFACTUAL, CHANGE

Note: these figures plot the second-period between-industry real log-wage variance, as a share of that in the data, implied by the model when changing one or more parameters at time as shown in Table 9 and in the upper part of Tables C.9 and C.10, for industry (total employment), routine and non-routine workers.



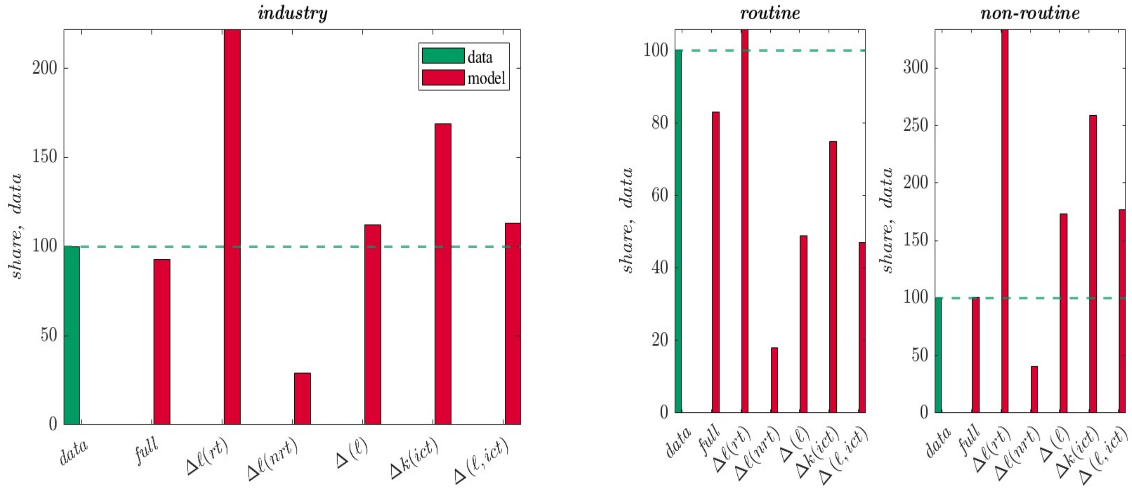
(A) INDUSTRY



(B) TASKS

FIGURE C.5: MODEL COUNTERFACTUAL, FIXING

Note: these figures plot the second-period between-industry real log-wage variance, as a share of that in the data, implied by the model when fixing one or more parameters at time as shown in Table C.8 and in the lower part of Tables C.9 and C.10, for industry (total employment), routine and non-routine workers.



(A) INDUSTRY (B) TASKS

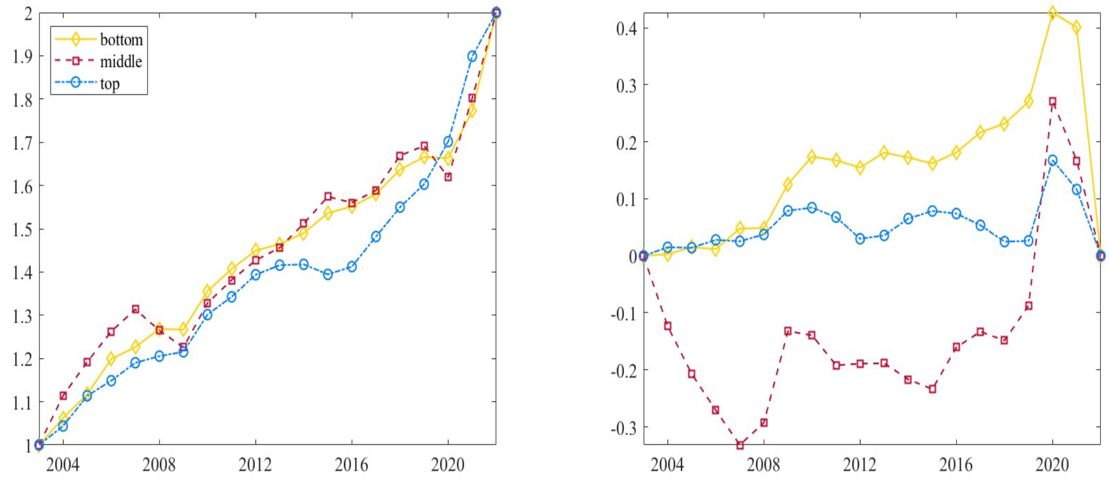
FIGURE C.6: MODEL COUNTERFACTUAL, SBTC

Note: these figures plot the second-period between-industry real log-wage variance, as a share of that in the data, implied by the model when changing one or more parameters at time while keeping fixed the all the parameters, as shown in Tables C.12 and C.13, for industry (total employment), routine and non-routine workers.

TABLE C.11: MODEL VS. DATA COUNTERFACTUAL, SERIES AND NEW PARAMETERS

		model $\mid \Delta\,\Phi(x)$						
							$\Delta\,(tfp)$	
	data	$\Delta\ell\,(rt)$	$\Delta\ell\,(nrt)$	$\Delta\ell$	$\Delta k\,(ict)$	$\Delta\,(tech)$	*newly*	*baseline*
WAGES, VARIANCE								
routine	2.285	.62	.41	.46	.54	.40	.57	.34
non-routine	2.314	.07	.02	.02	.04	.03	.02	.09
industry	2.299	.35	.22	.24	.29	.21	.30	.22

Quantification of Model C. Changes in variances in real log-wages in the model induced by variations in one or more series, keeping fixed the others, and imposing the newly estimated parameters to be homogeneous across industries. $\Delta\,(\ell)$ refers to joint variations in routine and non-routine series, $\Delta\,(tech)$ is associated to simultaneous changes in both ICT capital and non-routine workers, while changes in estimated industry-specific Hicks-neutral exogenous total factor productivity (TFP), given eq. (14), are captured by $\Delta\,(tfp)$; TFP series are taken under the same (newly estimated) parameters' values, or given the baseline calibration (Table 3.2). In all the columns, the variation is computed throughout the period-by-period percentage differential thus identifying overall changes in empirical trends implied both by the data and the model; same values differ in terms of decimals.

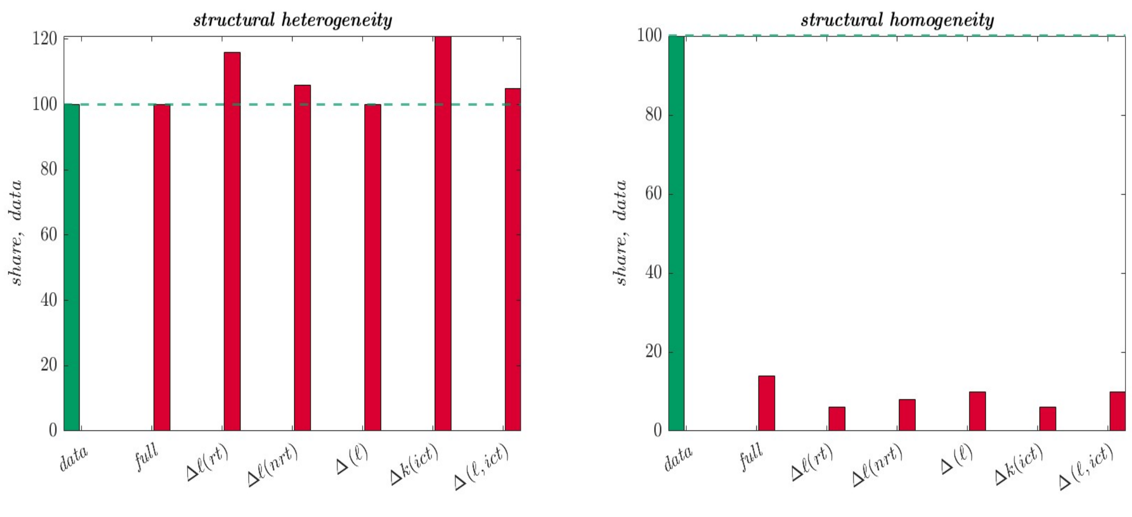


(A) Unique calibration, newly

(B) Difference with baseline

FIGURE C.7: ESTIMATED PRODUCTIVITIES UNDER NEW PARAMETERS

Note: the figure shows the estimated Hicks-neutral exogenous total factor productivity (TFP) measures estimated from the model. Panel C.7a plots the series given a calibration where all the parameters are evenly set at the same *newly estimated* values for all the industry-groups, while Panel C.7b shows the difference between such series and the estimated TFP measures using the baseline calibration reported in Table 6. Series are scaled to be in the same range for graphical comparison.



(A) Structural heterogeneity

(B) Homogeneous parameter

FIGURE C.8: MODEL COUNTERFACTUAL, SERIES

Note: these figures plot the between-industry real log-wage variance, as a share of the data, implied by the model when one or more series at time are changing, keeping fixed the parameters. Panel C.8a reports the result of Table C.7, where structural parameters are heterogeneous across industries. By contrast, Panel C.8b reports the outcome from Table 10, under industry-homogeneous structural parameters; note that a similar graph would appear if considering Table C.11.

(Comment for Skill-Biased Technological Change (SBTC) estimation) *Finally, the analysis turns to the systematic impact of Skill-Biased Technological Change (SBTC). The model can evaluate how simultaneous changes in capital-labour series-x affect between-industry wage inequality when the evolution of production technology parameters ($\alpha, \mu, \lambda, \rho, \sigma$) and labour market concentration (θ) is held fixed. Given the re-estimation procedure, consider the following counterfactual setup:*

$$\Delta var\Big(w\left(s\right)-\overline{w}\Big)=f\Big(\Phi_s\left(x,t_1\right),\Delta\Phi_s\left(x_{t_2},-x_{t_1}\right)\ \Big|\ \Theta_s\left(t_1\right)\Big) \qquad \text{(Model D)}$$

where the change in between-industry wage variance is a function of industry-specific factor series, $\Phi_s\left(x,t_1\right)$, and parameters, $\Theta_s\left(t_1\right)$, taken in period one, while considering one (or a combination of) series at the second period level while keeping fixed the others, $\Phi_s\left(x_{t_2},-x_{t_1}\right)$ to first period.

Table C.12 presents the results. Column (2) reports the between-industry variance in the second period (t_2) for both the data and the model, while columns (3)-(5) show, for each series in column (1), the implied wage variance, its share relative to the "all-series" model, and the fraction of the observed variance explained. Differences in routine workers alone substantially inflate wage inequality (2.62), with ICT capital contributing to a lesser extent (1.69). Combinations of both labour types, and in combination with ICT capital, account for more than the actual real log-wage variance (112-113%). Compared to Table 9, these "changing-series" shares of data-driven wage variance are considerably higher than when also including substitution elasticities and sorting/segregation effects. This implies that the "quantity effect" of the SBTC, by itself, tends to overstate between-industry wage inequality. Changes in capital and labour series are necessary but not sufficient to fully explain observed real log-wage variance: the key drivers are the heterogeneous structural parameters, which vary unevenly across industries. The next section illustrates this more clearly by recomputing between-industry wage inequality while fully neutralizing structural differences, isolating the contribution of factor input variations alone.

The analysis performed is conducted by considering industries as a whole, that is, keeping the total employment in terms of both routine and non-routine workers, in order to address the determinants of between-industry wage inequality. As noticed, these two exercises can be performed also by distinguish the effects of changing parameters on separate job tasks; to this end, Table C.13 reports the SBTC analysis for both routine and non-routine workers. The impact of SBTC seems to have major effect on non-routine workers: in fact, while the variance explained by shifts in the series explain substantially less data-share compared to the case in which shifts in structural parameters are also considered, the wage variance of non-routine workers explodes. Therefore, as in the model, major impact of only the series is on non-routine workers.

As argued, Skill-Biased Technological Change (SBTC) theory implies that wage differentials are shaped by different adoption rate of technology, so that dispersion in wages is mostly affected by how much an industry increases its technological capital and this, in turn, will increase its demand for high-skill or non-routine workers.

Thus, below I have performed an exercise where the main structural parameters in the model, namely $\{\rho_s, \sigma_s, \theta\}_{\forall s \in \{bot,mid,top\}}$, and the production function weighting parameters, namely $\{\alpha_s, \mu_s, \lambda_s\}_{\forall s \in \{bot,mid,top\}}$, are keeping fixed at the first period, and let to change the series related to routine and non-routine workers, and that of ICT capital.

TABLE C.12: MODEL COUNTERFACTUAL, SBTC

		Δ ***model*** $\mid \Delta\ m\Big(\Phi = \{x_{t_2}, -x_{t_1}\}\ \Big\vert\ \Theta\,(t_1)\Big)$		
industry wages	$var(w)_{t_2}$	*level*	*share, model*	*share, data*
DATA	1.18			
MODEL	1.09			
$\Delta\ell\,(rt)$		2.62	2.42	2.22
$\Delta\ell\,(nrt)$		.35	.32	.29
$\Delta\ell$		1.32	1.22	1.12
$\Delta k\,(ict)$		1.99	1.84	1.69
$\Delta(\ell, ict)$		1.34	1.23	1.13

Quantification of Model D. Model implied between-industry real log-wage variance changes between two time spans uniformly calibrated, with changes according to variations in some series; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,t_2),\Phi(-x,t_1)\,\middle|\,\Theta(t_1)\right]}{m\left[\Phi(x,t_2)\,\middle|\,\Theta(t_2)\right]}$*, where $\Theta\,(t_1)$ and $\Theta\,(t_2)$ identify the set of parameters in the first and second periods, $\Phi\,(x, t_2)$ the set of capital and labour series where some of them are taken in the second period, while $\Phi\,(-x, t_1)$ reflects the set of all the series in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

TABLE C.13: MODEL COUNTERFACTUAL, SBTC BY TASKS

		Δ ***model*** $\mid \Delta\, m\Big(\Phi = \{x_{t_2}, -x_{t_1}\} \;\Big|\, \Theta(t_1)\Big)$		
wages	$var(w)_{t_2}$	*level*	*share, model*	*share, data*
ROUTINE				
DATA	1.14			
MODEL	.95			
$\Delta\ell\,(rt)$		1.21	1.28	1.06
$\Delta\ell\,(nrt)$		.20	.21	.18
$\Delta\ell$		.56	.59	.49
$\Delta k\,(ict)$		.86	.91	.75
$\Delta(\ell, ict)$		.53	.56	.47
NON-ROUTINE				
DATA	1.21			
MODEL	1.22			
$\Delta\ell\,(rt)$		4.04	3.30	3.34
$\Delta\ell\,(nrt)$		.50	.40	.41
$\Delta\ell$		2.09	1.70	1.73
$\Delta k\,(ict)$		3.13	2.56	2.59
$\Delta(\ell, ict)$		2.14	1.75	1.77

Quantification of Model D for routine and non-routine workers separately. Model implied between-industry real log-wage variance changes between two time spans uniformly calibrated, with changes according to variations in some series; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,t_2),\Phi(-x,t_1)\,\middle|\,\Theta(t_1)\right]}{m\left[\Phi(x,t_2)\,\middle|\,\Theta(t_2)\right]}$*, where* $\Theta(t_1)$ *and* $\Theta(t_2)$ *identify the set of parameters in the first and second periods,* $\Phi(x, t_2)$ *the set of capital and labour series where some of them are taken in the second period, while* $\Phi(-x, t_1)$ *reflects the set of all the series in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

D. Case under monopsony power

(Discussion) *In this appendix I am going to replicate all the analysis in the main text (calibration, estimation and counterfactuals from Model A and Model B) under the case in which firms are assumed to be monopolistically competitive and take the labour supply of each task-a as given. In this case, frictions due to a concentration in the labour market suggest that wages' formation is decided at firm level so that, taking labour supplies and industry variables as given, the profit-maximization problem for firm-h in industry-s becomes*

$$\max_{p_h(s),k_h(j,s),w_h(a,s)} \left[\mathcal{D}_h(s) \mid y_h(s), \ell_h(a,s) \right]$$

where firms have wage-setting power over employees. In other words, the profit maximization implies that the Cobb Douglas-nested CES production function in eq. (9) is subject to the labour supply curves specified by eq. (7), and to the conditional firm demand in eq. (8), thus exploiting both monopolistic and monopsonistic power, respectively, in combination. Whether one decides to assume or not a monopsony power by the part of firms (i.e., maximizing taking $\ell_h(a,s)$ as a constraint) does not change substantially the results and the conclusion of the main text. In fact, the only difference between optimal wages for routine and non-routine workers for a monopolistically competitive firm only, as in eq. (10) and in Appendix B, and that of a monopsonistic-monopolistically competitive firm is just the wage-markdown element $\mathcal{M}^\theta = \frac{\theta}{1+\theta}$ in both the composite parameters $\chi(a,s)$, a feature of firms when considering also a monopsonistic environment.

Since θ, measuring the degree of sorting and segregation effects in the economy, is the same for all industries, changes in variances due to changes in θ have an even effect in each industries, thus not changing the conclusion of the main text. This is because, as already noticed, the specification of the households' sorting choices comprises in itself the definition of monopsony power since the measure of workers of type-a in firm-h, industry-s is mostly determined by relative wages as in eq. (7).

TABLE D.1: SUMMARY OF CALIBRATION UNDER MONOPSONY

	parameter	*bottom*	*middle*	*top*	*global*	***source***
		value				
α	*physical capital, share of $y(s)$*	0.263	0.195	0.514		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in $y(s)$*	0.676	0.490	0.333		*MSM*
λ	*ICT capital share in $Q(s)$*	0.455	0.456	0.439		*MSM*
θ	*households' productivities dispersion*				11.4	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.329	0.420	0.249		*estimation*
σ	*EoS, routine and ICT composite*	0.634	0.400	0.766		*estimation*

Parameters by industry groups. Under "data" the values are directly computed from the data sources, while "external" sets standard values from the literature. "MSM" refers to the Methods of Simulated Moments, and "estimation" reports the previously estimated values from Table 5.

TABLE D.2: METHOD OF SIMULATED MOMENTS UNDER MONOPSONY

	parameter	*value*	***moment to match***	*data*	*model*
				fit	
μ_{bot}	weight of routines in $y(bot)$	0.6760	routine share, bottom	.0858	.1420
μ_{mid}	weight of routines in $y(mid)$	0.4902	routine share, middle	.1357	.1596
μ_{top}	weight of routines in $y(top)$	0.3331	routine share, top	.0985	.0637
λ_{bot}	weight of ICT in $Q(bot)$	0.4552	ICT share, bottom	.3968	.3968
λ_{mid}	weight of ICT in $Q(mid)$	0.4585	ICT share, middle	.3042	.3042
λ_{top}	weight of ICT in $Q(top)$	0.4398	ICT share, top	.2990	.2990
θ	productivity dispersion	11.376	wage premium, $w(a, \{s, s'\})$	.9945	.9945

Estimated values and related matched moment using the Methods of Simulated Moment.

TABLE D.3: MODEL FIT UNDER MONOPSONY, UNTARGETED MOMENTS

moment	*data*	*model*
	fit	
aggregate task-premium	.001	.005
aggregate wage	-.008	-.073
routine wage, bottom	-.016	-.070
routine wage, middle	-.001	-.051
routine wage, top	-.007	-.080
non-routine wage, bottom	-.019	-.060
non-routine wage, middle	.003	-.075
non-routine wage, top	-.010	-.046

Untargeted moments to match to validate the calibration strategy. All moments, referred to real log-wages, are taken as percentage changes throughout the series.

TABLE D.4: METHOD OF SIMULATED MOMENTS UNDER MONOPSONY, SUB-PERIODS

		2003-2012			**2013-2022**		
	moment to match	*value*	*data*	*model*	*value*	*data*	*model*
μ_{bot}	routine share, bottom	0.225	0.087	0.103	0.415	0.085	0.051
μ_{mid}	routine share, middle	0.492	0.132	0.061	0.506	0.140	0.030
μ_{top}	routine share, top	0.906	0.100	0.013	0.608	0.097	0.083
λ_{bot}	ICT share, bottom	0.647	0.399	0.399	0.786	0.395	0.395
λ_{mid}	ICT share, middle	0.518	0.314	0.314	0.490	0.295	0.295
λ_{top}	ICT share, top	0.170	0.287	0.287	0.326	0.311	0.311
θ	wage premium, $w(a, \{s, s'\})$	7.255	0.994	0.994	7.787	0.995	0.995

Estimated values and related matched moment using the Methods of Simulated Moments.

TABLE D.5: SUMMARY OF CALIBRATION UNDER MONOPSONY, 2003-2012

		value				
	parameter	*bottom*	*middle*	*top*	*global*	***source***
α	*physical capital, share of* $y(s)$	0.131	0.091	0.254		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in* $y(s)$	0.225	0.492	0.906		*MSM*
λ	*ICT capital share in* $Q(s)$	0.647	0.518	0.170		*MSM*
θ	*households' productivities dispersion*				7.26	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.355	0.431	0.408		*estimation*
σ	*EoS, routine and ICT composite*	0.366	0.429	0.367		*estimation*

Parameters by industry groups, first-half of the sample.. Under "data" the values are directly computed from the data sources, while "external" sets standard values from the literature. "MSM" refers to the Methods of Simulated Moments, and "estimation" reports the previously estimated values from Table 8.

TABLE D.6: SUMMARY OF CALIBRATION UNDER MONOPSONY, 2013-2022

		value				
	parameter	*bottom*	*middle*	*top*	*global*	***source***
α	*physical capital, share of* $y(s)$	0.131	0.104	0.260		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in* $y(s)$	0.415	0.506	0.608		*MSM*
λ	*ICT capital share in* $Q(s)$	0.786	0.490	0.326		*MSM*
θ	*households' productivities dispersion*				7.79	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.819	0.345	0.508		*estimation*
σ	*EoS, routine and ICT composite*	0.326	0.438	0.357		*estimation*

Parameters by industry groups, second-half of the sample. Under "data" the values are directly computed from the data sources, while "external" sets standard values from the literature. "MSM" refers to the Methods of Simulated Moments, and "estimation" reports the previously estimated values from Table 8.

TABLE D.7: MODEL COUNTERFACTUALS, MONOPSONY

industry wages	$var\,(w)_{t_2}$	Δ ***model*** $\mid \Delta\, m\Big(\Phi\,(t_2)\,,\Theta = \{\Psi_{t_2}, -\Psi_{t_1}\}\Big)$ *level*	*share, model*	*share, data*
		CHANGE		
DATA	1.18			
MODEL	1.08			
$\Delta\sigma$		2.36	2.19	1.99
$\Delta\rho$		.88	.82	.75
$\Delta(\sigma,\rho)$		1.12	1.03	.94
$\Delta\theta$		1.34	1.24	1.13
$\Delta(\sigma,\rho,\theta)$		1.16	1.07	.98
		FIXING		
DATA	1.18			
MODEL	1.08			
$\Delta\Theta_{\mid\sigma}$		.94	.87	.79
$\Delta\Theta_{\mid\rho}$		2.21	2.04	1.87
$\Delta\Theta_{\mid(\sigma,\rho)}$		1.31	1.21	1.11
$\Delta\Theta_{\mid\theta}$		1.04	.96	.88
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		1.30	1.20	1.10

Quantification of Model A and Model B given both monopsonistic and monopolistic power by the part of firms. Model implied between-industry real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by fixing a specified parameter, $\frac{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2),\Theta(-\Psi,t_1)\right]}{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2)\right]}$*, where* $\Phi\,(t_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (Ψ, t_2)*, are taken in the second period, while* $(-\Psi, t_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Opposite notation for "fixing". Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

TABLE D.8: MODEL COUNTERFACTUALS FOR ROUTINE WORKERS, MONOPSONY

routine wages	*var* $(w)_{t_2}$	Δ ***model*** $\mid \Delta\, m\left(\Phi\,(t_2)\,, \Theta = \{\Psi_{t_2}, -\Psi_{t_1}\}\right)$ *level*	*share, model*	*share, data*
		CHANGE		
DATA	1.14			
MODEL	.94			
$\Delta\sigma$		.97	1.03	.85
$\Delta\rho$		.33	.35	.29
$\Delta(\sigma,\rho)$		.64	.68	.57
$\Delta\theta$		.55	.59	.49
$\Delta(\sigma,\rho,\theta)$		.70	.74	.61
		FIXING		
DATA	1.14			
MODEL	.94			
$\Delta\Theta_{\mid\sigma}$		.71	.75	.62
$\Delta\Theta_{\mid\rho}$		1.15	1.22	1.01
$\Delta\Theta_{\mid(\sigma,\rho)}$		.84	.89	.73
$\Delta\Theta_{\mid\theta}$		.89	.95	.78
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		.81	.86	.71

Quantification of Model A and Model B given both monopsonistic and monopolistic power by the part of firms. Model implied between-industry routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by fixing a specified parameter, $\frac{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2),\Theta(-\Psi,t_1)\right]}{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2)\right]}$, *where* $\Phi\,(t_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (Ψ, t_2), *are taken in the second period, while* $(-\Psi, t_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Opposite notation for "fixing". Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

TABLE D.9: MODEL COUNTERFACTUALS FOR NON-ROUTINE WORKERS, MONOPSONY

non-routine wages	*var* $(w)_{t_2}$	Δ ***model*** $\mid \Delta\, m\Big(\Phi\,(t_2)\,, \Theta = \{\Psi_{t_2}, -\Psi_{t_1}\}\Big)$ *level*	*share, model*	*share, data*
		CHANGE		
DATA	1.21			
MODEL	1.22			
$\Delta\sigma$		3.76	3.08	3.10
$\Delta\rho$		1.44	1.18	1.19
$\Delta(\sigma, \rho)$		1.59	1.30	1.31
$\Delta\theta$		2.12	1.74	1.75
$\Delta(\sigma, \rho, \theta)$		1.63	1.33	1.34
		FIXING		
DATA	1.21			
MODEL	1.22			
$\Delta\Theta_{\mid\sigma}$		1.17	.96	.97
$\Delta\Theta_{\mid\rho}$		3.27	2.68	2.70
$\Delta\Theta_{\mid(\sigma,\rho)}$		1.78	1.46	1.47
$\Delta\Theta_{\mid\theta}$		1.18	.97	.97
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		1.79	1.47	1.48

Quantification of Model A and Model B given both monopsonistic and monopolistic power by the part of firms. Model implied between-industry non-routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by fixing a specified parameter, $\frac{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2),\Theta(-\Psi,t_1)\right]}{m\left[\Phi(t_2)\,\middle|\,\Theta(\Psi,t_2)\right]}$*, where* $\Phi\,(t_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (Ψ, t_2)*, are taken in the second period, while* $(-\Psi, t_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Opposite notation for "fixing". Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

References in appendix

ARVAI, K. and K. MANN, “Consumption inequality in the digital age”, *Banque de France Working Paper Series n. 890*, 2022. 36.

EDEN, M. and P. GAGGL, “On the welfare implications of automation”, *Review of Economic Dynamics*, 2018, 19:15-43. 36.

HICKS, J. R., *The Theory of Wages*, Palgrave Macmillan, London 1932. 47.

HORVATH, M., “Sectoral shocks and aggregate fluctuations”, *Journal of Monetary Economics*, 2000, 45(1):69-106. 47.

KARABARBOUNIS, L. and B. NEIMAN, “The global decline of the labor share”, *Quarterly Journal of Economics*, 2014, 29(1):61-103. 45, 53.

KLEINMANN, B., “Wage inequality and the spatial expansion of firms”, *Job Market Paper*, 2023. 41.

KOH, D., R. SANTAEULÀLIA-LLOPIS, and Y. ZHENG, “Labor share decline and Intellectual Property Products capital”, *Econometrica*, 2020, 88(6):2609-2628. 36.